DISSERTATION MANUSCRIPT

# Systems Analysis and Intelligent Technologies for Consistent Modeling of Hierarchical Video Data Generation Processes

Reach-efficient context memory and activation-efficient temporal optimization

**Leonid Ruslanovych Kulyk · VNTU · 2026**

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# Front Matter

## 1. Title Page

- **Topic (UA, approved, verbatim — fixed):** «Системний аналіз та інтелектуальні технології консистентного моделювання процесів ієрархічної генерації відеоданих»
- **Topic (EN):** *Systems Analysis and Intelligent Technologies for Consistent Modeling of Hierarchical Video Data Generation Processes*
- Candidate: Kulyk Leonid Ruslanovych. Specialty 124 «System Analysis», field 12 «Information Technologies». Supervisor: Prof. O. B. Mokin, Dr. Sc. (Eng.). VNTU, Vinnytsia — 2026.
- UDC: 004.054:[004.032.26+004.85] **[AUTHOR: confirm the UDC classification and the title-page quotation against the approved institutional record.]**

## 2. Abstract

The purpose of the dissertation research is to improve the efficiency and the temporal and spatial consistency of hierarchical video data generation processes through the development of systems analysis methods, mathematical models, and intelligent information technologies. The dissertation addresses the actual scientific and technical problem of systems analysis, mathematical modeling, and uncertainty resolution in complex multi-step systems for the hierarchical generation of spatio-temporal data.

The systems contribution separates two resource bottlenecks. Hierarchical multi-rate context packaging reduces the state memory required to reach distant history; reversible temporal optimization reduces the activation tape required to train long computations. In conceptual form, these act on different terms of $M_{\mathrm{system}}=M_{\mathrm{parameters}}+M_{\mathrm{optimizer}}+M_{\mathrm{context/state}}+M_{\mathrm{activations}}+M_{\mathrm{transient}}$. They are complementary dissertation strands rather than one causally integrated pipeline. Their shared prescription is to compress what must be remembered and reconstruct what need not be stored.

The introduction presents the relevance and justification of the research topic in the context of systems analysis requirements; describes the purpose, object, subject, and tasks of the research; states the scientific novelty; shows the connection of the dissertation with research programs; states the practical significance of the results, their approbation and publications; and describes the personal contribution of the applicant.

## Chapter-by-Chapter Summary

**Chapter 1** performs a systems analysis of hierarchical video data generation processes and the criteria of their consistency. Generative video models are analyzed as open complex dynamic control systems; structural, hierarchical, and temporal decompositions of the system state space are performed, and the types of uncertainty (informational, structural-algorithmic, computational, and target) are classified. The analysis of existing approaches establishes the central systemic contradiction of the field: state-of-the-art video generators extend a *flat* (single-level) spatio-temporal context, which leads to a rapid growth of attention memory and computational cost with context length and, simultaneously, to the degradation of temporal consistency over long horizons — content that leaves the field of view is not re-generated consistently upon return. It is further shown, based on the author's own experimental findings, that hierarchical video models over discrete (vector-quantized) token spaces are unstable to train at practically reachable scale, which motivates hierarchical modeling in *compact continuous hierarchical latent state spaces*. Energy-based models are reviewed as literature, with their resource barrier noted (their capability boundary is later characterized experimentally in Chapter 4). The criteria apparatus is improved by introducing new types of consistency (conceptual, hierarchical, trajectory), with trajectory consistency operationalized through measurable autoregressive-rollout and re-entry-stratified metrics. The research tasks are defined.

**Chapter 2** develops mathematical methods for uncertainty estimation and resource-efficient modeling of multi-step video processes. A method for estimating spatio-temporal uncertainty based on entropy metrics (the V-BLT architecture) is proposed, enabling adaptive decomposition of video data. The mathematical apparatus of N-dimensional rotary positional embedding (RoPE-ND) is developed to preserve the structural coherence of data, including the alignment of coordinate systems across abstraction levels of hierarchical models. A method for optimizing the computational resources of a complex system based on reversible transformations (TempVerseFormer) is proposed, and the time-agnostic backpropagation algorithm is mathematically substantiated, ensuring constant (independent of the optimization-horizon length) consumption of activation memory. The chapter closes with the extension of this apparatus to stochastic multi-level recurrences: an exactly invertible stochastic recurrence (the "store-the-noise" scheme) is proved and measured — reverse-replay gradients coincide with backpropagation through time (cosine similarity 1.000000), the invertibility error is at numerical precision ($5.5\times10^{-15}$ in fp64; $8.4\times10^{-13}$ for the stochastic scheme), and activation memory in a $K$-step rollout is $O(1)$ in $K$ — together with a three-axis applicability taxonomy (reversibility across depth, time, and computation) that delimits where reversibility pays (long temporal rollouts) and where it cannot (inner optimization loops, where the achievable saving versus gradient checkpointing is 0.26% measured, with an analytic band of 0.6–1.0%).

**Chapter 3** develops a mathematical model of multi-stream video sequence generation that packages long-range context across temporal rate and spatial resolution. The state space is a compact two-level hierarchical latent representation (a frozen two-level convolutional autoencoder interface with per-level decodable coarse and fine latents); its coarse stream is the measured spatially compressed implementation of long-history packaging, not the only possible carrier of the memory horizon. Per-frame generation uses an iterative refinement (denoising diffusion) core, selected empirically by the matched-capacity, matched-NFE mechanism analysis of Chapter 4. The inter-level transfer of context signals is realized within a single joint attention mechanism with per-token level, role, and frame conditioning (generalizing the cross-level cross-attention of the earlier design), with unidirectional control enforced by the information-design asymmetry of the two levels and by gradient detachment at the representation and rollout boundaries; the causality of the system is enforced by sequence construction and the autoregressive recursion; the autonomous autoregressive video-prediction algorithm rolls the model out window-by-window with a frozen pixel decoder. The chapter also gives the formal definition of the *memory-horizon looping testbed* and its re-entry-stratified evaluation protocol — a criterion-driven measurement instrument in which the hardness of every predicted frame ("recoverable only from memory") is an exactly computable analytic label — and derives the memory and complexity estimates of the rollout algorithm.

**Chapter 4** presents the experimental research and practical application of the developed methods and technologies. The information base includes procedurally synthesized datasets (2D shapes, 3D polycubes) used in the reversible-transformer studies, the purpose-built looping memory-horizon testbed (the primary benchmark: a $192\times64$ toroidal world observed through a panning $64\times64$ viewport, with 67% of the world off-screen at any instant and a re-entry loop period of approximately 48 frames), and a measured bridge evaluation on the standard Moving MNIST suite (canonical 10→10 protocol, n = 10,000; the model halves the persistence baseline's perceptual error, and the flat-versus-hierarchical null on this memory-horizon-free benchmark serves as the predicted negative control). A matched-capacity, matched-NFE comparison of prediction mechanisms (regression, diffusion, energy-based descent) on a shared backbone shows that regression averages over possible futures while diffusion commits to sharp, correctly located content (foreground error 0.0512 versus 0.0664; on the hard, genuinely generative subset 0.1549 versus 0.1808), and includes a two-sided capability study of Energy-Based Transformers using the official implementation — a characterized negative result: one published recipe yields a stably mis-ordered energy landscape, the other repairs the ordering, yet gradient descent from the prescribed initialization does not reach structured outputs at the scale reachable in this work. On a hash-pinned untouched LoopMom holdout, the registered pair-specific reach effects at the 896-token operating point are +0.0980 and +0.0914 across two training seeds, and the whole-population effects are +0.0474 and +0.0436. A simultaneous max-statistic family

confirms hierarchical packaging at 896 and 1,152 context tokens, leaves 1,792 unresolved, and confirms flat full-resolution context at 2,432 and 3,072 tokens. The frozen latent interface therefore supplies the same nominal 48-frame reach with 2.7–3.4 times fewer context tokens in the two confirmed hierarchical packages, while the upper-budget reversals show that compression is not free. All 20 registered sampler/NFE point estimates retain their direction, with 18 simultaneous lower bounds above zero; an independently trained continuous codec retains all four registered directions, although the 2,432-token inversion becomes family-wise unresolved. A privileged pose-indexed raw-memory reference reaches foreground MSE 0.0494 and establishes a remaining explicit-access gap without constituting a token-matched competitor. On the persistent-object 48→300 Moving-MNIST boundary, hierarchy preserves spatial support and continuity better, whereas flat context preserves digit identity better. The constant-activation-memory capability is demonstrated separately: classical backpropagation through time exhausts GPU memory at rollout depth $\mathrm{K}=120$, while the reversible implementation remains at approximately 1.1–1.2 GB through $\mathrm{K}=160$. The practical application of the developed methods in the tasks of ORNETIKA LLC and in the educational process of VNTU is described (per the existing implementation acts, which cover the reversible-transformer method and entropy tokenization). [A2]'s published 2-D rotating-shape validation is reported as the receipt of the reversibility method; [A1] supplies the 2-D/3-D entropy-tokenization examples. Moving MNIST and TaxiBJ are separate dissertation bridges, while MMNIST-CIFAR and KTH remain explicitly unexecuted extensions. In the valid persistent-object 48→300 Moving-MNIST bridge, hierarchy lowers union error on all three seeds and improves trajectory/coverage cleanly on two, while flat context preserves digit identity on all three; foreground-only error is unresolved. The result is reported as a mixed external boundary rather than a universal model ranking.

The general conclusions present the main results of the dissertation research, confirming the fulfillment of all the set tasks, with honest scoping: the hierarchy is validated where its reach covers memory-relevant history under a bounded budget, while full-resolution fidelity can recover at larger tested budgets. A seed-0 late self-fed inversion did not replicate at two new training seeds and is therefore not stated as a general trust boundary. The energy-based mechanism is a characterized negative; reversibility is scoped by its applicability taxonomy. The four central reach–fidelity directions survive all registered NFE/sampler variants and an independently trained continuous codec at the point-estimate level; the upper inversion is explicitly retained as a family-wise precision boundary under alternate sampler and codec conditions. A separate privileged pose-indexed raw-memory reference reaches foreground MSE 0.0494 and beats all four registered neural references with positive simultaneous lower bounds, showing that geometric reach and reliable explicit addressing are distinct while making no token-matched model claim. The protected persistent-object long Moving-MNIST result further

separates the axes: coarse long-memory packaging preserves spatial support and continuity, whereas flat full-resolution context preserves fine digit identity.

## 7 items, mapped 1:1 to the 7 tasks

1. Characterized the process of hierarchical video data generation as a complex control system with inherent informational, structural-algorithmic, computational, and target uncertainties; performed structural, hierarchical, and temporal decompositions of the system and identified the central systemic contradiction — the resource blow-up and consistency degradation of flat long-context modeling — that the subsequent chapters resolve. *(Task 1)*

2. Improved the criteria apparatus of systems analysis of generative processes by introducing the criteria of conceptual, hierarchical, and trajectory consistency, and operationalized trajectory consistency through measurable autoregressive-rollout metrics stratified by an exactly computable re-entry hardness label. *(Task 2)*

3. Developed a method for estimating the state of uncertainty of spatio-temporal data through their adaptive decomposition based on entropy metrics (V-BLT) with N-dimensional rotary positional embedding, which allows dynamic allocation of computational resources to high-information zones of the scene and mathematical alignment of coordinate systems between abstraction levels of hierarchical models. *(Task 3)*

4. Developed a method for the computationally efficient optimization of multi-step iterative systems based on reversible transformations for time-agnostic backpropagation, ensuring constant activation-memory consumption independent of the temporal horizon; extended it to stochastic multi-level recurrences via an exactly invertible stochastic recurrence (store-the-noise; gradients coincide with backpropagation through time), and delimited its applicability domain by a three-axis taxonomy. *(Task 4)*

5. Developed a mathematical model of multi-stream video generation in compact hierarchical latent state spaces, supporting context streams at different temporal rates and spatial resolutions over a shared attention backbone, with unidirectional context transfer (gradient detachment at the representation and rollout boundaries) and a per-frame iterative-refinement generation core selected by controlled mechanism analysis; measured spatial and temporal packages restore content beyond a contiguous flat window at bounded token budgets. *(Task 5)*

6. Created an intelligent information technology and a software complex (Python/PyTorch) for the systems analysis of hierarchical video generation processes, integrating the developed methods of adaptive tokenization, resource-efficient optimization, and hierarchical generation, together with the

purpose-built memory-horizon looping testbed and its re-entry-stratified evaluation protocol as a first-class measurement instrument. *(Task 6)*

7. Conducted comprehensive experimental verification of the developed methods and technology:

a matched-capacity, matched-NFE comparison of prediction mechanisms (including the two-sided characterized study of energy-based descent), single-step and 152-frame autoregressive-rollout evaluations with clip-clustered paired-bootstrap confidence intervals, and memory-consumption measurements; the developed system outperforms the token-budget-matched flat control at the tested lower/interior points where its package alone reaches memory-relevant history, while tested larger budgets expose a full-resolution-fidelity inversion; the boundary conditions of every claim are stated explicitly. *(Task 7)*

The main results of the dissertation research are published in 10 scientific works: 3 articles in professional journals of Ukraine in specialty 124; 1 article indexed in the Scopus scientometric database; and 6 conference abstracts. The final bibliographic details of the fourth article and the six conference abstracts require the applicant's publication records.

## Scientific Novelty

1. **For the first time**, a methodology for the systems analysis and evaluation of long-range

memory in hierarchical video generation processes was developed — a purpose-built memory-horizon testbed whose re-entry hardness labels are exactly computable from recorded control tracks, with a re-entry-stratified evaluation protocol and a matched-context-token-budget comparison design — and it was demonstrated with its help that **long-span multi-stream context packaging** — a hierarchical allocation of a fixed context-token budget across spatial resolution and temporal rate, of which the coarse-level-memory paradigm of hierarchical latent video models (§1.3.2, adopted prior art) is the measured spatial form and strided full-resolution history the measured temporal form — restores content that re-enters the field of view beyond the horizon of a flat context window at the same context-token budget, with tested lower/interior and upper regimes bracketed by explicit token arithmetic; unlike existing evaluation practice, which aggregates prediction error without isolating memory-dependent content, the developed methodology makes the memory contribution of the packaging directly measurable and localizes the improvement of temporal consistency to registered reach-differential populations.

2. **For the first time**, a method for the computationally efficient optimization of processes in

multi-step iterative systems was developed based on reversible transformations for time-agnostic backpropagation, which, unlike classical backpropagation through time, ensures constant memory consumption regardless of the depth of the temporal horizon; the method is extended to stochastic

multi-level recurrences via an exactly invertible stochastic recurrence with noise storage, and its applicability domain is delimited by a three-axis taxonomy of reversibility (across depth, time, and computation).

3. **Improved** the criteria apparatus of systems analysis of the results of generative models by introducing the criteria of conceptual consistency (the ability to generate unknown combinations), hierarchical consistency (alignment of structural levels), and trajectory consistency, the latter operationalized through autoregressive-rollout degradation metrics stratified by re-entry hardness, which allows a more comprehensive and measurable assessment of the adequacy of complex generative systems.

4. **Improved** the method for assessing the state of uncertainty of spatio-temporal data through their adaptive decomposition using N-dimensional rotary positional embedding and per-pixel entropy maps, which allows dynamic allocation of computational resources during the analysis of complex visual scenes.

## implementation acts — kept as-is, they match reality

- practical activities of **ORNETIKA LLC** (Kyiv): the developed information technology was used in designing intelligent systems for the analysis of spatio-temporal video data and anomaly detection in complex technical environments; the resource-efficient methods (TempVerseFormer) and adaptive entropy tokenization substantially reduced the computational cost of modeling long temporal sequences (implementation act of October 14, 2025);

- the **educational process of VNTU** in the disciplines *Information Technologies for Monitoring and Analysing the State of Complex Systems *and* Internet of Things and Intelligent Data Analysis* for PhD students of specialty 124 (implementation act of September 19, 2025).

## Practical significance

The practical significance lies in the development of a complex of algorithms and software tools implementing the developed intelligent information technology for the systems analysis of hierarchical video generation processes (Python/PyTorch; repository modules for the hierarchical model, the reversible recurrence, and the testbed generator). Unique synthetic spatio-temporal datasets were created for testing generation and optimization systems: 2D shapes, 3D polycubes, and the looping memory-horizon testbed with recorded action tracks and analytic hardness labels (3,000 training clips × 100 frames; 500 test clips × 200 frames). The proposed technology can be applied to state estimation and prediction of future states of complex systems in technical, medical, and environmental monitoring.

### Keywords

systems analysis; complex system; mathematical modeling; intelligent information technology; decision making; neural network; deep learning; artificial intelligence; video data generation; **hierarchical video generation**; **compact hierarchical latent state spaces**; **memory horizon**; **temporal consistency**; **context packaging**; **reversible transformations**; **diffusion refinement core**; autoregressive prediction; state estimation; forecasting; consistency.

## 3. Applicant's Publications

The following stable labels are used throughout the dissertation:

- **[A1]** — Kulyk L. R., Mokin O. B. «Scaling Video Prediction with Spatio-Temporal Patches»,

*Visnyk of Vinnytsia Politechnical Institute*, 2025, No. 5, pp. 129–139, DOI 10.31649/1997-9266-2025-182-5-129-139.

- **[A2]** — Kulyk L. «Temporal modeling with reversible transformers» (TempVerseFormer),

*Neural Computing and Applications*, 2026, Vol. 38, article 30, DOI 10.1007/s00521-025-11707-3, sole author, Scopus-indexed.

- **[A3]** — Kulyk L. R., Mokin O. B. «Methods for Ensuring Consistent Generation in Diffusion Models», *Visnyk of Vinnytsia Politechnical Institute*, 2024, No. 4, pp. 75–85, DOI 10.31649/1997-9266-2024-175-4-75-85.

- **[A4]** — the fourth article (in finalization): a controlled causal study of long-span context packaging (rate × resolution × depth) for long-horizon video prediction — the memory-horizon testbed, the matched-token-budget frontier with measured edges, the shape study, the matched-capacity, matched-NFE mechanism study including the EBT characterized negative, and the supporting reversibility theory. **[AUTHOR: supply target venue and status.]**

- Six conference theses. **[AUTHOR: supply the six verified bibliographic entries.]**

## 4. Symbols and Abbreviations

| Abbreviation | Expansion |
|---|---|
| VNTU | Vinnytsia National Technical University |
| AE | Autoencoder |
| AR | Autoregressive |
| BPTT | Backpropagation Through Time |

| Abbreviation | Expansion |
|---|---|
| CI | Confidence Interval |
| CRN | Common Random Numbers |
| DDIM | Denoising Diffusion Implicit Models |
| DDPM | Denoising Diffusion Probabilistic Models |
| DiT | Diffusion Transformer |
| EBM | Energy-Based Model |
| EBT | Energy-Based Transformer |
| EMA | Exponential Moving Average |
| fg-MSE | Foreground Mean Squared Error |
| GPU | Graphics Processing Unit |
| LPIPS | Learned Perceptual Image Patch Similarity |
| MCMC | Markov Chain Monte Carlo |
| MSE | Mean Squared Error |
| NFE | Number of Function Evaluations |
| OOM | Out of Memory |
| RoPE-ND | N-dimensional Rotary Positional Embedding |
| SSIM | Structural Similarity Index Measure |
| V-BLT | Video Byte-Latent Transformer |
| VAE | Variational Autoencoder |

## 5. Table of Contents


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- General Conclusions
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OpenSTL/Moving-MNIST bridge

# Introduction

## Justification and Relevance

The rapid development of information technologies and artificial-intelligence methods over the last decade has produced new classes of mathematical models capable of solving complex problems of analysis, prediction, and synthesis of multidimensional data. A special place among them is occupied by generative video models, which have evolved from simple statistical tools into highly complex multi-level systems capable of forming high-quality spatio-temporal video data. Such systems are critically important for many applied fields: from medical diagnostics and environmental monitoring to the prediction of the state of technical systems and the construction of simulators for autonomous control.

However, despite significant progress, the control of the generation process remains insufficiently formalized from the standpoint of systems analysis methodology. Modern architectures — diffusion and autoregressive video models — are mostly treated as "black-box" deep-learning algorithms. Operating under high structural, informational, and target uncertainty, these systems are prone to generating anomalies — spatial, logical, or temporal inconsistencies — which makes their outputs unreliable for decision-making. The use of single-level (flat) latent spaces mixes the levels of abstraction of the visual scene (semantic composition, geometry, texture) within one representation, which leads to the loss of hierarchical control over the generation process.

The dominant engineering answer to the consistency problem — extending a *flat* spatio-temporal context window — runs into an objective systemic barrier. First, the computational cost and memory of the attention mechanism grow rapidly with the number of context tokens (quadratically in computation, linearly in the key–value cache), so long flat contexts are resource-prohibitive precisely at the horizons where they are most needed. Second, even where a long flat context is affordable, the temporal consistency of long generated sequences degrades: content that leaves the field of view is not restored consistently upon its return, and autoregressive error accumulation (exposure bias) destabilizes the trajectory of the generation process. Third, the identification of parameters of long-horizon models by backpropagation through time requires activation memory that grows linearly with the number of steps of the modeled process, which makes deep training trajectories unattainable on practically available computing systems.

These are two different memory bottlenecks and the dissertation treats them separately. For systems analysis, peak memory is decomposed conceptually as

$$M_{\mathrm{system}} = M_{\mathrm{parameters}} + M_{\mathrm{optimizer}} + M_{\mathrm{context/state}} + M_{\mathrm{activations}} + M_{\mathrm{transient}}.$$

The hierarchical context-packaging model of Chapters 3–4 reduces $M_{\mathrm{context/state}}$ for a required temporal reach by storing distant history at a lower spatial rate or a lower temporal rate. The reversible TempVerseFormer method of Chapter 2 reduces $M_{\mathrm{activations}}$ for long temporal computation by reconstructing intermediate states during the backward pass. These are complementary resource axes, not one integrated mechanism and not evidence that their savings automatically multiply. The common design principle is: **compress what must be remembered; reconstruct what need not be stored**.

A natural remedy is the hierarchical decomposition of the latent state space. Discrete vector-quantized hierarchies are an important reviewed family, but they also introduce codebook and assignment-specific optimization risks; this dissertation does not present a controlled VQ-versus-continuous comparison. The implemented direction is therefore stated constructively: hierarchical modeling of video generation processes in **compact continuous hierarchical latent state spaces**, in which multiple context streams can package long-range information at different temporal rates and spatial resolutions, context signals are transferred unidirectionally (with gradient detachment), per-frame generation is performed by an iterative refinement (denoising diffusion) core selected by controlled mechanism analysis, and the memory barrier of long-horizon activation-memory growth is bounded by reversible transformations that make backpropagation time-agnostic within their measured applicability domain. Chapter 1 explicitly marks the representation evidence boundary; Chapter 4 tests robustness to a second continuous codec seed rather than claiming representation-family universality. That intervention preserves all four registered directions; three contrasts remain simultaneous-CI clean, while the alternate-codec 2,432-token inversion remains positive but family-wise unresolved. A separately gated exact-pose raw-memory reference then measures the remaining explicit-access gap: its 48-frame mosaic attains foreground MSE 0.0494 and all four registered neural-minus-reference simultaneous lower bounds are positive. Because this reference receives raw pixels and exact future poses, it is reported as a privileged ceiling rather than a token-matched competitor.

Thus an objective systemic contradiction arises between the necessity of ensuring multi-level consistency of video data over long time horizons (which requires long memory horizons and deep hierarchies) and the strict resource constraints of modern computing systems (the attention-memory blow-up of flat contexts and the linear growth of training memory with horizon length). Resolving this contradiction requires the creation of new mathematical methods of decomposition, resource-efficient control, and intelligent information technologies of systems analysis, which determines the high relevance of this dissertation research.

## Relationship to Research Programmes, Plans and Topics

The dissertation was carried out in accordance with the research plans of Vinnytsia National Technical University, in particular within the state-budget research topics:

- Research topic No. **[AUTHOR: research-project number 1]**, *Development of Information

Technologies and Mathematical Methods for Intelligent Analysis of Spatio-Temporal Data in Complex Systems* (2021–2023), in which the applicant developed the method of adaptive estimation of spatio-temporal uncertainty and entropy-based tokenization of video data;

- Research topic No. **[AUTHOR: research-project number 2]**, *Systems Analysis and Modelling of

Multi-Step Processes under Resource Constraints and Uncertainty *(2024–2026), in which the applicant developed the method of computational-resource optimization based on reversible transformations and the hierarchical generative model of video data. [AUTHOR: verify both approved project titles and the applicant's role against the official project reports.]**

## Aim and Research Tasks

**The purpose of the dissertation research** is to improve the efficiency and the temporal and spatial consistency of hierarchical video data generation processes through the development of systems analysis methods, mathematical models, and intelligent information technologies.

The dissertation's components serve **one backbone question**: *how can hierarchical decomposition, temporal abstraction, and resource-aware rollout improve the consistency of long-horizon video generation under bounded memory and computation?* Each contribution has a defined role in answering it: adaptive entropy-based decomposition and RoPE-ND (Chapter 2) — how to represent uncertain visual information efficiently; reversible transformations and time-agnostic backpropagation (Chapter 2) — how to train and roll out long processes within memory limits; the multi-stream hierarchical predictor (Chapter 3) — the demonstrated system; the memory-horizon testbed and re-entry-stratified protocol (Chapters 3–4) — the causal instrument for measuring whether context memory actually matters; the frontier and shape experiments (Chapter 4) — the evidence for how to allocate a fixed context budget; the energy-based study (Chapter 4) — the measured mechanism-selection negative.

For the fourth article, context packaging is the manipulated design space. **Reach** asks whether the package contains the memory-relevant past; **fidelity** asks what useful information survives compression or subsampling; reliable reuse under self-fed generation is treated as a robustness question because the proposed late trust boundary did not replicate. The terminal empirical summary is conditional: **hierarchy buys reach; flat context buys fidelity**.

To achieve this purpose, the following scientific and practical tasks had to be solved:

1. Analyze the state and problems of generative video-data models as complex systems; classify the types of uncertainty and identify the systemic causes of anomalies (inconsistencies) in the generation process, including the resource and consistency limitations of flat long-context modeling.

2. Improve the criteria apparatus of systems analysis of generative processes by extending the taxonomy of consistency types (conceptual, hierarchical, trajectory) and operationalizing trajectory consistency through measurable rollout metrics stratified by re-entry hardness.

3. Develop a method for estimating the state of uncertainty of spatio-temporal data through their adaptive decomposition using entropy metrics and the mathematical apparatus of multidimensional rotary positional embedding.

4. Develop a method for optimizing the computational resources of multi-step iterative models that ensures constant, process-length-independent consumption of memory on the basis of reversible mathematical transformations; extend it to stochastic multi-level recurrences and delimit its domain of applicability.

5. Develop a mathematical model of multi-stream video generation in compact hierarchical latent state spaces, supporting context streams at different temporal rates and spatial resolutions over a shared attention backbone, with unidirectional transfer of context signals (gradient detachment at the representation and rollout boundaries) and a per-frame iterative-refinement generation core selected by controlled mechanism analysis.

6. Create an intelligent information technology and a software complex for the systems analysis of hierarchical video generation processes, integrating the developed methods of adaptive tokenization, resource optimization, and hierarchical generation, together with a purpose-built memory-horizon testbed and re-entry-stratified evaluation protocol as its measurement instrument.

7. Conduct comprehensive experimental studies and approbation of the developed methods and technology on synthetic and open datasets — including a matched-capacity, matched-NFE comparison of prediction mechanisms — and compare their efficiency and consistency with modern analogues.

## Object and Subject of Research

**The object of research** is the processes of systems analysis, mathematical modeling, and control of complex multi-level systems of video data generation under structural, informational, and computational uncertainty.

**The subject of research** is the mathematical models, methods, and intelligent information technologies of systems analysis, uncertainty estimation, criteria-based evaluation, and control in hierarchical generative systems.

## Research Methods

To solve the set tasks, a complex apparatus of modern science was used: the methodology of systems analysis for the decomposition of generative processes and the classification of uncertainty and consistency types; probability theory, mathematical statistics, and information theory for entropy estimation and the assessment of spatio-temporal uncertainty of data; the theory of stochastic processes for the mathematical description of the iterative refinement (denoising diffusion) generation core, with the Markov-chain formalism applied insofar as autoregressive generation constitutes a Markovian state transition and, within the experimental mechanism study, for the description of the gradient-descent (Langevin) sampling of energy-based models; optimization methods and matrix/tensor algebra for the formalization of training objectives and the development of reversible transformations and attention mechanisms; machine- and deep-learning methods, including backpropagation, for the identification of parameters of the mathematical models; elements of decision-making theory for the construction of criteria-based decision procedures — the pre-registered decision rules of the experimental program and the re-entry-stratified evaluation protocol, which operationalizes the acceptance or rejection of model configurations as a formal criteria-based choice; and methods of statistical inference (paired bootstrap confidence intervals, common random numbers) for the experimental evaluation protocol.

- In **[A1]** (hierarchy/BLT-adaptation article), the applicant proposed and implemented the

method of adaptive decomposition of spatio-temporal data based on entropy metrics (the V-BLT architecture) with multidimensional rotary positional embedding, and the procedural synthesis of video datasets. The two-stage V-BLT architecture (local byte-level encoder/decoder around a global latent transformer) is the applicant's **first realization of the hierarchical adaptation of video generation** — the two-level decomposition whose mature, memory-carrying form Chapter 3 develops; the dissertation's hierarchical line thus runs [A1] (two-stage entropy-adaptive hierarchy) → [A4] (two-level latent hierarchy with coarse-level long-range memory).

- In **[A2]** (TempVerseFormer, Springer *Neural Computing and Applications* 2026, Scopus, sole

author), the applicant developed and mathematically substantiated the algorithm of time-agnostic backpropagation based on reversible transformations, which resolved the computational uncertainty of long-horizon training. The journal is described only as Scopus-indexed; no unverified quartile claim is used.

- In **[A3]** (consistency-criteria article), the applicant improved the criteria apparatus for

the evaluation of complex generative systems by formalizing the notion of «conceptual consistency».

- In **[A4]** (the fourth article, in finalization) and within the dissertation research, the

applicant developed the hierarchical latent video generation model with long-range coarse-level memory; constructed the memory-horizon looping testbed and the re-entry-stratified evaluation protocol; conducted the matched-capacity, matched-NFE study of prediction mechanisms, including the two-sided capability study of energy-based transformers (a characterized negative result); developed the theory of the exactly invertible stochastic recurrence with the three-axis reversibility taxonomy; and performed all complex experimental studies.

## Validation and Presentation of Results

The main provisions, results, and conclusions of the dissertation were reported and discussed at 6 scientific conferences: **[AUTHOR: verify every event below against the applicant's records and bind each event to the corresponding conference publication.]**

- 50th Scientific and Technical Conference of VNTU Departments, Vinnytsia, 2021;
- 51st Scientific and Technical Conference of VNTU Departments, Vinnytsia, 2022;
- *Problems of Higher Mathematical Education: Contemporary Challenges*, VNTU, Vinnytsia, 2022;
- 52nd Scientific and Technical Conference of VNTU Departments, Vinnytsia, 2023;
- 53rd Scientific and Technical Conference of VNTU Departments, Vinnytsia, 2024;
- Dynamical System Modeling and Stability Investigation (DSMSI-2023), Kyiv, 2023.

## Publications

Based on the dissertation topic, 10 scientific works were published, including: 3 articles in professional journals of Ukraine included in the Ministry of Education and Science list for specialty 124 ([A1], [A3], [A4]); 1 article in an international scientific journal indexed by the Scopus scientometric database ([A2]); and 6 abstracts of reports at international and all-Ukrainian scientific-practical conferences. **[AUTHOR: supply the venue/status of [A4] and the six conference entries, then confirm that the 3+1+6 count is correct.]**

## Dissertation Structure and Length

The dissertation consists of an introduction, four chapters, conclusions to each chapter, general conclusions, a list of references, and appendices.

- **Chapter 1** performs the systems analysis of hierarchical video generation processes and the

criteria of their consistency: generative video models as complex dynamic control systems; structural, hierarchical, and temporal decomposition; the classification of uncertainty types; the analysis of existing approaches (flat long-context diffusion and autoregressive generators and their resource/consistency barrier; hierarchical latent modeling as the remedy; the instability of discrete hierarchical video models; energy-based models as reviewed literature); the improved consistency-criteria apparatus; and the statement of the research tasks.

- **Chapter 2** develops the mathematical methods for uncertainty estimation and

resource-efficient modeling of multi-step video processes: the entropy-based V-BLT method, RoPE-ND, reversible transformations with temporal chaining (TempVerseFormer), time-agnostic backpropagation, and the exactly invertible stochastic recurrence with the three-axis reversibility taxonomy.

- **Chapter 3** develops the multi-stream mathematical model of video sequence generation with

context packaged across temporal rate and spatial resolution in compact hierarchical latent state spaces: the two-level latent state space, the iterative-refinement generation core, joint inter-level attention with unidirectional control (information-design asymmetry plus gradient detachment at the representation and rollout boundaries), the autoregressive rollout algorithm with its memory and complexity estimates, and the formal definition of the memory-horizon testbed with the re-entry-stratified evaluation protocol.

- **Chapter 4** presents the experimental research and practical application: the information

base; the matched-capacity, matched-NFE mechanism study (including the two-sided characterized study of energy-based descent); the single-step and 152-frame autoregressive-rollout (after a 48-frame ground-truth warm-up) evaluations of the hierarchical model; the memory-consumption measurements; and the implementations.

The total volume of the work is **[AUTHOR: final page count]** pages; the main part comprises **[AUTHOR: final main-text page count]** pages, with the final figure and table counts obtained mechanically during typesetting. The bound scientific bibliography currently contains 79 external sources. The appendices contain copies of the implementation acts, the preregistration trail, and the software and experiment-configuration map of the developed information technology.

# Chapter 1 — Systems Analysis of Hierarchical Video Generation and Its Consistency Criteria

## 1.1 Generative video models as complex control systems

The development of artificial-intelligence and machine-learning methods over the last decade has produced new classes of mathematical models capable of solving complex problems of data analysis, prediction, and synthesis. A special place among them is occupied by generative models, which have transformed from simple statistical tools into extremely complex multi-level information systems. Modern generative models — in particular those operating on spatio-temporal video data — hold billions of parameters and demand enormous computational resources [1]. Yet despite impressive visual results, the control of the generation process in such models remains insufficiently formalized. For the most part these models are treated as "black-box" deep-learning algorithms, which complicates guaranteeing the stability of their operation, the predictability of their results, and their resource efficiency [2, 3].

To address these problems it is necessary to change the paradigm of perceiving generative algorithms and to consider them as open complex dynamic systems. In this formulation the process of video data generation is not merely a mathematical transformation of information but a purposeful process of changing the state of a complex system under control actions and the influence of the external environment. Iterative generative models — diffusion models [4], autoregressive systems, and energy-based models employing Markov chain Monte Carlo — fall squarely under the definition of a complex dynamic control system.

A complex system controlling the process of video data generation is characterized by input actions, a state space, and output data. The input actions of such a system are the initial noise distribution, which plays the role of an entropy source, and a set of conditioning vectors (for example, textual descriptions, semantic masks, previous frames of the video sequence, or recorded control/action tracks), which play the role of control signals. The state space is the multidimensional latent or pixel space in which the system resides at every discrete moment of time. The output data is the final generated object, which must satisfy the specified criteria of quality and consistency. The scheme of a modern generative video model as an example of an open complex dynamic control system is shown in Fig. 1.1.

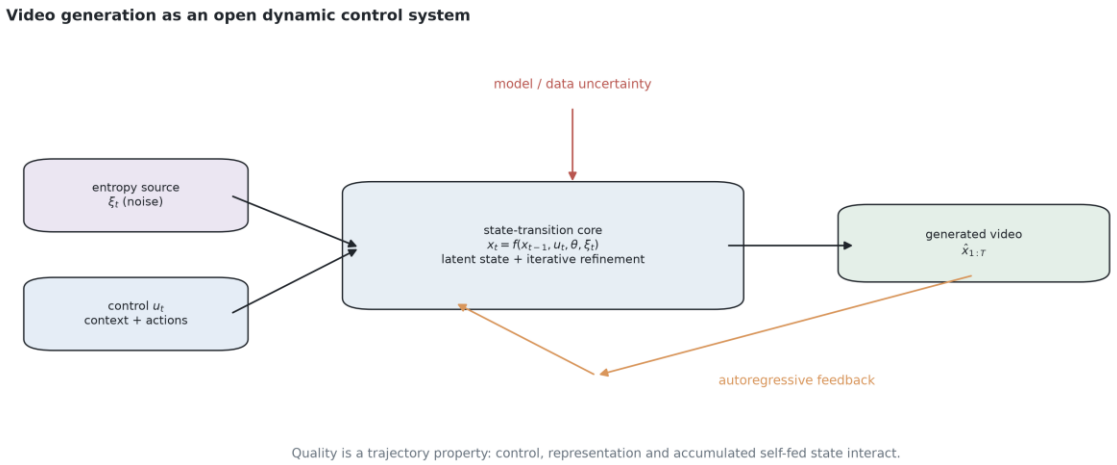



**Figure 1.1 — Video generation as an open dynamic control system.** Noise acts as an entropy source, previous frames and actions as control signals, and autoregressive feedback makes output quality a property of the complete self-fed trajectory. Source: author's schematic.

In accordance with classical systems theory [5], the object under study is characterized by a large number of interrelated elements, nonlinear interactions, emergence, and a hierarchical structure. In the context of computer vision the elements of the system are individual pixels or latent features (tokens). Their interaction is provided by attention mechanisms and convolutional operations, which model spatio-temporal correlations. Nonlinearity is achieved through nonlinear activation functions in the artificial neural networks that form the core of the system.

An important characteristic of such a system is the iterative nature of its state transitions. Unlike generative adversarial networks (GAN) [6] or variational autoencoders (VAE) [7], which attempt to map an input noise vector to an output image in a single computation step, iterative models such as denoising diffusion probabilistic models (DDPM) [4] decompose this process into a multitude of discrete steps. This approach fundamentally changes the mathematical nature of the system: the state evolves in time (or in the pseudo-time of optimization steps), and every next state depends on the previous one, the current parameters, and the control action.

Mathematically, the evolution of the state of such a system is described by a recurrent equation of Markov type:

$$x_t = f(x_\{t-1\}, u_t, \theta, \xi_t) \tag{1.1}$$

where x_t is the system state (a multidimensional tensor of video data or its latent representations) at step t; u_t is the vector of control actions (generation conditions); \theta is the parameter matrix of the system (the neural network weights); \xi_t is the stochastic component (the random number generator); and f is the nonlinear transition function describing the law of state change.

The presence of the stochastic component \xi_t indicates that the system operates under uncertainty. Every generation step is a probabilistic process, and the system produces not a deterministic result but a sample from a specified multidimensional probability distribution. According to research on complex systems under risk and uncertainty [8, 9], such operating conditions make the system extremely sensitive to initial conditions and create the threat of anomalous states. In generative models this manifests as artifacts in images or violations of physical logic in video.

A complex system always possesses a hierarchical structure [10], where every level describes the object with a certain degree of detail or abstraction. In video models this hierarchy is expressed with particular clarity. At the **macro level — the highest level of abstraction** — the system operates with semantic concepts (global style, the mutual arrangement of objects, scene composition). At the **meso level**, structural contours and geometry are determined. At the **micro level — the lowest level of abstraction, the highest level of detail** — high-frequency textures, lighting, and individual pixels are formed.

It is the hierarchical structure that gives rise to such a systemic property as emergence: a meaningful image arises only thanks to complex spatial and temporal correlations between elements at different levels of the hierarchy. If the system controlling the generation process cannot ensure coherence between these levels, the emergent property is destroyed (for example, an object receives a correct texture imposed on a distorted global shape). Hence the task of ensuring stable generation reduces to the task of ensuring the stability of a complex hierarchical system.

### 1.1.1 Generation as an optimal-control problem

Traditional deep-learning approaches attempt to solve the stability problem extensively, by increasing the number of parameters \theta and the volume of the training sample. This path, however, has physical limits and leads to exponential growth of computational complexity. A need therefore arises for mathematical methods of optimal control of the generation trajectory.

Consider the generation process as an optimal-control problem. At the initial moment $t = 0$ the system state $x_0$ is fully chaotic (a pure-noise distribution), corresponding to the state of maximal informational uncertainty. At the final moment $t = T$ the system must arrive at a state $x_T$ belonging to the target distribution of real video data (a state of high internal structural organization).

The formalization of this process requires a systemic criteria apparatus. In the present research the quality of the terminal state and of the trajectory leading to it is described by a **consistency criterion functional** $J(x, u)$, which quantitatively expresses the degree of non-correspondence between the current state $x$ and the target concept under conditions $u$. Generation then becomes a constrained optimization problem over the state space:

$$x^{*} = \arg\min_{x} J(x, u) \tag{1.2}$$

subject to the state-transition law (1.1). The smaller the value of the criterion, the more consistent and realistic the generated object.

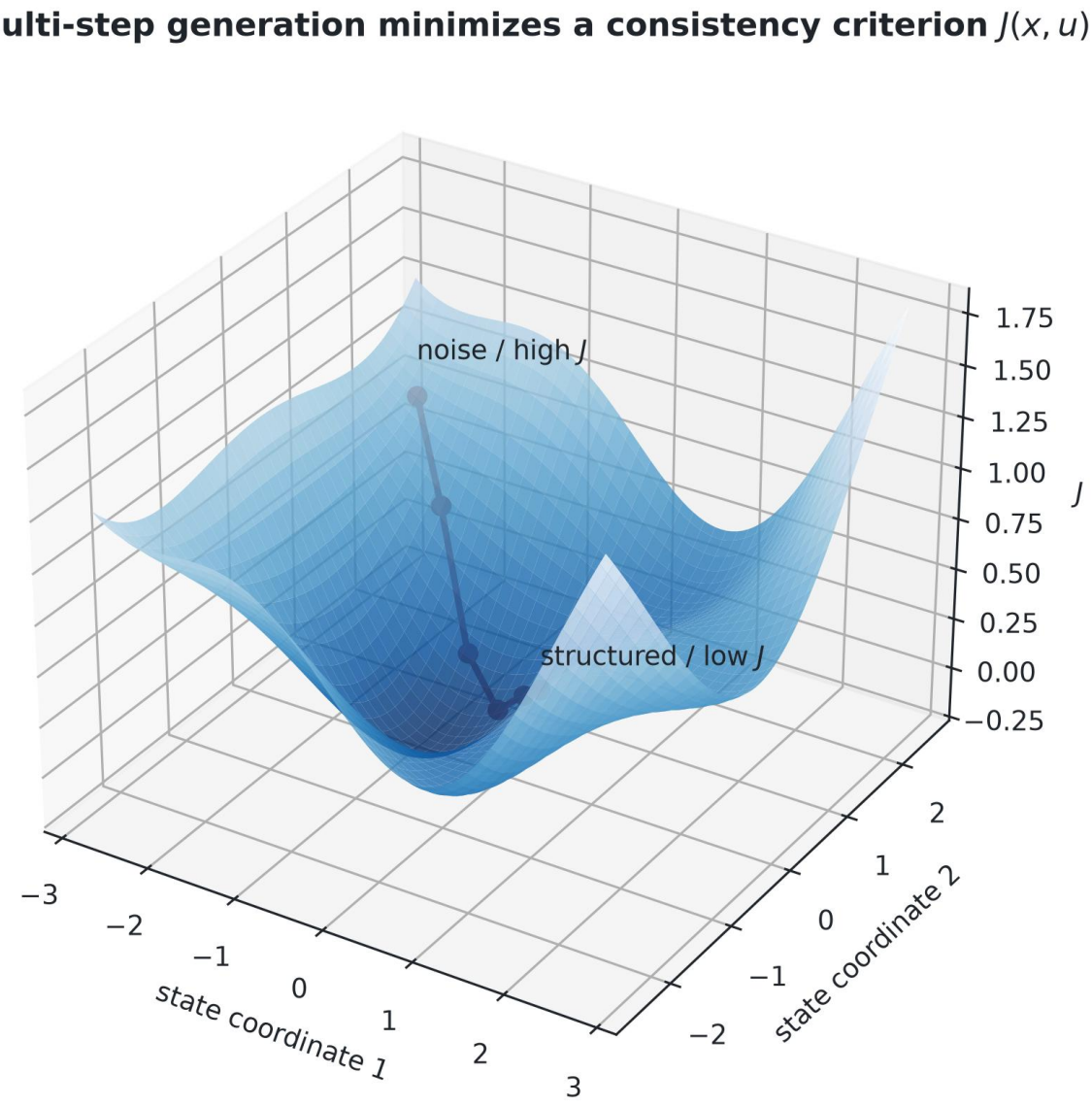



**Figure 1.2 — Conceptual multi-step generation on the landscape of a consistency criterion $J(x,u)$.** The trajectory moves from a high-criterion noise state toward a structured low-criterion region. This is a systems-level abstraction; only energy-based models instantiate the landscape as an explicit learned scalar energy. Source: author's schematic.

In the context of the intelligent information technology being developed, it is important to clearly delimit the roles of two fundamental system concepts: the **global consistency criterion** and **entropy**. The global criterion (1.2) is trajectory-level: it is a functional the modeling system strives to minimize along the entire multi-step generation trajectory in order to reach integral consistency of the object; in

this dissertation it is operationalized not by a learned scalar energy but by the measurable consistency criteria of §1.4 — in particular the re-entry-stratified rollout metrics of §1.4.5, which the hierarchical model of Chapter 3 is built to optimize. Entropy, by contrast, acts as a **local diagnostic indicator**: it quantifies the level of structural informational uncertainty at specific spatio-temporal coordinates of the data. Entropy maps allow the system to "understand" where complex moving contours are located and where there is static background, and accordingly to adaptively redistribute limited computational resources (this apparatus is realized in the V-BLT subsystem in Chapter 2). Together these two system concepts form a closed loop of analysis and control.

Controlling this system runs into **computational uncertainty** on two fronts. First, at the stage of parameter identification (training): since the transition function is realized by a deep neural network, identifying the system parameters along a long chain of iterations requires storing all intermediate states (activations) in operating memory, so the memory of backpropagation through time grows linearly with the trajectory length. This creates a resource barrier that rules out long control trajectories on practically available computing systems. Second, at the stage of operation (inference): extending the temporal context of a flat attention mechanism inflates its key–value memory linearly and its computation quadratically with context length (analyzed quantitatively in §1.3.1). Overcoming the first front requires special methods of structural optimization — in particular the introduction of reversible transformations that reduce the memory-consumption complexity (developed in Chapter 2); overcoming the second requires hierarchical decomposition of the state space with a compact long-memory level (developed in Chapter 3).

### 1.1.2 Applied domains

Beyond its theoretical significance, understanding generative video models as complex control systems has enormous practical importance for various applied fields. The tasks solved by such systems intertwine closely with the objects of study of the specialty "System Analysis".

The first important application domain is medical imaging and the development of intelligent systems for supporting clinical decision-making. Complex multidimensional spatial data (MRI, CT) often require reconstruction, super-resolution, or cross-modal generation to minimize the radiation load on the patient. In this domain any generation anomaly (a "hallucination" in the form of non-existent tumors, or the loss of real anatomical structures) has fatal consequences, so the requirement of absolute structural consistency is critical.

The second domain is ecological and technical monitoring using remote sensing data. Satellite data often contain gaps due to cloud cover or technical failures. Generative systems are used for spatial or temporal interpolation of missing data and for predicting the development of ecological catastrophes

(for example, the spread of forest fires). In this case the generated intermediate states must strictly obey the laws of physics, thermodynamics, and the topology of the terrain.

The third domain is the creation of educational and training simulators (for example, for autonomous vehicle control systems). Training robots requires millions of hours of synthetic video data with diverse scenarios (including extreme ones). The requirement imposed on the generative system is long-term temporal coherence: objects cannot suddenly disappear, deform over time, or violate the logic of physical interactions. In particular, an object that leaves the camera's field of view must reappear in a consistent state and position when the camera returns to it — a requirement that, as will be shown in §1.3.1 and §1.4.5, directly demands a long memory horizon from the modeling system.

The analysis of the listed practical domains shows that modern generative systems are subject to the following strict systemic requirements:

1. High visual quality and detail of the target data distribution;
2. Controllability of the system and the ability to respond to control actions;
3. Hierarchical representations without loss of coupling between levels;
4. Absence of spatial, temporal, and logical anomalies — including over long horizons, when

content leaves and re-enters the field of view;

5. Resource efficiency under constraints of computational memory;
6. Interpretability of the process.

Summarizing, it can be stated that generative video models have reached the stage of evolution at which their further development by empirical architecture search exhausts its potential. What is required is the treatment of these algorithms as complex systems controlling dynamic processes, which demands the development of a corresponding mathematical apparatus: methods of decomposition, identification of uncertainty types, introduction of new consistency criteria, and creation of algorithms for optimal resource control. The following subsections of this research are devoted precisely to these questions.

## 1.2 Decomposition of the video generation system

The principal instrument of systems analysis in the study of complex systems is the method of decomposition — the division of an integral system into interrelated subsystems, elements, and couplings, with the aim of simplifying analysis and revealing deep properties and sources of uncertainty. For the multidimensional system controlling the process of video data generation it is expedient to apply several orthogonal kinds of decomposition: structural (functional), hierarchical (semantic), and temporal.

### 1.2.1 Structural decomposition

Structural (functional) decomposition treats the generative system as a set of functional information-processing blocks. In most modern architectures, irrespective of their mathematical basis, three basic macro-subsystems are distinguished [11].

The first subsystem is the **Encoder** — the block of condition preprocessing and encoding. Its function is to transform "raw" control signals (for example, a textual description, semantic masks, or the pixels of context video frames) into a formalized multidimensional feature space.

The second subsystem, the **Iterative Generator / Denoiser**, is the central block of iterative transformation. It is the mathematical core of the system, directly executing the state-transition equation and performing optimization or denoising steps in the feature space. This subsystem is responsible for eliminating the entropy of the initial state and forming a consistent structure.

The third subsystem, the **Decoder**, is the block of decoding and reconstruction, which translates the optimized latent state of the system back into the space of human perception (the pixel space of an image or video stream).

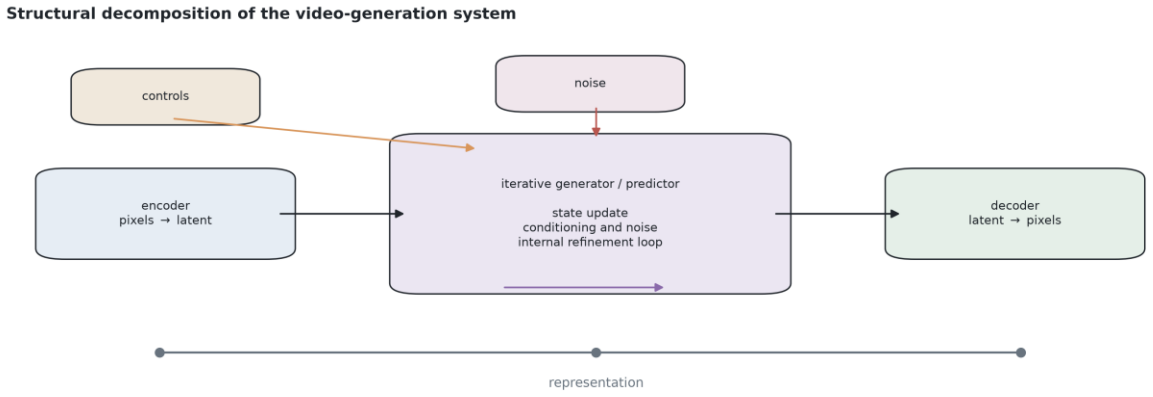



**Figure 1.3 — Structural decomposition of the video-generation system.** The encoder and decoder define the representation interface; the central predictor carries the controlled state transition and iterative refinement. Source: author’s schematic.

Of greatest interest for systems analysis is the central subsystem, since it is where the state-control process takes place and the principal work of overcoming uncertainty is concentrated. Its detailed study requires the deeper kinds of decomposition that follow.

### 1.2.2 Hierarchical decomposition of the state space

Complex systems have a pronounced hierarchical nature [10]. Accordingly, the state space of the central generation subsystem cannot be "flat" — it must be decomposed into macro, meso, and micro levels.

At the **macro level** (the level of highest abstraction) the system operates with semantic vectors of low spatial resolution but high semantic saturation. Here the questions of conceptual composition are decided: the presence of objects, their mutual arrangement, and correspondence to the specified condition.

At the **meso level** of abstraction (medium resolution) the system forms structural contours, geometry, and the spatial relations of object parts.

At the **micro level** (high resolution) the questions of local pixel organization are decided: the coherence of textures, lighting, and micro-details.

The problem of many modern generative systems — for example, standard diffusion models built over variational autoencoders [12, 7] — is that they operate in a single-level latent space, ignoring the necessity of explicit hierarchical decomposition [13, 14, 15]. This produces an internal control conflict in which macro- and micro-decisions are taken simultaneously. The mixing of levels is a direct source of generation anomalies: a correct object texture may be imposed on a distorted global shape.

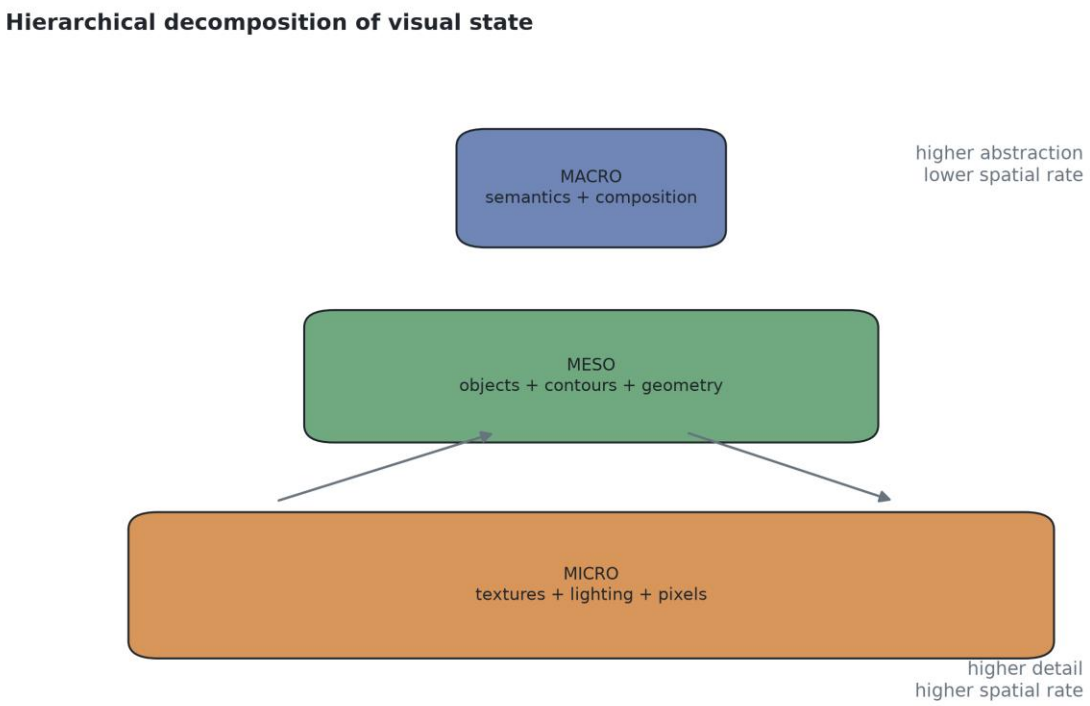



**Figure 1.4 — Hierarchical decomposition of visual state.** Macro denotes the highest abstraction and lowest spatial rate; micro denotes the pixel-proximal level and highest detail. Source: author's schematic.

The systems approach requires the development of mathematical models of explicit hierarchical decomposition with mechanisms of strict coherence between these orthogonal levels of representation. This decomposition is the core of the present dissertation's topic. Moreover — a property central to the direction developed in Chapters 2–4 — an explicitly decomposed hierarchy allows the levels to run at *different temporal rates and context lengths*: a compact upper (coarse) level with a low information rate can afford to retain a long temporal context at the same memory budget for which a flat single-level representation affords only a short one, and can therefore serve as the carrier of the system's long-range temporal memory (quantified in §1.3.1 and developed in Chapter 3).

### 1.2.3 Temporal decomposition

The third necessary kind of stratification is temporal decomposition. When modeling systems that generate video data, time must be considered in two entirely different, orthogonal dimensions, which creates the so-called "two-dimensional temporal grid".

The first dimension concerns the internal **pseudo-time** of the generation process — the partition of the continuous process of state change into discrete refinement steps $\mathrm{k} = 1, \backslash\mathrm{dots}, \mathrm{K}$ *per generated frame*: the steps of the reverse diffusion process, the token steps of an autoregressive decoder, or the descent

steps of an energy-based sampler. Every such step is a micro-transition of the system in phase space, during which the total entropy of the state decreases and its consistency increases.

The second dimension is introduced upon the transition to processing video sequences and reflects the **physical time axis** $t = 1, \dots, T$, where every discrete slice is a separate frame. The complexity of system control lies in the fact that the generative model must simultaneously ensure stability in both temporal dimensions: the system state must evolve correctly along the optimization axis $k$ for every individual frame, and at the same time all generated frames must be spatio-temporally coherent along the physical axis $t$, forming smooth motion kinematics.

The physical axis, in turn, gives rise to the key quantitative characteristic of this dissertation: the **memory horizon** — the maximal temporal distance $\Delta t$ at which information about the past state of the modeled process can still influence the generated present state. For a system whose central subsystem attends to a flat context window of $w$ frames, the memory horizon is bounded by $w$; events separated by more than $w$ frames (for example, content that left the field of view and returns later) are informationally inaccessible to it.

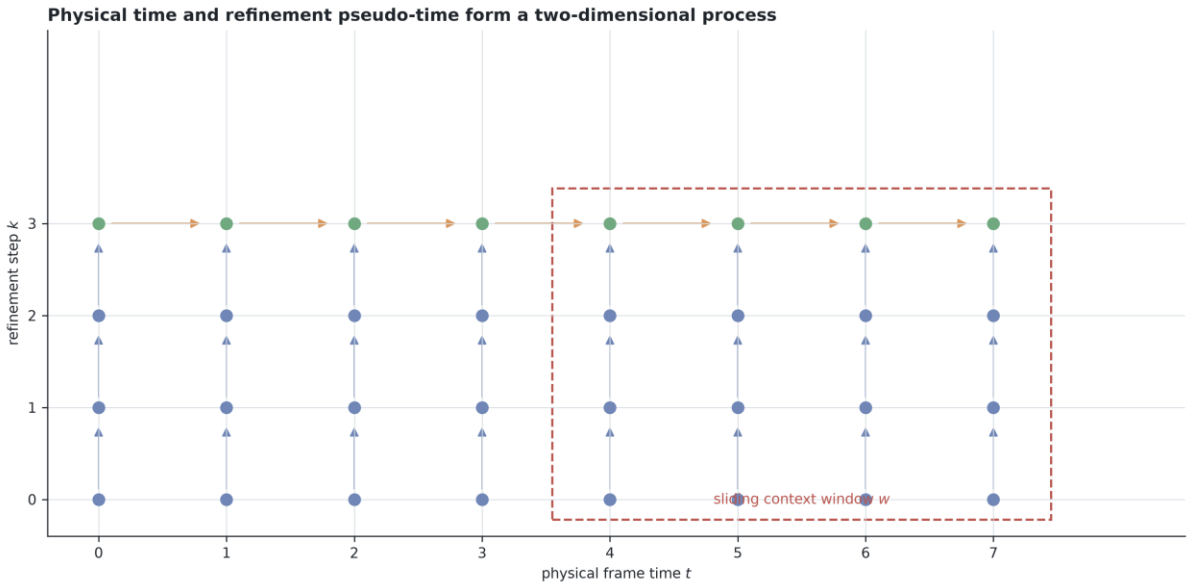



**Figure 1.5 — Two-dimensional temporal decomposition of video generation.** Horizontal transitions advance physical frame time, vertical transitions refine each predicted state, and the sliding context window bounds directly addressable history. Source: author's schematic.

### 1.2.4 Classification of uncertainty types

The performed decomposition of the complex system permits a classification of the kinds of uncertainty arising in it. In accordance with the theory of systems analysis of complex systems under risk [8, 9] and system methodology [8, 9], uncertainties are classified by source and nature. In the video generation

system under study, four principal kinds of uncertainty are distinguished: informational, structural-algorithmic, computational, and target uncertainty.

**Informational uncertainty** is conditioned by the very nature of the iterative generative process, which begins from a state of pure noise (the state of maximal entropy). Moreover, informational uncertainty is inevitably present in the control actions. For example, the textual description "a car drives along a road" or a coarse semantic mask is always incomplete: it specifies only boundary macro-conditions, leaving a multitude of micro-details (the shape of the headlights, the texture of the asphalt, the lighting) undescribed. The task of the complex system is the correct and plausible resolution of this uncertainty during generation. If the system completes the missing information incorrectly, "hallucinations" and anomalies arise, rendering the result inconsistent [16].

**Structural-algorithmic uncertainty** is connected with the rigid limitations of the mathematical apparatus itself (the neural network). Models always have finite capacity and use architectural inductive biases. A separate and extremely destructive manifestation of this uncertainty is the problem of spatio-temporal tokenization. Standard transformer architectures (ViT, ViViT) partition the image into a regular grid of fixed-size patches, ignoring the fact that the visual signal has non-uniform information density.

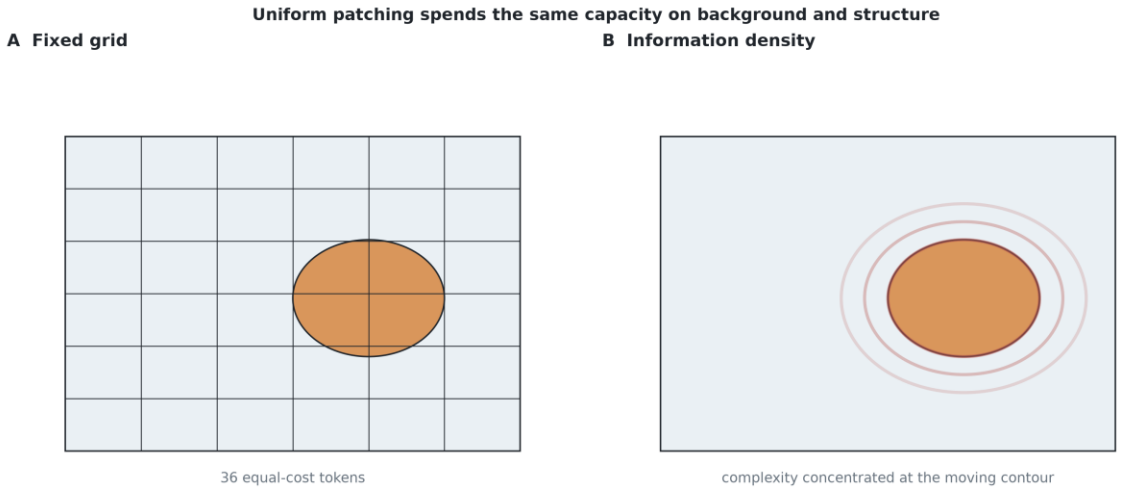



**Figure 1.6 — Fixed-grid tokenization under non-uniform information density.** Equal-size patches assign equal token cost to static background and to high-information moving contours. Source: author’s schematic.

As follows from the principle of optimal control, a uniform distribution of resources across monotonous background and complex moving objects is suboptimal. This gives rise to structural uncertainty: computational resources are lacking where they are genuinely needed and are wasted where there are no objects or dynamics. Solving this problem requires the creation of adaptive tokenization systems based on the measurement of local entropy (developed in Chapter 2).

**Computational uncertainty** is the most critical for iterative multi-step models. It arises on two fronts, corresponding to the two temporal axes of §1.2.3. On the training front (the identification of model parameters), the backpropagation-through-time algorithm (BPTT) requires storing in operating memory all intermediate states for every step of the modeled process; since the state space of visual data is enormous, the linear growth $O(N \cdot K)$ of memory requirements quickly exceeds the physical limits of the hardware. The easiest "solution" — cutting the optimization trajectories down to a few steps — renders the system incapable of modeling long-term dynamics. On the operation front, the flat attention context over the physical time axis inflates the key–value memory linearly and the computation quadratically with the number of context tokens (§1.3.1, eq. (1.7)), so the memory horizon of a flat system is capped by the hardware precisely in the regimes where a long horizon is required. The task of radically reducing memory consumption — for gradient computation along deep trajectories and for context retention along long horizons — is a critically necessary condition for the viability of multi-step generative systems.

**Target uncertainty** arises when the criterion optimized by the model during training does not coincide with the global goal during generation. For example, diffusion models optimize the local error of noise prediction at one isolated step, but during generation these steps are chained into a long sequence in which errors accumulate. The absence of a single global objective functional of the form (1.2) makes the control process "blind" to the final result.

The performed multidimensional decomposition of the video generation control system and the classification of the identified kinds of uncertainty form a clear methodological basis for the further research. Systems analysis shows incontrovertibly that to ensure high generation consistency and to solve the anomaly problem it is insufficient to improve individual architectural elements empirically.

What is required is a comprehensive mathematical approach comprising: methods of adaptive entropy-based state estimation for resolving spatial uncertainty; resource-saving algorithms based on reversible computation for overcoming computational uncertainty; and the creation of a new **hierarchical control apparatus over compact continuous hierarchical latent state spaces** for eliminating structural and target uncertainty.

The next subsection conducts a detailed critical analysis of existing approaches to the mathematical modeling of visual data — diffusion, hierarchical latent and autoregressive, and energy-based models — with the aim of identifying their strengths and their fundamental systemic limitations in the context of the requirements formulated above.

## 1.3 Analysis of existing approaches

To solve the posed problems of systems analysis and to construct an intelligent information technology of video data generation, a thorough critical review of the existing mathematical approaches to modeling complex multimodal spatio-temporal distributions is necessary. The modern scientific landscape offers several competing paradigms, each with its own architectural features, advantages, and fundamental systemic limitations. This subsection reviews three classes of models **in the order dictated by the argument of this dissertation**: (i) diffusion models — the state of the art in generation quality, whose systemic limitation is the monolithic flat long-context attention; (ii) hierarchical latent and autoregressive approaches — the remedy direction, within which the choice between discrete and continuous latent spaces is examined; (iii) energy-based models — reviewed as literature together with the resource-efficiency methods they motivated. The focus criteria are systemic consistency and resource control.

### 1.3.1 Diffusion models: generation quality and the flat long-context barrier

Denoising diffusion probabilistic models (DDPM) [4] and their non-Markovian generalizations (DDIM) [17] are today the dominant standard in high-quality image and video generation. Unlike traditional generative approaches such as GAN [6], which attempt to map random noise directly onto the data, diffusion models implement a more effective and mathematically grounded iterative strategy.

The mathematical basis of the diffusion approach rests on concepts of non-equilibrium thermodynamics. The modeling process is decomposed into two mutually inverse processes: forward and reverse diffusion.

The forward diffusion stage is fixed and consists in the gradual addition of Gaussian noise to the original "clean" data sample $x_0$, progressively driving it to the state of random noise $x_T$. The process is modeled as a Markov chain:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\!\left(x_t;\; \sqrt{1-\beta_t}\, x_{t-1},\; \beta_t \mathbf{I}\right) \tag{1.3}$$

where $\beta_t$ is a predetermined deterministic schedule of noise variance at step t. A mathematical advantage of this construction is the closed-form availability of the noised state at an arbitrary step directly from the original state:

$$q(x_t \mid x_0) = \mathcal{N}\!\left(x_t;\; \sqrt{\bar\alpha_t}\, x_0,\; (1-\bar\alpha_t)\mathbf{I}\right) \tag{1.4}$$

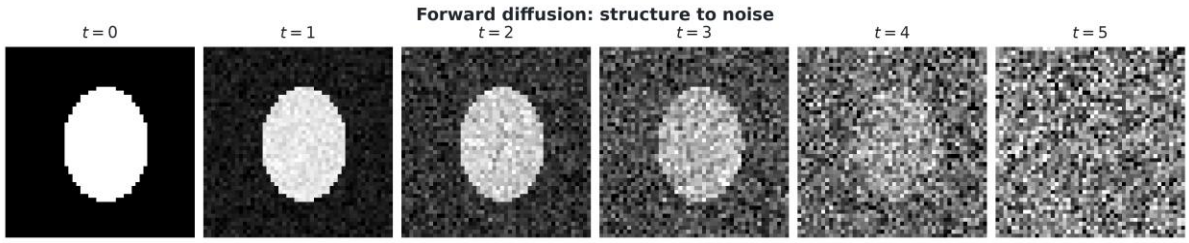



**Figure 1.7 — Forward diffusion.** Progressive Gaussian corruption destroys the information structure of a sample from x_0 toward x_T. Source: author's schematic.

At the reverse diffusion stage the system learns to invert this process, denoising the corrupted sample and restoring the data. Unlike forward diffusion, the reverse process constitutes a complex computational problem solved by deep neural networks:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\!\left(x_{t-1};\; \mu_\theta(x_t, t),\; \Sigma_\theta(x_t, t)\right) \tag{1.5}$$

In practice the model is optimized to predict the added noise \epsilon via the simplified loss:

$$L_{\text{simple}} = \mathbb{E}_{x_0, \epsilon, t}\left[\,\bigl\|\epsilon - \epsilon_\theta\bigl(\sqrt{\bar\alpha_t}\,x_0 + \sqrt{1-\bar\alpha_t}\,\epsilon,\; t\bigr)\bigr\|^2\,\right] \tag{1.6}$$

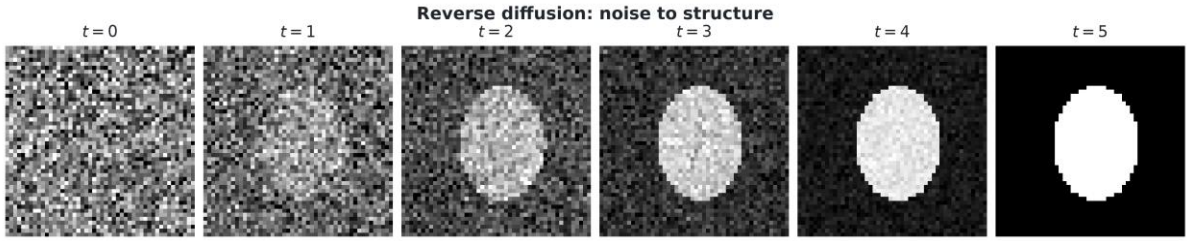



**Figure 1.8 — Reverse diffusion.** A learned denoising field iteratively restores structure from noise. The sequence is conceptual and does not depict an experimental sample. Source: author's schematic.

This "forward–reverse" paradigm makes it possible to generate images of extremely high quality: the decomposition of denoising into steps naturally decomposes the learning problem and simplifies the model's acquisition of features. Classical pixel-space diffusion models, however, face colossal memory deficits, which led to latent diffusion models (LDM) [12] operating in the compressed space of a variational autoencoder. The architectural evolution then displayed a stable trend away from convolutional U-Nets toward transformers: Diffusion Transformers (DiT) [18] demonstrated that processing noised images as sequences of patches scales better, and in the video domain this approach evolved into systems of the Sora class [19], operating on spatio-temporal latent patches and generating video up to a minute long.

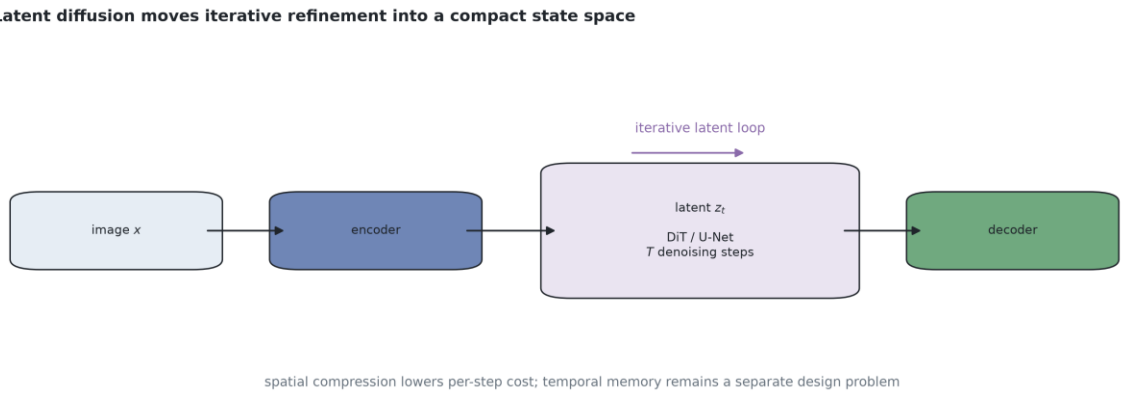



**Figure 1.9 — Latent-space diffusion.** Spatial compression reduces the per-step denoising cost, but it does not by itself solve the temporal-memory allocation problem. Source: author's schematic.

Despite their dominant generation quality, modern diffusion video systems possess systemic limitations that are the direct object of this dissertation's analysis. **The paradigm-level strengths of iterative refinement are not in question here** — on the contrary, Chapter 4's matched-capacity, matched-NFE mechanism study confirms empirically that an iterative refinement (diffusion) core commits to sharp, correctly located content where a regression core averages over futures; the per-frame generation core of the model developed in Chapter 3 is itself a diffusion core, selected by exactly that controlled analysis. The limitations lie elsewhere — in *how the temporal context is represented and scaled*.

The first limitation is the **flat, single-level latent space**. The latent space of the underlying autoencoder is single-level; the system is forced to solve macro-problems (global composition) and micro-problems (textures) simultaneously within one tensor, which precludes independent control of abstraction levels and, per §1.2.2, is a direct source of hierarchical inconsistency.

The second limitation is the **resource blow-up of flat long-context attention**. To lengthen the memory horizon, a flat video transformer must lengthen its spatio-temporal token context. For a context of $T_w$ frames tokenized at $N_s$ spatial tokens per frame, the attention context length is $L = T_w \cdot N_s$, and the principal costs of the attention mechanism scale as

$$C_{\text{compute}} = O(L^2 \cdot D) = O(T_w^2 N_s^2 D), \qquad M_{\text{KV}} = O(L \cdot D) = O(T_w N_s D) \tag{1.7}$$

where $D$ is the model width: computation grows quadratically and key–value cache memory linearly with the context length. Doubling the memory horizon of a flat system quadruples its attention computation and doubles its inference memory — the resource cost is paid *at every generation step, for every frame*, precisely at the horizons where long memory is most needed. This is the central resource contradiction of flat long-context modeling.

The third limitation is the **degradation of long-horizon temporal consistency** even where a long flat context is affordable. Empirical evaluations of long-video generators and world models show that content which leaves the field of view is not restored consistently upon its return, and that consistency metrics degrade with the distance to the relevant past [20, 21, 22, 23]. In systems-analysis terms, the flat context window imposes a hard memory horizon (§1.2.3): information older than the window is not merely expensive — it is absent from the system state. Related to this is **target uncertainty**: during training the model optimizes the local error of one isolated step, while during generation errors accumulate along the whole autoregressive chain — the phenomenon of exposure bias [24, 25].

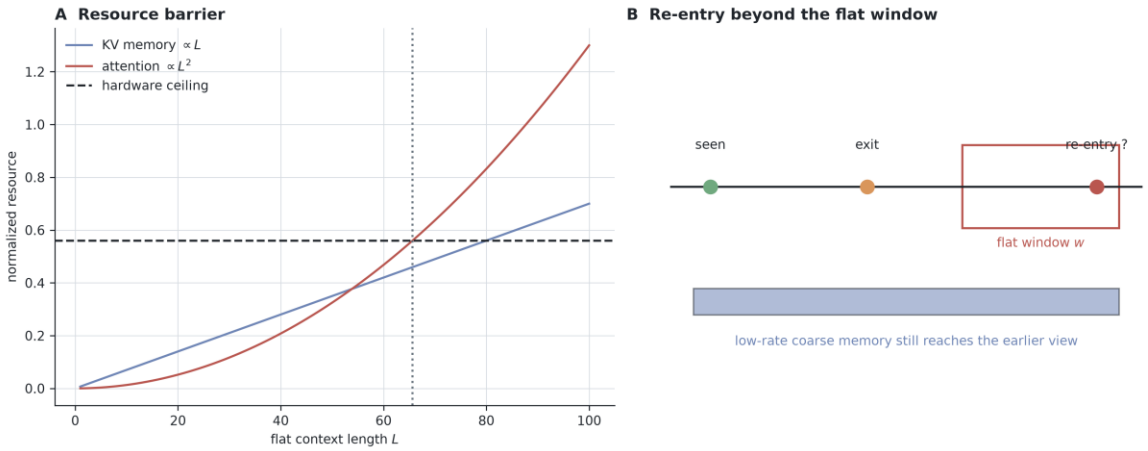



**Figure 1.10 — Flat long-context resource and re-entry barrier.** Panel A is schematic rather than a hardware benchmark: key–value memory grows linearly and attention work quadratically with sequence length. Panel B shows why a low-rate long-memory stream can remain addressable after the relevant content has left a short flat window. Source: author's schematic. Source: NEW (draw)

### 1.3.2 Hierarchical latent approaches and autoregressive prediction

The remedy direction for the contradictions of §1.3.1 is formed by approaches that (i) treat video prediction autoregressively along the physical time axis and (ii) decompose the latent state space instead of flattening it. This subsection reviews the components of that direction: patch-based autoregressive transformers, discrete-token models, and hierarchical latent models — and motivates, by the author's own measurements, the choice of *continuous* hierarchical latent spaces made in this work.

**Patch-based autoregressive transformers.** This line of work originates in the transfer of natural-language-processing principles to computer vision. The standard approach, popularized by the Vision Transformer (ViT) [26] and its video adaptations ViViT [27] and Video Swin Transformer [28], partitions the video into a grid of non-overlapping spatio-temporal patches that are linearized and fed to a transformer. The principal advantage of transformer models is their capacity to model long-range dependencies between distant regions of the video; their principal cost is exactly the quadratic attention

complexity (1.7). Although Video Swin partially mitigates it with local attention windows, the fundamental limitation — a single-level latent space with fixed patching — remains. Fixed-grid tokenization is a form of rigid, content-blind discretization: static low-information background and dynamic scene regions receive equal computational budgets, which §1.2.4 identified as a source of structural-algorithmic uncertainty. An analogous problem in language modeling was resolved by the Byte Latent Transformer (BLT) [29], which groups bytes into latent patches of variable size governed by information density (entropy). Transferring this paradigm to the visual modality resolves the rigid-tokenization component of structural uncertainty; the V-BLT architecture developed in this dissertation realizes that transfer and is presented in Chapter 2. *(Preserved from old §1.3.2.)*

**Discrete-token autoregressive models.** A prominent branch of autoregressive video modeling operates over *discrete* latent spaces: a vector-quantized autoencoder (VQ-VAE) [30] compresses frames into grids of codebook indices, over which an autoregressive transformer learns the video distribution — the VideoGPT line [31] and its successors, including long-video architectures with compressed memory [15]. The attraction of this paradigm is genuine: discrete tokens give a clean autoregressive likelihood, natural compatibility with language-model tooling, and strong compression.

Discrete bottlenecks also introduce specific optimization risks—straight-through assignment, codebook underutilization, dead codes and level collapse—that are documented in the VQ literature [30, 31]. The implemented fourth-article system therefore uses **compact continuous hierarchical latent state spaces**, in which both levels are end-to-end differentiable and independently decodable. This is an implementation choice, not an empirical conclusion that continuous representations are universally superior: the present repository contains no controlled hierarchical-VQ-versus-continuous comparison and makes no such claim. Representation sensitivity is instead tested within the continuous-codec family by the pre-registered alternate-codec-seed study in Chapter 4.

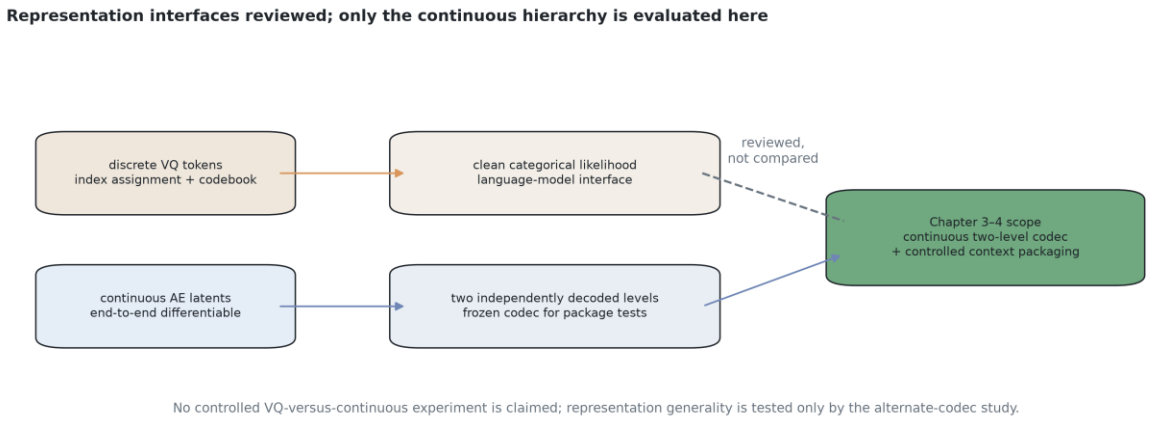



**Figure 1.11 — Representation interfaces and the evidence boundary of this dissertation.** Discrete VQ models are reviewed as an important autoregressive family; the controlled experiments use a

continuous two-level codec. The dashed path denotes a comparison not performed here. Source: author's schematic.

**Hierarchical latent video models.** The direct remedy for the flat-latent limitation is the class of models whose latent state space is explicitly multi-level, with distinct temporal or spatial rates per level. Early representatives use multi-rate recurrent latents — clockwork variational hierarchies [13] — where upper levels tick more slowly and thereby retain longer temporal context. The modern form of this idea in the diffusion family is the hierarchical latent video diffusion model of the MilliVid class [32]: a compact hierarchy of continuous latents (a low-rate *coarse* level and a high-rate *fine* level, each independently decodable to pixels), a diffusion core that denoises the levels jointly with cross-level attention, and autoregressive rollout along physical time. The systemic significance of this spatial construction follows directly from eq. (1.7): because the coarse level carries far fewer tokens per frame than the fine level, a *heterogeneous* context — long in coarse tokens, short in fine tokens — covers a memory horizon many times longer than a flat context of the same total token budget. The hierarchy converts the quadratic cost of horizon extension into a near-linear one along the cheap level, making the coarse level one economical carrier of long-range temporal memory. Temporal subsampling provides a distinct route to the same span and is compared directly in Chapter 4. *(Chapter 3 develops the spatial form with a two-level continuous latent interface and unidirectional context transfer; Chapter 4 treats it as one point in the broader context-packaging design space.)*

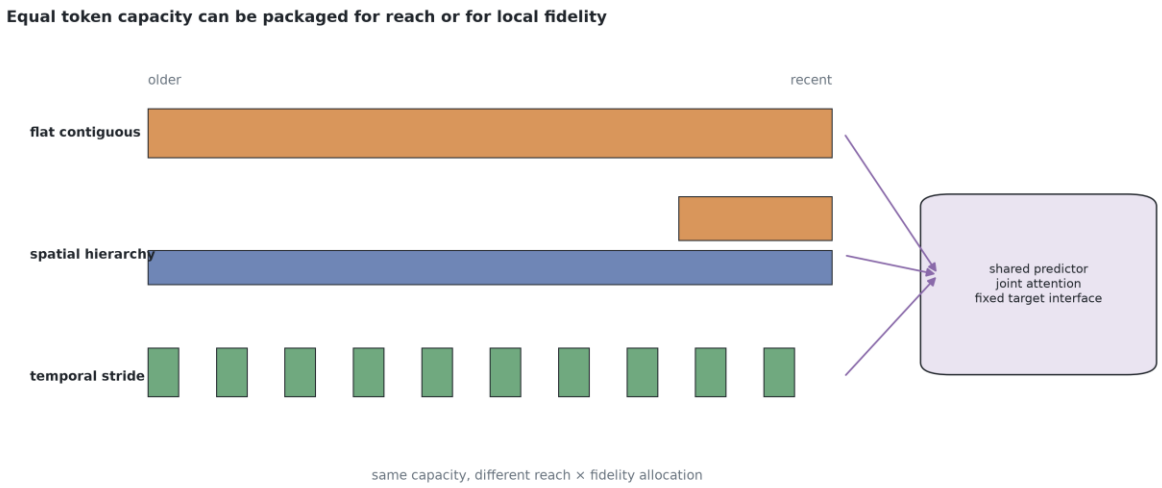



**Figure 1.12 — Context packages allocate equal capacity differently across reach and fidelity.** Flat contiguous, spatially hierarchical and temporally strided packages feed the same predictor family; the exact experimental token counts and target interfaces are reported in Chapters 3–4. Source: author's schematic. Formally, the hierarchical factorization of the video distribution replaces the flat conditional $p_\theta(x_t \mid x_{t-w:t-1})$ with a level-decomposed conditional of the form

$$p_\theta\!\left(z^{c}_{t}, z^{f}_{t} \,\middle|\, z^{c}_{t-w_c:t-1},\; z^{f}_{t-w_f:t-1},\; u_t\right), \qquad w_c \gg w_f \tag{1.8}$$

where $z^{c}$ and $z^{f}$ are the coarse- and fine-level latent states, $w_c$ and $w_f$ their context windows, and $u_t$ the control signals; the asymmetry $w_c \gg w_f$ at a matched token budget $w_c N_s^{c} + w_f N_s^{f} \approx \mathrm{const}$ is what extends the memory horizon without the flat blow-up (1.7).

**Methodology for comparing hierarchical configurations.** The canonical comparison methodology of this class is the Clockwork-VAE benchmark [13]: one configuration table over several datasets and metrics, quality-retention curves over long open-loop rollouts, and level-content probes (an upper level following its prior while lower levels see the context). The dissertation adopts this *format* in §4.4 while tightening three controls the original lacks — a systems-analysis point in its own right. First, capacity: the compared CW-VAE-class models differ in parameter count (their RSSM baseline 5M vs CW-VAE 12M; only the NoTmpAbs ablation is matched), whereas in the token-stream architecture of Chapter 3 depth adds tokens to a shared backbone with only per-level I/O heads (+1.4% parameters, disclosed) — the depth axis is backbone-capacity neutral. Second, context cost: a recurrent state has no token-budget axis, so hierarchy-versus-flat comparisons in the recurrent family cannot hold context cost fixed; the token interface makes every configuration's cost an exact count, enabling matched-budget comparison. Third, metrics: the CW-VAE evaluation aggregates SSIM/PSNR over full frames, and the author's own measurement (§3.4.3) shows a constant all-black frame outperforming a trained CW-VAE on such aggregates on sparse-foreground video — hence the foreground-masked, re-entry-stratified protocol of §3.4. It is also noted that the CW-VAE paper's depth evidence is an information-allocation (KL-per-level) analysis rather than a prediction-quality dose–response; the quality-per-depth cells are measured directly in §4.4.5. The temporal-abstraction relatives — VTA [33] (whose lower level resets when the upper level ticks) and the clockwork recurrent lineage [34] — complete the review of hierarchy *shapes*: spatial and temporal abstraction, depth and budget allocation are four axes of the design space. Which form of hierarchy pays and where is therefore a measurement question (§4.4), not a postulate.

Summarizing the hierarchical-latent review: this paradigm inherits the generation quality of the diffusion core (§1.3.1) while remedying its two systemic limitations — the flat latent space and the flat context window — and, per the author's measured finding, its *continuous* form is the one trainable at practically reachable scale. It is therefore adopted as the base paradigm of this dissertation.

### 1.3.3 Energy-based models and resource efficiency

The third class of the review is that of energy-based models (EBM). The EBM paradigm originates in statistical physics [35, 36]. Its basic principle is that the probability of a data configuration $x$ is defined by the Gibbs/Boltzmann distribution:

$$p_\theta(x) = \frac{\exp\left(-E_\theta(x)\right)}{Z(\theta)} \tag{1.9}$$

where $E_\theta(x)$ is a scalar energy function parameterized by a neural network and $Z(\theta)$ is the partition function. The lower the energy, the higher the probability of the state and the more consistent it is deemed.

An important theoretical bridge connects EBM to diffusion models through score matching [37, 38]: the score function approximated by a diffusion model is the negative gradient of an (implicit) energy,

$$\nabla_x \log p(x) = -\nabla_x E(x) \tag{1.10}$$

From the standpoint of systems analysis the attraction of this paradigm is evident: the energy is a single explicit optimization criterion; the system always moves toward equilibrium; and the energy value serves as a universal indicator of anomalies in a generated frame — if an artifact is generated, the local energy grows. Transformer-parameterized realizations of this idea — Energy-Based Transformers (EBT) [39] — extend it to autoregressive video modeling and add a systemically appealing property: the number of descent steps (1.11) is a natural "think-longer" control knob, allowing the system to spend more computation on harder predictions.

However, the broad application of EBM is critically limited by computational uncertainty. To train such a model, backpropagation must be threaded through $K$ MCMC iterations of the forward pass; BPTT memory grows as $O(K \cdot L)$, where $L$ is the network depth, and since every MCMC iteration requires a gradient with respect to the input (double backward), memory overflow occurs rapidly [40, 41], precluding deep optimization trajectories for high-quality video data.

Approaches to this barrier constitute the resource-efficiency toolbox of multi-step systems. Gradient checkpointing [42, 43] trades computation for memory by storing only a subset of activations and recomputing the rest, reducing memory to the order of the square root of the trajectory length. Neural ODEs [44] offer constant memory through adjoint integration of differential equations but suffer from numerical instability. The most promising direction is discrete reversible neural networks — RevNet [45] and its transformer form RevViT [46] — which restore activations mathematically instead of caching them. It is this apparatus of reversible computation, applied to removing the resource constraints of long-horizon *temporal* modeling, that forms the core of the TempVerseFormer method presented in Chapter 2 and underlies the constant-memory autoregressive rollout of the hierarchical model of Chapter 3.

An honest forward reference completes this review. In the present research the energy-based mechanism was not adopted a priori but *examined experimentally* under matched conditions: the matched-capacity, matched-NFE mechanism study of Chapter 4 (§4.2) includes a two-sided capability study of Energy-Based Transformers, run with the official published implementation and both published training recipes at the largest scale reachable on the available hardware. Its outcome is a **characterized**

**negative** at that scale: one published recipe converges to a stably *mis-ordered* energy landscape (the mean of the data attains lower energy than the true target); the other recipe repairs the ordering — the true target becomes the energy minimum among probed candidates — yet gradient descent from the prescribed initialization still does not reach structured outputs, and inference-time computation does not buy the capability back; meanwhile a diffusion core of roughly half the parameter count, on the same latents and protocol, commits to sharp and correctly located predictions. The study is scoped explicitly: it tests a capability the EBT publication does not claim at a scale it did not run, and contradicts none of the published claims. For this dissertation its systemic significance is that the choice of the per-frame generation core in Chapter 3 is made *by controlled measurement, not by paradigm preference* — and the measurement selects diffusion.

Summarizing the critical analysis of existing approaches: diffusion models deliver the highest generation quality, but their monolithic flat realization runs into the long-context resource blow-up (1.7) and the memory-horizon consistency degradation; patch-based transformer models capture global context well but require entropy-based adaptation of their rigid tokenization; hierarchical latent models remedy the flat-latent and flat-context limitations, and in their continuous form are trainable at reachable scale (whereas the author's experiments found hierarchical VQ configurations unstable); energy-based models offer an attractive explicit criterion but are resource-barred in training, and their descent mechanism is experimentally characterized as a negative at this work's reachable scale (Chapter 4). The synthesis defining this dissertation is therefore: **a hierarchical continuous-latent video model with a diffusion refinement core, whose coarse level carries the long memory horizon (Chapter 3), supported by entropy-adaptive decomposition (Chapter 2), reversible-computation resource optimization (Chapter 2), and a criterion-driven measurement instrument (Chapters 3–4)**.

## 1.4 Consistency criteria and methods

In the theory of complex systems one of the fundamental notions is the stability of a system [47] — its capacity to maintain a specified state or target trajectory under external disturbances and internal stochasticity. In the context of generative video models the notion of stability is most closely tied to the term "consistency" (coherence, non-contradiction). Traditionally, the evaluation of generation results in machine learning reduces to the subjective criterion of visual attractiveness or to statistical metrics of proximity to the training sample, such as FID [48]. From the standpoint of systems analysis such an approach is severely limited.

Systems analysis requires the transition from abstract aesthetic evaluation to a formalized criteria apparatus. The generative model is treated as a system transforming the input information signal of the condition $u$ into the output state $x$. If the state $x$ contains internal logical, spatial, or physical contradictions, it is classified as anomalous. Accordingly, **consistency is the criterion of the absence of**

**anomalies in the generated state of the complex system**; ensuring consistency becomes synonymous with ensuring the reliability and stability of the information technology of generation.

Since video data have a deep multidimensional structure, consistency cannot be described by one scalar parameter. It is a complex systemic phenomenon requiring an extended taxonomy. This subsection analyzes the classical kinds of consistency — spatial, object, and temporal — integrates the notion of conceptual consistency, and formalizes hierarchical and trajectory consistency, which are critically important for hierarchical multi-step generative systems.

### 1.4.1 Spatial and object consistency

The basic level of coherence for any visual signal is **spatial consistency**. This criterion determines the structural integrity of the generated frame at the level of geometry and the mutual arrangement of elements. Spatial consistency guarantees that objects in the image have correct proportions, that boundaries are sharp, and that perspective and lighting obey uniform physical laws across the whole scene.

The next level is **object consistency**. In many practical tasks the system must generate a series of states of one and the same object (from different viewpoints, under different lighting), preserving its unique identifying features. Modern generative algorithms approach this task in several ways.

The simplest method of achieving object consistency in diffusion models is generation by the image-to-image approach [49]: an existing reference image is used as the starting point to steer the generation toward a specified result. The diffusion model transfers the information of the image into its latent space, subtly influencing the direction of the process.

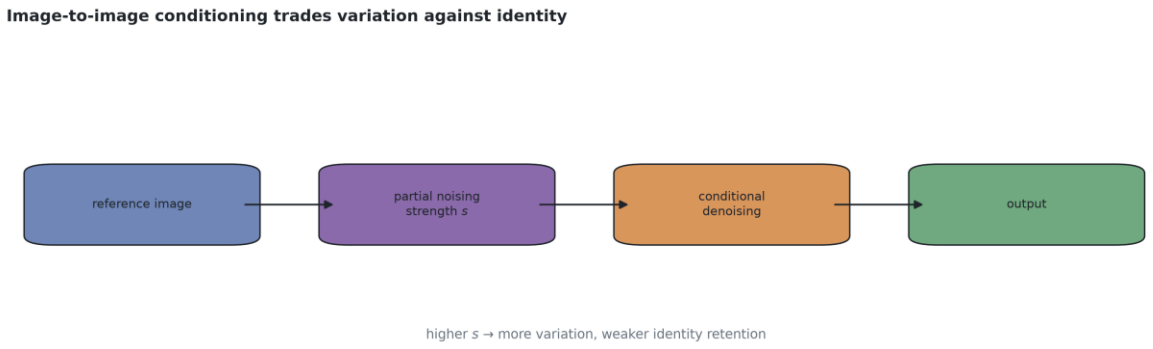



**Figure 1.13 — Image-to-image conditioning.** Increasing noising strength expands variation but weakens direct preservation of the reference identity. Source: author’s schematic.

This approach, however, has rigid systemic limitations — the difficulty of precise control: a higher degree of noising of the reference increases variability and loses the consistency of fine details (textures, scratches).

A deeper method is fine-tuning of the model itself by approaches of the DreamBooth family [50] and its low-rank variation LoRA [51]. This method uses a set of reference images of a specific object to adjust the internal representation (the weight coefficients $\theta$) of the model. Although DreamBooth can secure the consistency of specific features (for example, facial features or the texture of a particular object), it is computationally heavy and works exclusively within the "learned" object.

For enforcing strict spatial consistency of shape there exists the ControlNet architecture [52], which injects a special control signal (depth maps, contours, segmentation masks) into the reverse diffusion process. Methods of the ControlNet type perform the function of hard boundary control of the admissible state space: the input image acts as a guiding form constraining the evolution of the system.

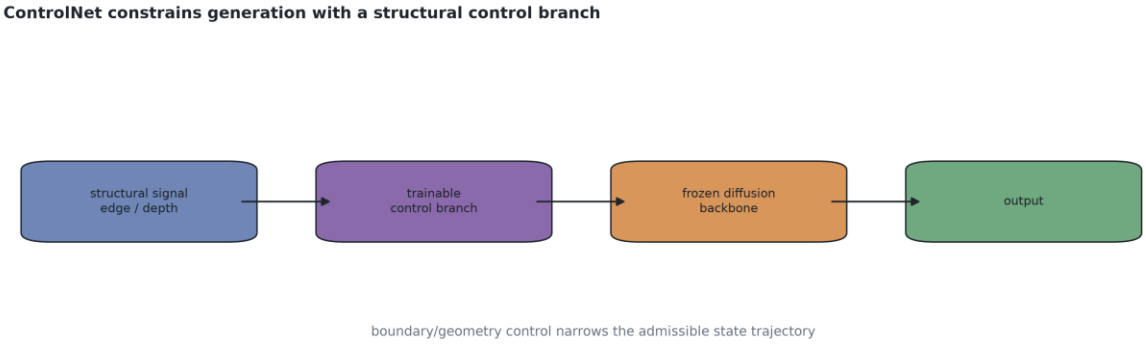



**Figure 1.14 — Structural control with a ControlNet-class branch.** A trainable branch maps edges, depth or segmentation into residual constraints for a frozen diffusion backbone. Source: author’s schematic after the ControlNet architecture [52].

Despite the effectiveness of ControlNet, it is not a complete solution for object consistency, since it focuses only on the form ("how") and not on the properties ("what"). A fully autonomous system must be able to maintain object identity through internal control mechanisms, without mandatory external structural prompts. *(This requirement — identity maintenance by internal memory rather than external prompts — is exactly the long-memory requirement that the coarse level of the Chapter 3 model serves.)*

### 1.4.2 Temporal consistency

Upon the transition from static images to video data the key criterion becomes **temporal consistency**. It requires that the sequence of generated states $x_1, x_2, \dots, x_T$ form a logical and smooth video

stream. Anomalies of temporal consistency manifest as flicker, sudden background color changes, or deformation of objects during motion.

The mathematically strict local requirement of temporal consistency is formalized through optical flow or a warping function $W$: if the vector field of motion between frames $t-1$ and $t$ is known, the generated state $x_t$ must maximally correspond to the warped previous state $W(x_{t-1})$. The local temporal-consistency error is defined as:

$$L_{\text{temp}} = \left\|\left(x_t - W(x_{t-1})\right)\right\|^2 \odot M \tag{1.12}$$

where $M$ is a binary mask excluding occlusion regions.

It must be emphasized, however, that the warp criterion (1.12) is **local**: it couples only adjacent frames and therefore certifies only short-range temporal consistency. It is entirely blind to the long-range regime identified in §1.3.1: content that leaves the field of view exits the domain of the mask $M$, and its consistent restoration upon re-entry many frames later is not constrained by any chain of local warp terms — it is constrained only by the memory of the generating system. Long-range temporal consistency is therefore operationalized separately, as part of trajectory consistency (§1.4.5), through metrics stratified by re-entry hardness. For ensuring temporal consistency in the developed information technology, the following mechanisms are used: multidimensional positional encoding 3D-RoPE, block-causal attention masks, and temporal chaining mechanisms (Chapter 2), together with the coarse-level long memory of the hierarchical model (Chapter 3).

### 1.4.3 Style and conceptual consistency

Besides object consistency, complex systems must maintain a unified **stylistic** appearance of a series of generated data. Modern approaches such as StyleAligned [53] solve this problem through intervention in the attention mechanism (shared attention): target images receive information about the reference image during generation, copying its global stylistic characteristics (color palettes, brushwork) without interfering with spatial geometry.

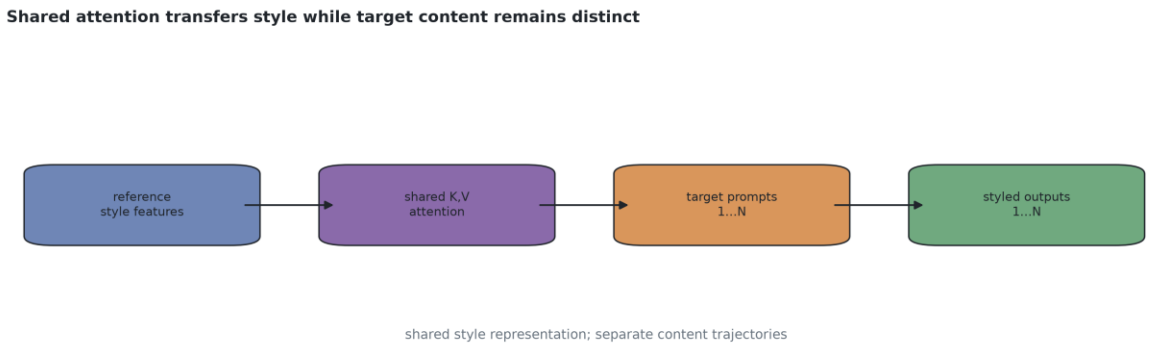



**Figure 1.15 — Style consistency through shared attention.** Reference attention features are shared across distinct target trajectories while their content conditions remain separate. Source: author's schematic after StyleAligned [53].

For a more thorough and qualitative analysis of generative models, within this dissertation research — published in **[A3]** (the consistency-criteria article) — the notion of **conceptual consistency** was formalized.

Marr's multi-stage account of visual representation — progressing from image evidence toward structured descriptions of the scene [54] — motivates a broad engineering analogy, but does not itself prescribe the taxonomy used here. The dissertation's narrower claim is that the traditional division into style and object alone is incomplete. Conceptual consistency is the capacity of the generative system for compositionality and extrapolation. It evaluates the ability of the model not merely to reproduce existing objects but to generate entirely new visual ideas (out-of-distribution), never encountered during training, while strictly observing the logic of the control conditions u.

For example, generating the image "a skyscraper built of skyscrapers" requires the system to perfectly disentangle the vectors of "shape" and "material". An anomaly of conceptual consistency is the loss of compositionality — when the system ignores part of the textual condition or generates objects side by side instead of combining them.

As the metric for evaluating conceptual consistency, the cosine similarity between the semantic vectors of the generated image and the textual condition in the joint space of a contrastive model (CLIP score) [55] is proposed. Let $E_I$ be the visual encoder and $E_T$ the textual encoder; then the measure of conceptual consistency is defined as:

$$C_{\text{concept}}(x, u) = \frac{E_I(x) \cdot E_T(u)}{\left\|E_I(x)\right\| \left\|E_T(u)\right\|} \tag{1.13}$$

High values of this indicator testify that the system has successfully resolved the target uncertainty and learned the deep causal relations of the visual space.

### 1.4.4 Hierarchical consistency

For the transition to systems of a higher order of complexity — multi-level generative models — this research introduces the notion of **hierarchical consistency**, developed within the construction of the hierarchical generation model of Chapter 3. This criterion follows directly from the principles of systems analysis of multi-level objects.

Hierarchical consistency is defined as the strict informational and structural coherence between the representations of one and the same visual object at different levels of abstraction: the macro level (semantics), the meso level (structure), and the micro level (pixels or textures). An anomaly of hierarchical consistency is a situation of internal system conflict: for example, the macro signal forms the contour of a brick building while at the micro level the system generates organic wood patterns.

Ensuring this property requires the introduction of an inter-level coherence operator. From the standpoint of system control, the essential requirement is that the junior (fine/detailed) level must not be able to contradict the senior (coarse/abstract) level's decisions at long range. In the Chapter 3 model this requirement is satisfied by two complementary mechanisms (§3.1.2–3.1.3). First, by **information-design asymmetry**: within a prediction step both levels are denoised jointly in one shared attention pass, but only the coarse level's history extends over the full memory horizon — long-range structure can therefore *originate only from the coarse channel*, and the fine level physically lacks the inputs to contradict it at long range; the control is unidirectional in the information-theoretic sense (coarse → fine for long-horizon signals), not in the scheduling sense. Second, by **gradient detachment at the system boundaries** where it is exact and load-bearing: between the frozen representation subsystem and the prediction core (the state coordinate system is a constant of the control problem), and between past and future along the autoregressive rollout (previously generated context enters as a constant, excluding destructive gradient interference between successive windows).

### 1.4.5 Trajectory consistency

The last — and, for multi-step generative systems, fundamental — criterion is **trajectory consistency**: the stability of the generation process itself as a trajectory of the complex system. In the original, single-frame reading it concerned the stability of the iterative refinement process in pseudo-time $k$; in the present research it is generalized and operationalized on the physical time axis, where the dominant practical instability of video generation lives: the degradation of **autoregressive rollout**, in which the system consumes its own outputs as context and errors compound along $t$ (the realization of the target uncertainty of §1.2.4 and the exposure bias of §1.3.1).

A trajectory-consistent system must satisfy two requirements. First, **bounded degradation**: the prediction error along an autoregressive rollout must grow in a controlled manner, without divergence or collapse of the generated content. Second — the requirement that separates a long-memory system from a flat one — **consistent re-entry**: content that left the field of view must be restored correctly when it returns, at any temporal distance within the modeled process.

The operationalization of this criterion, developed in Chapters 3–4, stratifies rollout error by an exactly computable hardness label. For a generated frame $x_t$ with context window $C$, let the **novel-foreground fraction** $\nu_t(C) \in [0, 1]$ be the fraction of the frame's foreground content that was informationally inaccessible from $C$ (in the purpose-built testbed of Chapter 3 this quantity is computed *exactly* from the recorded control-action track, rather than estimated). Trajectory consistency is then measured as the family of stratified degradation curves

$$\mathcal{E}_{b}(h) = \mathbb{E}\!\left[\, e\!\left(x_{t+h},\, \hat{x}_{t+h}\right) \,\middle|\, \nu_{t+h} \in b \,\right] \tag{1.14}$$

— the expected prediction error $e$ at rollout horizon $h$ within each hardness bin $b$ (easy: $\nu = 0$; medium: small $\nu$; hard re-entry: substantial $\nu$) — rather than as a single aggregate number. This stratification is what makes the criterion *diagnostic*: a flat-context system and a long-memory hierarchical system can be indistinguishable on aggregate error while differing qualitatively on the hard re-entry bin beyond the flat window's horizon; conversely, an anomaly of trajectory consistency (divergence, mean-collapse, identity loss at re-entry) localizes to specific bins and horizons.

For energy-based mechanisms the trajectory criterion has a specialized classical diagnostic in pseudo-time: the divergence of the sampling chain (1.11) is monitored by the **energy gap** between the initial and final states of the descent,

$$\Delta E = \frac{E(x_0) - E(x_K)}{E(x_0)} \tag{1.15}$$

which must be strictly positive, and by the ordering of the trained landscape (whether the true target attains lower energy than degenerate candidates). Finally, the practical attainability of long consistent trajectories depends on resolving the computational uncertainty of §1.2.4: removing the memory limits on trajectory depth by the time-agnostic backpropagation developed in **[A2]** and Chapter 2.

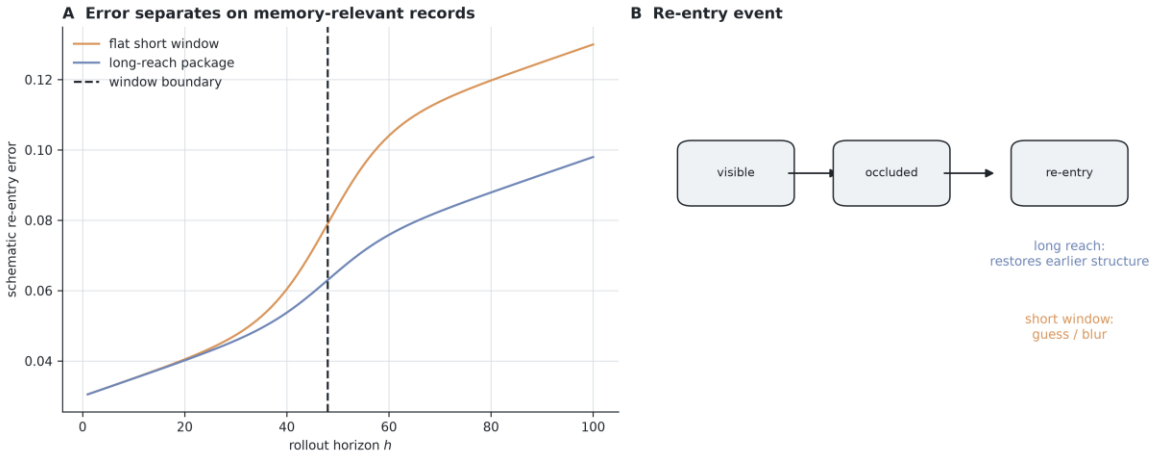



**Figure 1.16 — Conceptual operationalization of trajectory consistency.** The schematic predicts separation specifically on memory-relevant re-entry records beyond a short window; Chapter 4 tests this prediction with exact generator-derived observability labels and paired rollouts. Source: author's schematic; curves are illustrative, not measured.

Systems analysis shows that none of the kinds of consistency can be considered in isolation: they form an interdependent hierarchy of criteria. For example, without trajectory consistency it is impossible to attain high spatial and temporal consistency of a long generated sequence; and without hierarchical consistency, the macro-level memory that trajectory consistency relies on at re-entry cannot be trusted by the lower levels.

Classical approaches — adding a ControlNet or a LoRA — attempt to solve these problems locally, by patching individual criteria with external mechanisms. The systems approach instead requires an architecture in which the observance of the consistency criteria follows from the structure of the process itself: **an explicit multi-stream decomposition of context across temporal rate and spatial resolution with unidirectional context transfer** — so that hierarchical consistency is enforced by construction (stop-gradient control, §1.4.4), long-range temporal and trajectory consistency are supplied with the memory they informationally require (§§1.3.1–1.3.2), and the whole is trainable within realistic resource constraints thanks to reversible computation (Chapter 2). Constructing and validating exactly such an architecture is the subject of Chapters 3 and 4.

## 1.5 Conclusions and statement of the research tasks

In the first chapter of the dissertation a thorough systems analysis of the processes of hierarchical video data generation has been carried out. On the basis of complex-systems theory, generative video models have been characterized as open dynamic multi-level control systems operating under substantial structural, informational, computational, and target uncertainty.

The decomposition of the generation system revealed the key problems of contemporary approaches. The analysis of diffusion models showed that, while they define the state of the art in generation quality, their monolithic realization suffers from two systemic defects: a single-level (flat) latent space that mixes abstraction levels and loses hierarchical control, and a flat spatio-temporal attention context whose computation grows quadratically and whose memory grows linearly with context length — so that extending the memory horizon of a flat system is resource-prohibitive precisely where it is most needed, and the temporal consistency of long sequences degrades: content leaving the field of view is not restored consistently upon re-entry. In addition, the divergence between the local training objective and the long autoregressive generation trajectory (exposure bias) undermines long-horizon stability.

The review of patch-based transformer architectures demonstrated their advantages in capturing global contextual dependencies but revealed the critical problem of irrational computational resource allocation due to fixed spatio-temporal tokenization, which substantiates the necessity of methods for adaptive decomposition of visual data based on measuring their local entropy (developed in Chapter 2).

The study of hierarchical latent approaches established the remedy direction: an explicitly multi-level latent state space in which a compact upper (coarse) level, carrying few tokens per frame, can retain a long temporal context at a fixed token budget and thereby serve as the carrier of the long-range memory of the process. Discrete-token hierarchies remain an important reviewed alternative with codebook-specific optimization risks. The implemented and evaluated scope of this dissertation is a **compact continuous hierarchical latent state space**; because no controlled VQ-versus-continuous receipts are present, representation-family superiority is not claimed.

The examination of energy-based models (EBM) recorded their systemic attraction — an explicit scalar optimization criterion doubling as an anomaly indicator — alongside their fundamental computational barrier: gradient identification through long MCMC trajectories with linear memory growth. In this dissertation the energy-based mechanism is treated as reviewed literature whose capability boundary at reachable scale is characterized experimentally in Chapter 4; the survey of resource-efficiency methods it motivated (checkpointing, Neural ODE, reversible networks) identified reversible transformations as the most promising apparatus, which Chapter 2 develops into the time-agnostic backpropagation method and its extension to stochastic multi-level recurrences.

Furthermore, the chapter extended and improved the criteria apparatus for evaluating the results of generative models. The notion of consistency was reconceived as the systemic property of the absence of anomalies and the stability of the generation trajectory. To the classical criteria (spatial, object, temporal consistency) new kinds were added: conceptual, hierarchical, and trajectory consistency — the last operationalized through autoregressive-rollout degradation metrics stratified by an exactly

computable re-entry hardness label, which converts a previously declarative criterion into a measurable instrument of systems analysis.

The identified systemic contradiction — between the necessity of ensuring multi-level consistency of video data over long time horizons (which requires long memory horizons and deep hierarchies) and the strict resource constraints of modern computing systems (the attention-memory blow-up of flat contexts and the linear growth of training memory with horizon length) — determines the relevance of this dissertation research and forms its purpose.

**The purpose of the dissertation research** is to improve the efficiency and the temporal and spatial consistency of hierarchical video data generation processes through the development of systems analysis methods, mathematical models, and intelligent information technologies.

To achieve this purpose and to resolve the identified systemic contradictions, the following research tasks must be solved:

1. Analyze the state and problems of generative video-data models as complex systems; classify the types of uncertainty and identify the systemic causes of anomalies (inconsistencies) in the generation process, including the resource and consistency limitations of flat long-context modeling *(performed in Chapter 1)*.

2. Improve the criteria apparatus of systems analysis of generative processes by extending the taxonomy of consistency types (conceptual, hierarchical, trajectory) and operationalizing trajectory consistency through measurable rollout metrics stratified by re-entry hardness *(the taxonomy is constructed in Chapter 1; the operational metrics and their measurement instrument are developed in Chapters 3–4)*.

3. Develop a method for estimating the state of uncertainty of spatio-temporal data through their adaptive decomposition using entropy metrics and the mathematical apparatus of multidimensional rotary positional embedding.

4. Develop a method for optimizing the computational resources of multi-step iterative models that ensures constant, process-length-independent consumption of memory on the basis of reversible mathematical transformations; extend it to stochastic multi-level recurrences and delimit its domain of applicability.

5. Develop a mathematical model of multi-stream video data generation in compact hierarchical latent state spaces, supporting context streams at different temporal rates and spatial resolutions over a shared attention backbone, with unidirectional context-signal transfer and a per-frame iterative-refinement generation core selected by controlled mechanism analysis.

6. Create an intelligent information technology and a software complex for the systems analysis of hierarchical video generation processes, integrating the developed methods of adaptive tokenization, resource optimization, and hierarchical generation, together with a purpose-built memory-horizon testbed and re-entry-stratified evaluation protocol as its measurement instrument.

7. Conduct comprehensive experimental studies and approbation of the developed methods and technology on synthetic and open datasets — including a matched-capacity, matched-NFE comparison of prediction mechanisms — and compare their efficiency and consistency with modern analogues.

The solution of tasks 3–7 is the subject of the following chapters of the dissertation. Chapter 2 presents the mathematical apparatus of adaptive data tokenization and computational-resource optimization; Chapter 3 develops the hierarchical mathematical model with long-range coarse-level memory and the memory-horizon measurement instrument; Chapter 4 presents the experimental research and practical application.

The results of this chapter are published in the author's work [A3] and in the conference publications to be identified in the final author-supplied publication list.

# Chapter 2 — Mathematical Methods for Uncertainty Estimation and Resource-Efficient Modelling of Multi-Step Video Processes

## 2.1 Entropy-Based Estimation of Spatio-Temporal Video Uncertainty

As established in the previous chapter during the decomposition of the complex system, the processes of hierarchical video data generation operate under a high level of multidimensional uncertainty. To build an effective control system for such processes, it is necessary to develop a mathematical apparatus for resolving the informational, structural-algorithmic, and computational uncertainty at every stage of information processing. One of the fundamental problems that generates a significant share of the structural-algorithmic and computational uncertainty is the problem of inefficient decomposition of the input spatio-temporal data.

In classical systems analysis, the decomposition of a complex system must reflect its internal information saturation and structure. However, the traditional approaches to the mathematical modeling of video data, popularized by architectures of the Vision Transformer (ViT) type [26] and their spatio-temporal extensions ViViT [27] and Video Swin Transformer [28], are based on a rigid, a-priori-fixed partition of the image or video stream into a regular grid of non-overlapping patches of fixed size, e.g., 16×16 pixels.

From the standpoint of optimal control and resource-allocation theory, such an approach is severely suboptimal. The visual signal, especially video data, is characterized by an extremely non-uniform distribution of information density. Large areas of the frame are often occupied by a static, low-information background (monotonous sky, walls), while the key dynamics and high-frequency details (moving objects, complex kinematics) are concentrated in small local zones. Using a fixed grid forces the system to distribute its limited computational resource uniformly across all regions of the frame, wasting it on the processing of redundant information.

To solve this problem, it is necessary to move from a rigid geometric decomposition to an adaptive decomposition of the state space that is driven by the data themselves. The conceptual foundation for this transition is the new paradigm of byte-level language models that emerged in natural language processing, in particular the Byte Latent Transformer (BLT) [29]. These systems abandoned the paradigm of fixed token vocabularies in favor of dynamic grouping of elementary information units (bytes) into latent patches whose size is inversely proportional to the local complexity of the signal.

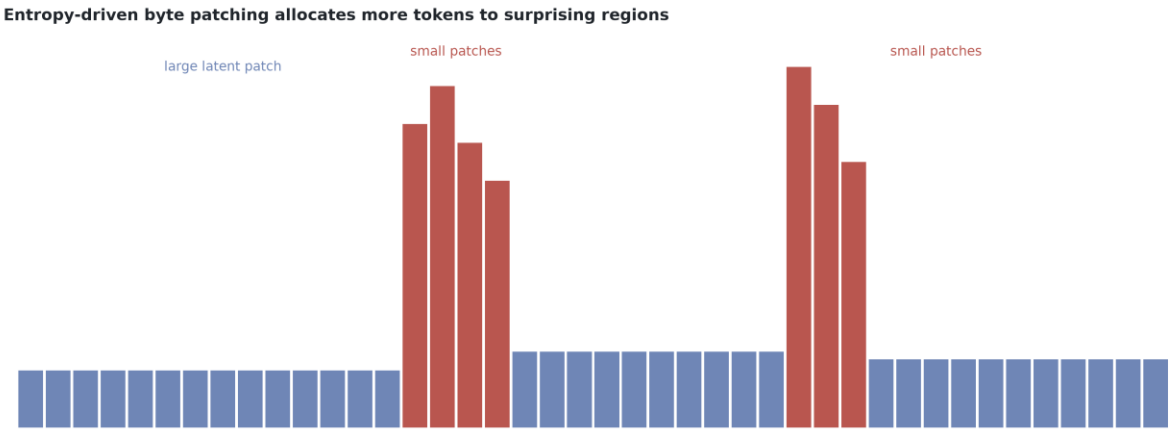



**Figure 2.1 — Conceptual operating principle of entropy-driven byte patching.** Long low-entropy regions are grouped into larger latent patches, whereas locally surprising regions receive finer allocation. The values are schematic, not empirical measurements. Source: author's schematic after BLT [29].

Within the dissertation research, the results of which are published in **[A1]**, the hypothesis was put forward that transferring the principles of adaptive entropy-based tokenization to the visual modality opens fundamental opportunities for optimizing and redistributing the computational resources of a generative system. On the basis of this approach, the author developed the method for estimating spatio-temporal uncertainty and a new mathematical architecture — the Vision Byte Latent Transformer (V-BLT).

Working at the level of "raw" bytes of video data, where each pixel of an RGB channel is represented by a discrete integer from 0 to 255, allows the system to avoid the loss of information that is unavoidable when heuristic pre-compression or vector-quantization (VQ-VAE) algorithms are used [30]. The developed V-BLT architecture is a three-tier hierarchical information-transformation system consisting of a Local Encoder, a Global Latent Transformer, and a Local Decoder. In the arc of this dissertation, V-BLT is the author's **first realization of the hierarchical adaptation of video generation** — a two-stage decomposition in which local byte-level stages surround a global latent stage and an entropy criterion governs the allocation of capacity between them; the mature form of this hierarchical decomposition, in which the upper level carries the long-range temporal memory of the process, is developed in Chapter 3.

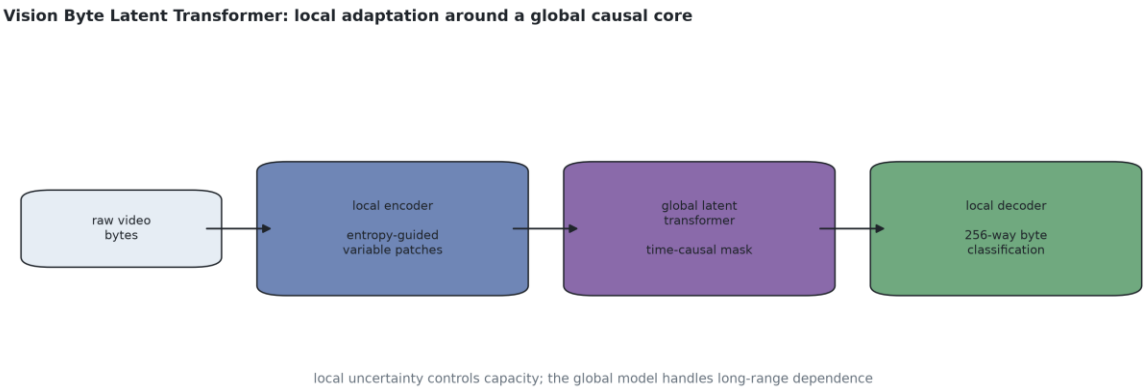



**Figure 2.2 — Vision Byte Latent Transformer architecture.** Local entropy-guided aggregation and byte classification surround the global time-causal latent transformer. Source: author's schematic of the [A1] method.

The first subsystem — the **Local Encoder** — performs the function of primary adaptive aggregation. The input is a video sequence as a multidimensional tensor of raw bytes. The task of the Local Encoder is the dynamic grouping of locally correlated bytes into compact latent representations in the form of spatio-temporal patches $Z = \{z_1, z_2, \dots, z_M\}$. Mathematically it is realized as a lightweight convolutional-transformer network capable of determining the optimal boundaries of information aggregation.

The second subsystem — the **Global Latent Transformer** — is the computational core and is responsible for modeling the long-term dynamics in the compressed space. Since this block performs autoregressive prediction for time-series modeling, its architecture requires a specific design of the attention-masking system in order to avoid structural uncertainty in the form of information leakage from the future [56], while fully resolving the spatial uncertainty within the current frame.

For this purpose, a spatio-temporal attention-masking matrix is used. Let the latent space have dimensionality $T' \times S'$, where $T'$ is the number of time steps and $S'$ is the number of spatial patches per frame. The attention matrix $M$ is computed so that for any token $z_i$ from frame $t_i$, the attention weights toward a token $z_j$ from frame $t_j$ obey the rule:

$$M_{i,j} = \begin{cases} 0, & \text{if } t_j \le t_i \\ -\infty, & \text{if } t_j > t_i \end{cases} \tag{2.1}$$

This strictly time-causal but spatially fully-connected mask guarantees the physical and logical plausibility of the generation process.

The third subsystem — the **Local Decoder** — projects the predicted latent macro-states back into the micro-space of image bytes. The key mathematical innovation of the developed method, which distinguishes it from standard pixel-regression approaches of the MSE type, is the use of a classification

model for byte generation. At the output, the decoder forms a tensor that yields, for each pixel, not a scalar brightness value but a full probability distribution over the 256 possible byte states.

Formally, for each byte (sub-pixel) $x_i$ the system computes a logit vector $L_i \in \mathbb{R}^{256}$. The probability estimate that the byte takes a specific discrete value $k \in \{0, \dots, 255\}$ is determined by the softmax activation function:

$$p(x_i = k) = \frac{\exp L_{i,k}}{\sum_{j=0}^{255} \exp L_{i,j}} \tag{2.2}$$

Such a mathematical formulation transforms the problem of minimizing the mean squared error (MSE), which by its nature is prone to smoothing and to the loss of high-frequency details, into a classification problem using the cross-entropy loss function [57]. This makes it possible to correctly model multimodal distributions: for example, the contour of a moving object may shift one pixel to the left or to the right with equal probability, which cannot be described by a single mean value.

The main advantage of the classification approach for systems analysis lies in the possibility of direct mathematical computation of a measure of spatio-temporal uncertainty. Having a discrete probability distribution for each pixel, the classical apparatus of Shannon information theory is applied. The information-entropy measure $H_i$ for each predicted pixel $i$ is computed as:

$$H_i = -\sum_{k=0}^{255} p(x_i = k)\,\log_2 p(x_i = k) \tag{2.3}$$

The entropy value lies in the range from 0 (absolute determinism and confidence of the system in its prediction) to 8 bits (maximum informational uncertainty, a uniform distribution). By computing this value for all pixels of a frame, the generative system forms a two-dimensional per-pixel entropy map.

Entropy maps are a direct indicator of the state of structural uncertainty and serve as a powerful diagnostic instrument. Empirical studies, the results of which are presented in **[A1]** and are considered in detail in Chapter 4, confirm that the highest entropy values systematically concentrate on the contours of fast-moving objects and in occlusion zones with complex kinematics, whereas the static, predictable background exhibits entropy rapidly approaching zero. Figures 2.3 and 2.4 show prediction examples for a simple 2D and a complex 3D sequence respectively, where the first row contains the expected frames of the sequence, the second the frames predicted by the V-BLT model, and the third the entropy maps corresponding to the predicted frames (black indicates high entropy, white low entropy). A key aspect of the V-BLT architecture is its ability to operate on raw bytes, which makes it possible to compute per-pixel entropy for every prediction. Entropy in this context is a measure of the model's uncertainty: the higher the entropy, the less "confident" the model is in its prediction for the given pixel.

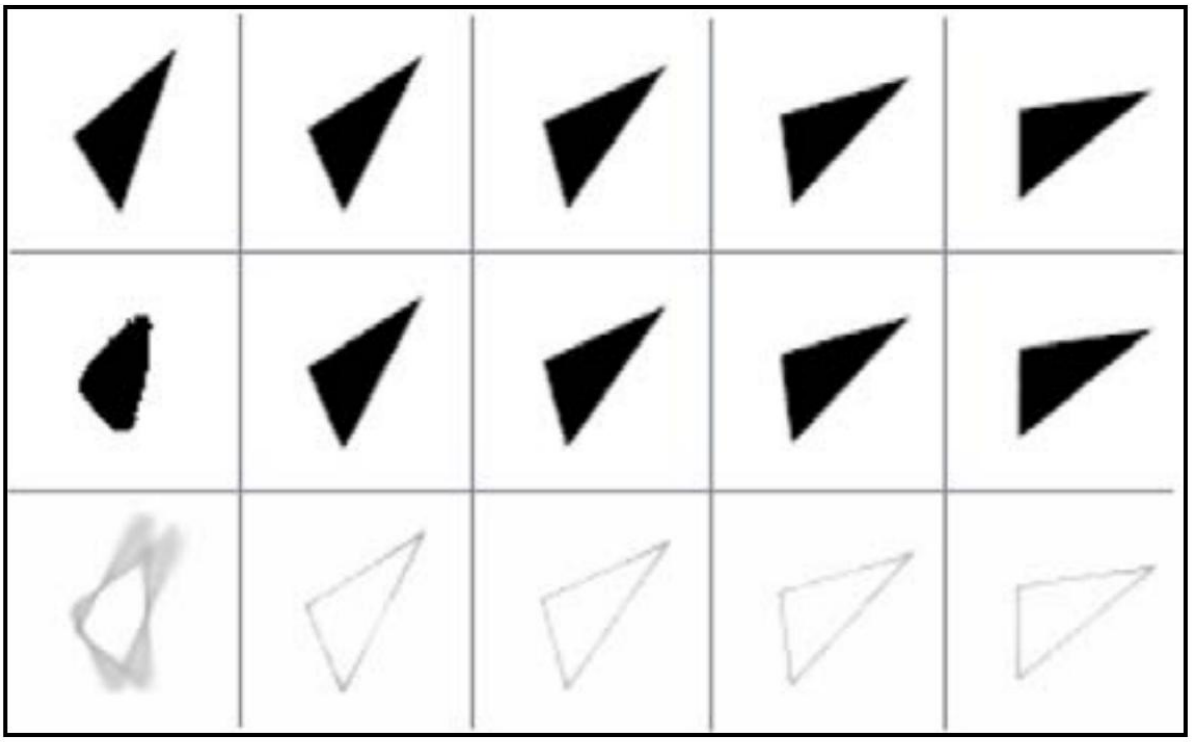


**Figure 2.3 — V-BLT prediction and entropy maps for a rotating 2D shape.** Rows show, respectively, ground truth, V-BLT predictions and per-pixel entropy maps over five future frames. The first prediction has the largest shape error and the broadest high-entropy region; subsequent predictions recover the rotating contour while entropy concentrates near its boundary. Source: Kulyk and Mokin, *Scaling Video Prediction with Spatio-Temporal Patches*, published Fig. 7, DOI 10.31649/1997-9266-2025-182-5-129-139.

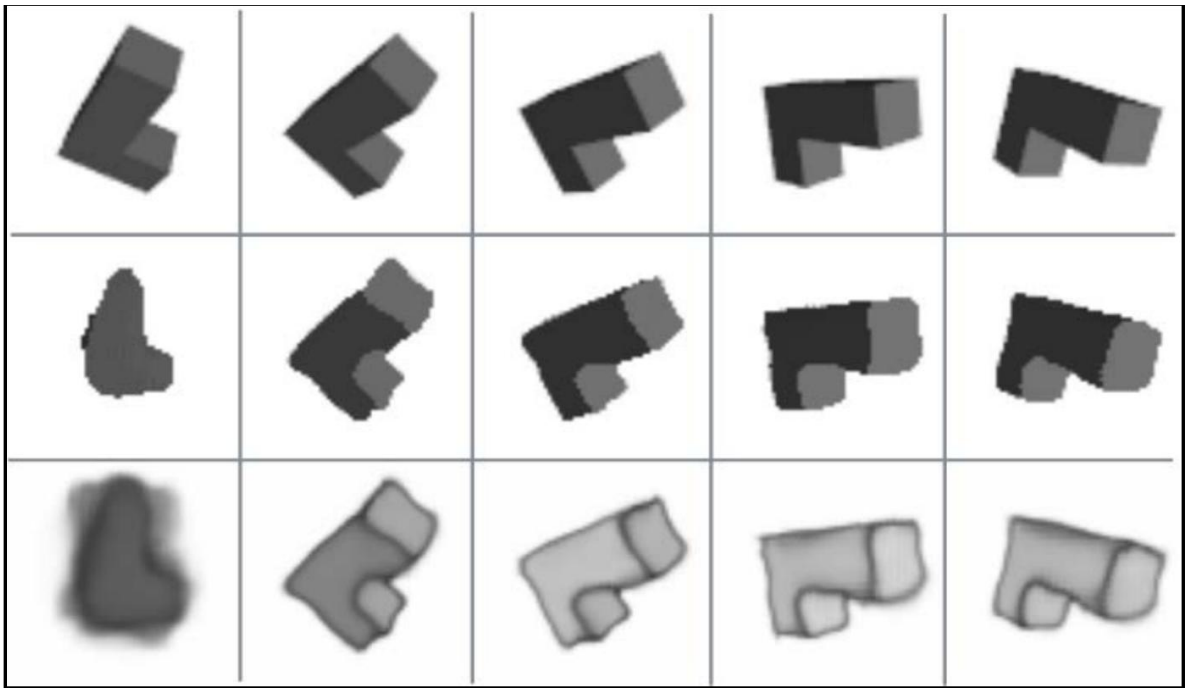


**Figure 2.4 — V-BLT prediction and entropy maps for a rotating 3D polycube.** Rows show ground truth, V-BLT predictions and per-pixel entropy maps. The first low-context forecast is visibly degraded; later frames preserve the coarse polycube structure, while uncertainty remains strongest along moving and newly exposed boundaries. Source: Kulyk and Mokin, *Scaling Video Prediction with Spatio-Temporal Patches*, published Fig. 8, DOI 10.31649/1997-9266-2025-182-5-129-139.

From the standpoint of complex-system control, the entropy map is an ideal feedback signal. It allows the problem of structural-algorithmic uncertainty to be solved by dynamically forming latent patches of variable size. Low-entropy regions (background) can be aggregated into large spatial blocks, saving the $O(L^2)$ resources of the global attention mechanisms. High-entropy regions (motion, fine details) can be split into small patches, concentrating the maximum computational power of the transformer exactly on the zones where the uncertainty is highest.

It is important to note the place of this instrument within the overall system developed in this dissertation. The per-pixel entropy is a *local* measure of informational complexity: it answers the question "where exactly is prediction hardest for the system". It thereby complements the *global* consistency criteria of the generation process — the hierarchical- and trajectory-consistency criteria introduced in Chapter 1, which for the hierarchical generation model developed in Chapter 3 are operationalized in Chapter 4 through autoregressive-rollout degradation metrics stratified by re-entry hardness. The local entropy signal governs the allocation of computational resources within a frame, while the global criteria evaluate the consistency of the generated process as a whole.

The successful substantiation and approbation of the entropy-based method for estimating spatio-temporal uncertainty opens the path to building fully resource-efficient video-generation models. However, the process of transforming a multidimensional spatio-temporal data array into a linear sequence of patches for transformer processing carries a serious mathematical problem — the loss of structural topology. The next subsection is devoted to solving this problem by adapting and integrating multidimensional rotary positional encoding.

## 2.2 Structural Coherence through Multidimensional Positional Encoding

During the transformation of a multidimensional spatio-temporal information array into a form suitable for processing by the model, the primary topology of the complex system is destroyed. Therefore, an important step is to ensure structural coherence before feeding the data into the system.

The multidimensional tensor of visual data, mathematically described as $X \in \mathbb{R}^{T \times H \times W \times C}$, where $T$ is the temporal dimension, $H$ and $W$ are the spatial coordinates, and $C$ the information channels, necessarily undergoes a linearization operation. As a result, a one-dimensional sequence of latent vectors $Z = \{z_1, z_2, \dots, z_L\}$ is formed, with total sequence length $L = T \cdot H \cdot W$. This change of dimensionality is a necessary condition for applying global self-attention mechanisms, but it completely erases the information about the mutual spatio-temporal arrangement of the elements.

In the theory of complex systems, such a loss of information is classified as a critical growth of structural uncertainty. The transformer architecture is, by its mathematical nature, invariant to permutations of the elements of a set. This means that without additional control signals the system cannot distinguish whether two latent patches were physically adjacent horizontally within one frame, or belonged to entirely different frames of the video sequence in time. The absence of a precise coordinate anchoring makes it impossible to model adequate kinematics and geometry of the visual scene.

The traditional approach to preserving positional information is absolute positional encoding [56]. This method adds a special coordinate vector $p_i$ to each latent token $z_i$. However, absolute encoding has fundamental drawbacks in this application: it destroys translation invariance, which is a natural property of the visual world, since the system must recognize and generate an object equally well regardless of its absolute coordinates in the frame.

An alternative approach is relative positional encoding (Relative Positional Bias) [58, 28], where coordinate information is added directly to the attention-weight matrix in the form of biases. Although this method models relative distances more correctly, it is accompanied by a significant growth of computational complexity and requires $O(L^2)$ memory for storing multidimensional bias matrices,

which directly contradicts one of the main tasks of this research — the optimization of resource consumption.

To overcome these systemic contradictions, within the development of the information technology it is proposed to apply the mathematical apparatus of Rotary Positional Embedding (RoPE) [59] with its subsequent generalization to N-dimensional space. The RoPE method combines the advantages of absolute encoding — the absence of quadratic memory costs for storing positions — with the translation-invariance property of relative encoding. The core idea is to represent the state vectors of the system as complex numbers and to apply to them an orthogonal rotation operation in the complex space.

Let the attention subsystem operate on query vectors $q$ and key vectors $k$ of dimensionality $D$. For a one-dimensional sequence, the position of element $m$ is encoded by multiplying the vector by an orthogonal rotation matrix $R_{\Theta,m}$. Applying this operation to the query and key vectors before computing their scalar product makes the attention weight a function of their relative distance only. In the complex representation this is proved by the following equation:

$$\left\langle R_{\Theta,m}\,q,\; R_{\Theta,n}\,k \right\rangle = \operatorname{Re}\!\left[ \left(q\,e^{im\theta}\right) \left(k\,e^{in\theta}\right)^{*} \right] = \operatorname{Re}\!\left[ q\,k^{*}\,e^{i(m-n)\theta} \right] \tag{2.4}$$

As the equation shows, the absolute coordinates $m$ and $n$ cancel algebraically, leaving only their difference $m - n$. This mathematically guarantees that the attention system reacts exclusively to the relative arrangement of elements.

However, the original RoPE method [59] was developed exclusively for one-dimensional sequences such as text and cannot be applied directly to spatio-temporal video data without losing the multidimensional topology. To solve this problem, within the research underlying the results of **[A1]**, the method of N-dimensional rotary positional embedding (RoPE-ND) [59] is used, which allows the system to simultaneously account for the mutual arrangement of elements along several independent coordinate axes.

The essence of the generalization is the decomposition of the latent token's feature space into $N$ independent subspaces, each responsible for its own physical coordinate. For video data the dimensionality of the coordinate system is $N = 3$: the time axis $t$, the vertical axis $y$, and the horizontal axis $x$. Accordingly, the state vectors $q$ and $k$ of total dimensionality $D$ are split into three equal non-overlapping segments of dimensionality $d = D/3$.

For the mathematical rotation operation in the complex plane to be possible, the dimensionality of each subspace must be an even number. This yields a hard architectural requirement on the generative

system: the total dimensionality of the latent feature space $D$ must be divisible by $2N$, i.e., divisible by 6 in the case of video data.

For each of the three coordinates, an independent set of basis rotation frequencies \theta_j is formed. The frequencies are typically defined by a geometric progression:

$$\theta_j = 10000^{-2j/d}, \qquad j \in \{0, \dots, d/2 - 1\} \tag{2.5}$$

Let a visual token of the system be located at the spatio-temporal point with physical coordinates (t_c, y_c, x_c). The rotation angles for each segment are computed as the product of the coordinate and the corresponding frequencies: t_c\theta_j, y_c\theta_j, x_c\theta_j. The resulting unitary transformation of the whole state vector q is formed as the concatenation of three independently rotated sub-vectors:

$$q'_{\text{rotated}} = \left[\; R^{d}_{\Theta,t_c} q_t \;\big\|\; R^{d}_{\Theta,y_c} q_y \;\big\|\; R^{d}_{\Theta,x_c} q_x \;\right] \tag{2.6}$$

where $R^{d}_{\Theta,\text{pos}}$ denotes the $d \times d$ rotation matrix for the corresponding axis. The same operation is applied to the key vector $k$.

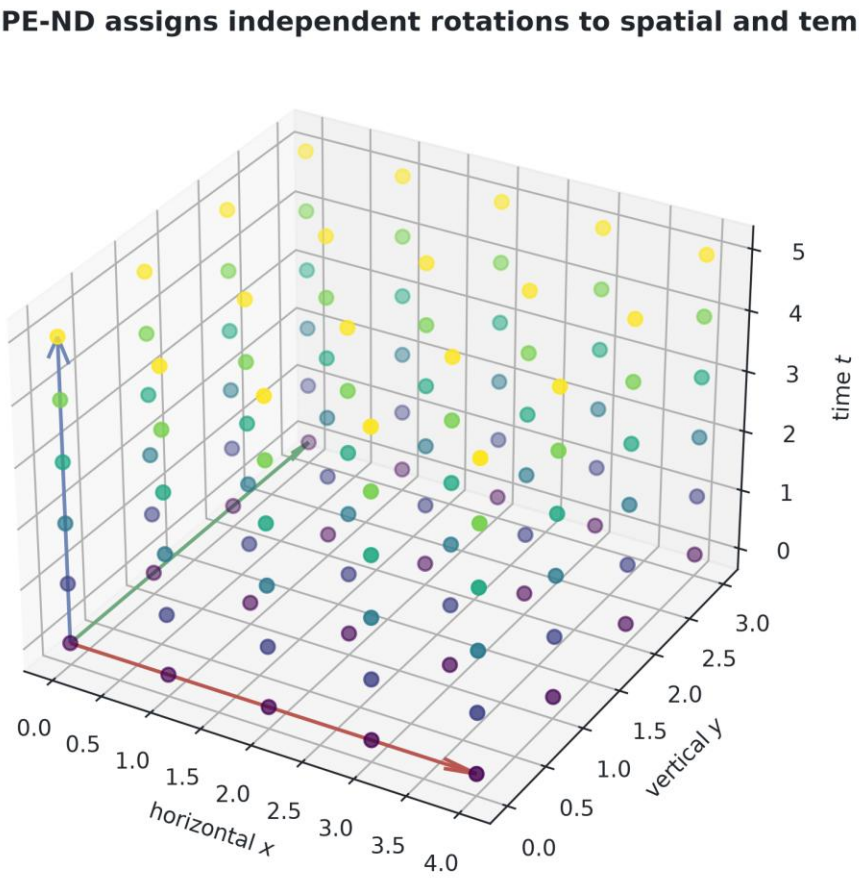



**Figure 2.5 — N-dimensional rotary positional embedding.** Independent rotations encode horizontal, vertical and temporal coordinates so attention depends on relative displacement along all three axes. Source: author's schematic of the [A1] method.

Applying such a multidimensional transformation restores the structural coherence of the complex system. When the attention-score matrix between two arbitrary tokens in the video sequence is computed, the scalar product decomposes into three independent sums, each depending exclusively on the coordinate difference along the corresponding axis:

$$q'^{\top}_{\text{rotated}}\, k'_{\text{rotated}} = f(\Delta t, \Delta y, \Delta x) \tag{2.7}$$

where $\Delta t = t_q - t_k$, $\Delta y = y_q - y_k$, $\Delta x = x_q - x_k$. This proves that the system mathematically identifies not the absolute position of objects in space but their relative spatio-temporal topology, which is critically important for modeling physical motion.

**Systemic significance of the method for hierarchical architectures.** Beyond the plain modeling of temporal dynamics in patch-based systems, the RoPE-ND method is a universal instrument and plays a decisive role in ensuring *hierarchical consistency* in the hierarchical video generation model developed in Chapter 3. In multi-level architectures, a problem of resolving structural uncertainty arises between

different levels of abstraction: for example, how to correctly transfer information between a fine level with a dense token grid (e.g., $32 \times 32$ tokens) and a coarse level with a sparser grid (e.g., $16 \times 16$ tokens).

For the correct routing of cross-attention signals, the coordinate systems of these levels must be mathematically aligned. The RoPE-ND method offers an elegant systemic solution through affine scaling of the coordinates. A parent coarse-level token physically covers a cluster of several child fine-level tokens; after the coordinates of both levels are brought to a common scale, the attention mechanism is forced to center precisely on the geometrically relevant zones. This eliminates the risk of chaotic mixing of semantic features between distant objects and ensures that the global composition governs exclusively its own local details — the mathematical basis of the unidirectional inter-level context-signal transfer developed in Chapter 3.

From the standpoint of resource efficiency, the computational complexity of applying RoPE-ND is strictly linear, $O(D)$, with respect to the feature-space dimensionality, and it requires no additional memory for storing large bias matrices. This makes the developed mathematical apparatus an ideal candidate for integration into architectures with time-agnostic backpropagation, where memory is the most valuable resource. The description of such an architecture and the corresponding optimization method is given in the following subsections.

## 2.3 Resource Optimization for Multi-Step Temporal Modelling

The improvement of the mathematical methods for uncertainty estimation (V-BLT) and structural coherence (RoPE-ND) considered in the previous subsections substantially raises the quality of spatio-temporal data modeling. However, the practical implementation of iterative generative models inevitably runs into the exhaustion of available computational resources as a fundamental systemic constraint. From the standpoint of systems analysis, the effectiveness of controlling any process is determined not only by the adequacy of the mathematical model but also by the feasibility of its implementation under strict resource constraints.

In the modeling of multi-step temporal processes, the main bottleneck is not so much the computational throughput of the processors (the number of floating-point operations, FLOPs) as the amount of operating memory of the graphics accelerators [60, 61]. This problem arises from the architectural features of modern artificial neural networks and their optimization algorithms.

The standard Backpropagation Through Time (BPTT) algorithm [62] is the basis for identifying the parameters of dynamic systems. To compute the control gradients, the algorithm requires the mandatory storage of all intermediate states of the system — the activations — at every layer and every time step during the forward pass.

Mathematically, the amount of operating memory $M$ required to store the full computation graph is linearly dependent on the depth of the temporal unrolling and the number of layers in the system:

$$M = O(N \cdot T \cdot L \cdot D) \tag{2.8}$$

where $N$ is the dimensionality of the spatial state (the number of tokens per frame), $T$ the number of time steps or optimization iterations, $L$ the number of model layers, and $D$ the dimensionality of the latent feature space. Since for video data the spatial parameter $N$ is extremely large, modeling long sequences $T$ very quickly leads to the exhaustion of the available physical memory (Out of Memory, OOM).

The presence of such computational uncertainty forces various heuristic compromises. The most common approaches are truncated backpropagation (Truncated BPTT) [63] and gradient checkpointing [42, 43]. Truncated backpropagation artificially limits the gradient horizon — e.g., optimizing over only the last 5 frames — which inevitably loses the memory of long-range cause-and-effect relations. The checkpointing method reduces memory consumption (to $O(\sqrt{T} \cdot L \cdot D)$ with the standard segment schedule; see Table 2.1), but the price is a significant slowdown of training due to redundant recomputation.

For the effective functioning of the generative model, a fundamental revision of its architecture is necessary. The systemic task is to develop a mathematical method of organizing the computations that would provide constant memory consumption, $O(1)$ with respect to the number of time steps $T$, without losing information about global long-range dependencies.

To solve this systemic problem, the dissertation research developed the Temporal Reversible Transformer (TempVerseFormer) architecture. The developed architecture and the mathematical method of computational-resource optimization were published by the author as the sole author in an international scientific journal indexed by the Scopus scientometric database (**[A2]**).

The method is based on the transition from traditional directed acyclic computation graphs to discrete reversible architectures. The core idea of reversible computation is to build state-transition functions that are bijective (one-to-one). If the transition function from layer $l$ to layer $l+1$ has an exact analytic inverse, the system can restore its previous state purely from the current state, without storing it in memory during the forward pass.

The introduction of the reversibility principle into deep neural networks was previously studied in continuous models of the Neural ODE type [44], but these suffer from numerical instability during integration. Discrete reversible models, e.g., RevViT [45], have shown high stability. The scientific novelty of TempVerseFormer lies in combining structural reversibility at the level of individual transformer layers with an iterative temporal-chaining mechanism for modeling multi-step processes.

The structural decomposition of the reversible block in the TempVerseFormer architecture is based on splitting the input information flow into two independent branches. Let the system operate on a latent state of total dimensionality $D$. At the input of the reversible block, this state is forcibly split into two mutually exclusive sub-vectors $x_1$ and $x_2$. The functioning of the reversible block is described by a system of two sequential nonlinear transformations using the attention mechanism and a multilayer perceptron (MLP).

The mathematical formalization of the forward pass through reversible block $l$ is described by the following system of algebraic equations.

Update equation of the first branch:

$$y_1^{(l)} = x_1^{(l)} + \operatorname{Attention}\!\left(x_2^{(l)}\right) \tag{2.9}$$

Update equation of the second branch:

$$y_2^{(l)} = x_2^{(l)} + \operatorname{MLP}\!\left(y_1^{(l)}\right) \tag{2.10}$$

This specific algebraic structure guarantees that only $x_2$ — which remains unchanged during the first stage — is used to compute the additive update of $x_1$. Accordingly, the input information can be reconstructed absolutely without loss by performing the operations in reverse order and replacing addition with subtraction.

Reverse reconstruction of the second branch:

$$x_2^{(l)} = y_2^{(l)} - \operatorname{MLP}\!\left(y_1^{(l)}\right) \tag{2.11}$$

Reverse reconstruction of the first branch:

$$x_1^{(l)} = y_1^{(l)} - \operatorname{Attention}\!\left(x_2^{(l)}\right) \tag{2.12}$$

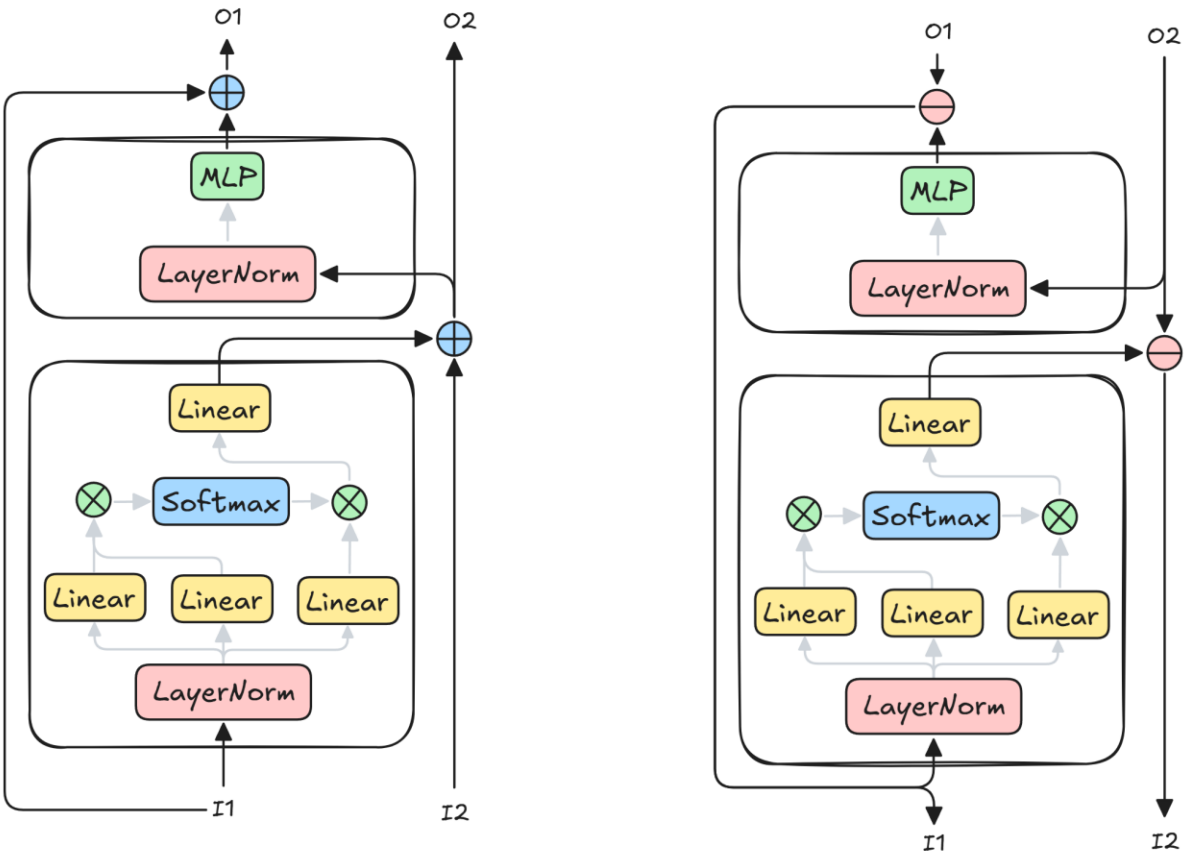



**Figure 2.6 — Forward and reverse information flow in a reversible transformer block.** The additive coupling structure permits exact reconstruction of block inputs from its outputs. Source: author's published TempVerseFormer project asset [A2].

For the reversible blocks to solve the task of controlling generation in time, they are integrated into a higher structural level of the system through the **temporal chaining** mechanism. Instead of concatenating all video frames into a single enormous tensor — which leads to the quadratic memory collapse $O((N \cdot T)^2)$ in the attention mechanisms — the system moves to an iterative autoregressive paradigm of state generation.

The temporal chaining mechanism works as follows. The system receives an initial context set $V_{\text{ctx}}$. At step t, the system applies the transition functional — a stack of L reversible blocks — and generates the time-shifted sequence $V_{\text{out}}$. This output tensor is fed back directly and becomes the input $V_{\text{in}}$ for the next generation step $t+1$. Mathematically this is described by the recurrence relation:

$$V_t = \Phi(V_{t-1}, \theta) \tag{2.13}$$

where $\Phi$ is the transition functor built exclusively from algebraically reversible transformer blocks.

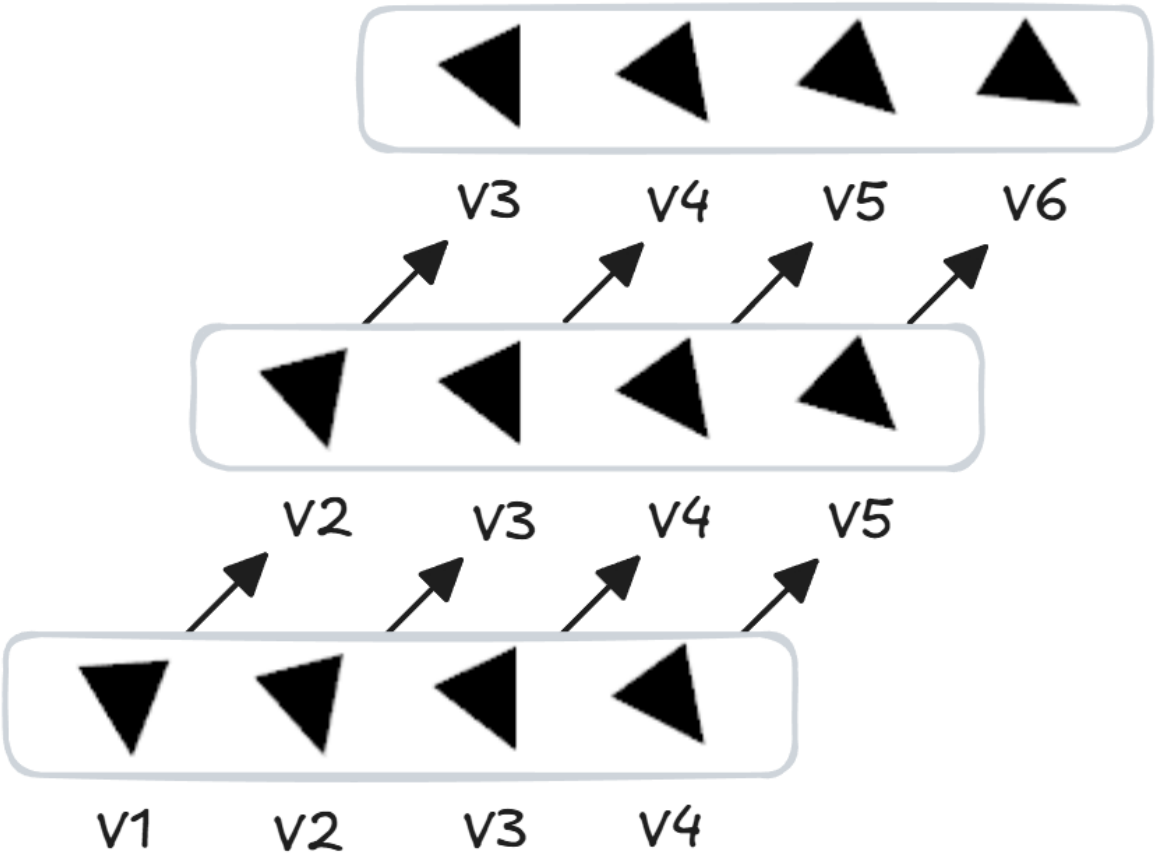



**Figure 2.7 — Temporal chaining for autoregressive time-series modeling.** The same reversible transition is applied along physical time while the realized state is passed to the next step. Source: author's published TempVerseFormer project asset [A2].

Since the functor $\Phi$ consists of reversible elements, the entire multi-step trajectory $V_1 \to V_2 \to \dots \to V_T$ preserves the property of global reversibility. Knowing only the final state $V_T$, the system has the mathematical ability to reconstruct, step by step, all intermediate values without storing them in memory.

To focus the computational resource of the reversible transformer exclusively on the analysis of complex temporal dynamics, the global architecture of the TempVerseFormer information technology uses an auxiliary spatial-compression subsystem — a variational autoencoder (VAE). Its function is to resolve the spatial redundancy of pixels: each video frame is projected into a compact latent vector before processing by the reversible blocks. After generation is completed, the latent states are unfolded back by the VAE decoder. This architectural modularity separates the task of spatial compression from the task of temporal control.

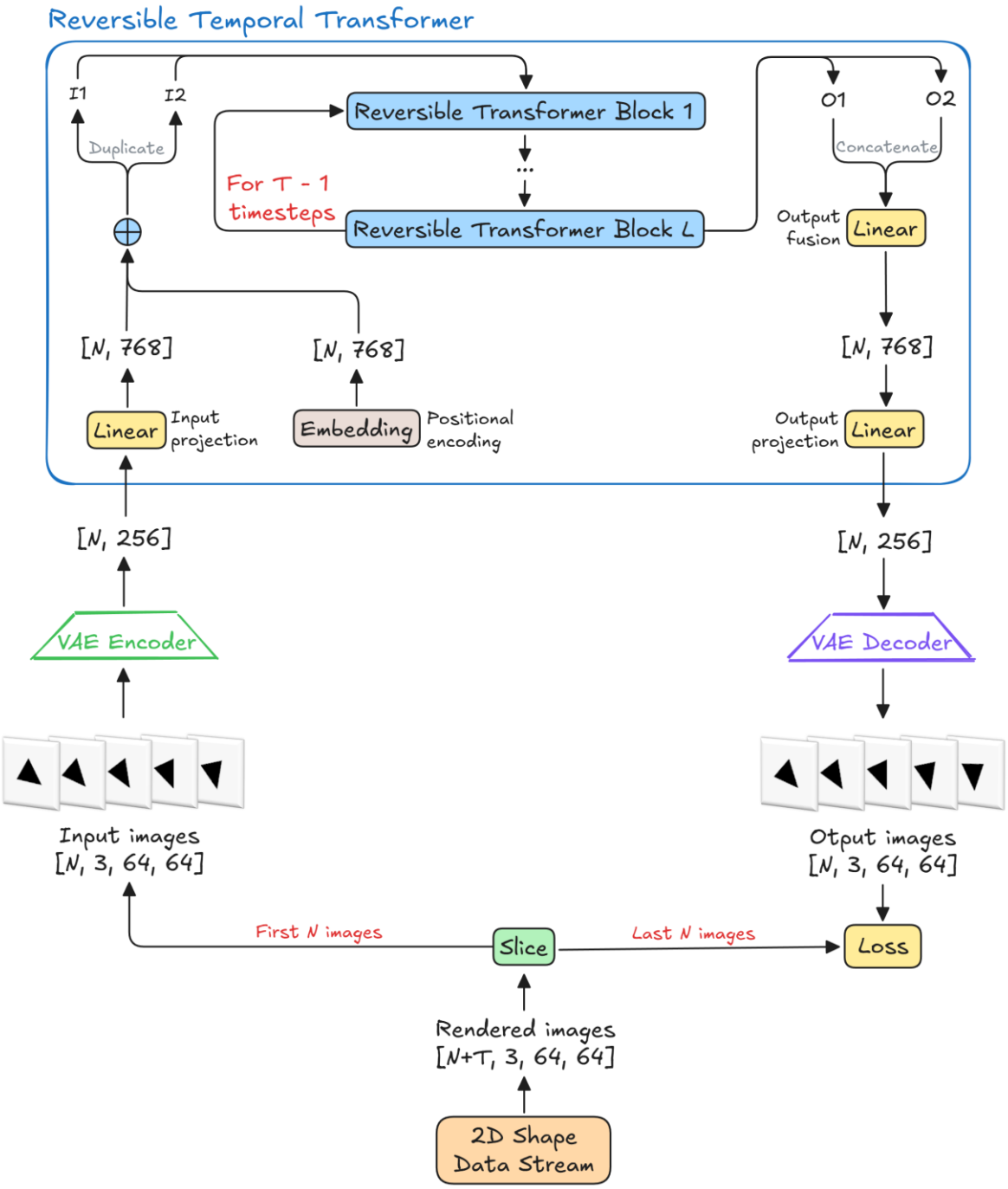



**Figure 2.8 — Global TempVerseFormer architecture.** A spatial VAE interfaces the video frames with the reversible temporal core. Source: author's published TempVerseFormer project asset [A2].

The integration of reversible blocks into a single system with temporal chaining creates an extremely powerful mathematical instrument: structural reversibility guarantees that information is not lost. Yet the main systemic challenge remains — converting this theoretical structural reversibility into an effective training algorithm for computing the control gradients of the loss function. Exactly how the

reversibility property is exploited by the algorithm to remove the computational uncertainty is considered formally in the next subsection.

## 2.4 Time-Agnostic Backpropagation through Reversible Transformations

Solving the problem of computational uncertainty — the critical deficit of operating memory in the optimization of iterative and temporal systems — requires a transition from classical training algorithms to new mathematical paradigms. The traditional BPTT method [62] is the de-facto standard for identifying the parameters of dynamic systems modeled by neural networks, but its mathematical architecture contains a fundamental deficiency from the standpoint of resource control.

The BPTT algorithm is based on the chain rule of differentiation of a composite function. To compute the control gradient of the loss function $L_{\text{loss}}$ with respect to the weight parameters $\theta$ of the system at the current time step, one must know not only the derivative of the state at the next step but also the exact values of the internal activations of the system at every step. Mathematically this means that the forward pass must be accompanied by the caching of absolutely all intermediate vectors of the multidimensional state space, for all steps, in the operating memory of the computing device.

In the modeling of complex spatio-temporal video data, this requirement leads to a catastrophic growth of resource consumption. The control of the system reaches a dead end: to increase the temporal consistency of generation, the optimization trajectory $T$ must be lengthened, but lengthening the trajectory exhausts the memory and makes the gradient computation impossible.

Within this dissertation research, the method of **time-agnostic backpropagation** was developed to resolve this systemic contradiction. The method is based on integrating the mathematical apparatus of discrete reversible transformations directly into the gradient-computation process along the temporal chain.

The main idea of the developed method is a change in the paradigm of computation-graph construction. Instead of statically caching all system states during the forward pass, they are reconstructed dynamically during the backward pass. This reconstruction is mathematically exact and requires no additional training or approximation, since it relies on the algebraic properties of the inverse transition functions $R^{-1}$ built into the architecture of the reversible blocks.

The training strategy of the complex generative system under this algorithm splits into two phases.

**Phase one — the forward pass.** The system sequentially computes the evolution of the state. For every time step $t$ from $1$ to $T$ and every layer $l$ from $1$ to $L$, the forward reversible transformation $R$ is applied to the current inputs $(x_1, x_2)$. After the outputs $(y_1, y_2)$ are computed, the system immediately

deletes all internal activations from memory, keeping only the result. Upon reaching the final step T, the system stores only one final macro-state in memory.

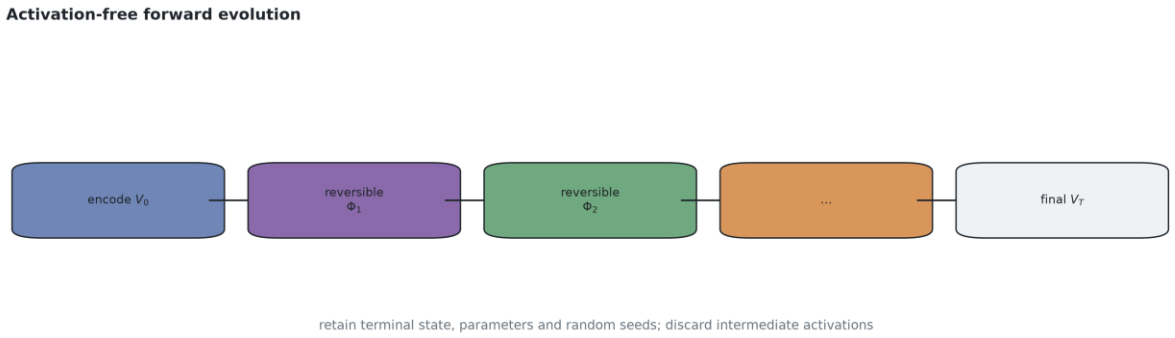



**Figure 2.9 — Activation-free forward evolution.** Intermediate temporal states are discarded; the terminal state, parameters and stochastic replay information are retained. Source: author's algorithmic schematic.

**Phase two — the reverse pass.** This phase is the conceptual core of the method. The reverse pass starts with the computation of the gradient of the loss function with respect to the final state of the system, $\nabla_{y_T} L_{\text{loss}}$. Then, for every time step t from T down to 1, the algorithm moves in the reverse direction: it reconstructs the layer inputs by the inverse transformations (2.11)–(2.12), locally rebuilds the computation graph of a single transition, computes the local gradients, and immediately destroys the local graph.

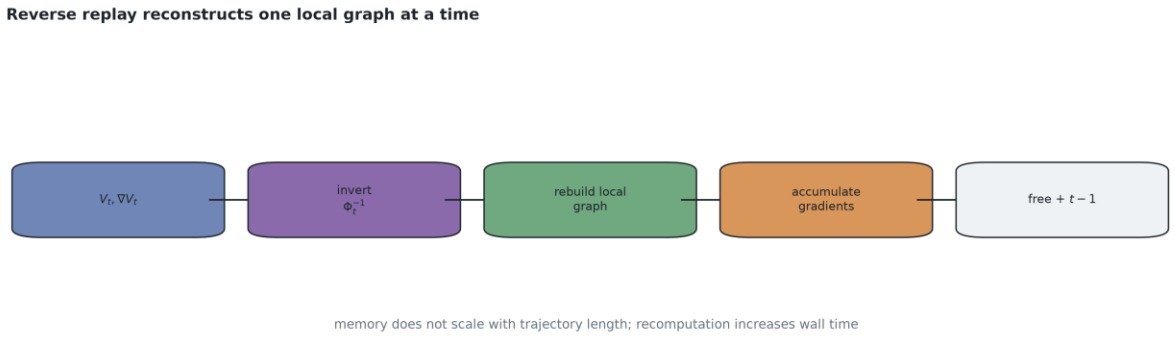



**Figure 2.10 — Reverse-pass local replay.** Each state is reconstructed, its local graph is rebuilt, gradients are accumulated, and the graph is released before moving to the preceding time step. Source: author's algorithmic schematic.

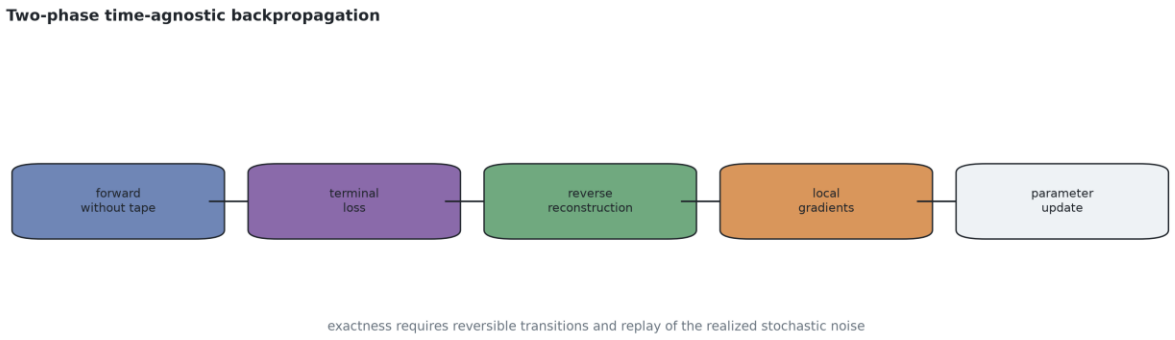



**Figure 2.11 — Two-phase time-agnostic backpropagation.** A tape-free forward phase is followed by exact reverse reconstruction and local gradient accumulation. Source: author's algorithmic schematic.

For a rigorous confirmation of the effectiveness of the developed method, we formulate and prove a lemma on the optimization of operating-memory consumption.

**Lemma 1 (on the computational complexity of time-agnostic backpropagation).** *Let a discrete dynamic system be given with a trajectory length of $T$ steps, where the transition functor is parameterized by a reversible neural network with $L$ layers and per-layer state dimensionality $D$. Then the peak amount of operating memory required to compute the gradients of the objective function with respect to the system parameters is $O(L \cdot D)$ and is constant, $O(1)$, with respect to the length $T$ of the temporal trajectory.*

**Proof.**

1. Assume the dimensionality of the state tensor at one layer is $\mathrm{D}$. Under classical BPTT the algorithm caches all inputs for every layer $\mathrm{l} \in \{1, \dots, \mathrm{L}\}$ and every time step $\mathrm{t} \in \{1, \dots, \mathrm{T}\}$; hence the peak memory is $\mathrm{M}_{\text{BPTT}} = \mathrm{T} \cdot \mathrm{L} \cdot \mathrm{D} = \mathrm{O}(\mathrm{T} \cdot \mathrm{L} \cdot \mathrm{D})$.

2. The proposed algorithm stores, during the forward pass, only the final state at step $\mathrm{T}$. The memory for this is $\mathrm{O}(\mathrm{D})$. Activations for $\mathrm{t} < \mathrm{T}$ and $\mathrm{l} < \mathrm{L}$ are deleted immediately after they are used for the next step. Hence the forward-pass memory does not depend on $\mathrm{T}$.

3. During the reverse pass the algorithm processes the steps sequentially: $\mathrm{t} = \mathrm{T}, \mathrm{T}\text{-}1, \dots, 1$. At the current step $\mathrm{t}$ the algorithm reconstructs the inputs and unrolls the local computation graph of a single transition. The size of this graph is $\mathrm{O}(\mathrm{L} \cdot \mathrm{D})$.

4. Immediately after the gradients $\nabla_{\theta} \mathrm{L}_{\text{loss}}$ and $\nabla_{x_{t-1}} \mathrm{L}_{\text{loss}}$ are computed, this local graph is completely destroyed and its memory freed before the algorithm proceeds to step $\mathrm{t}\text{-}1$.

5. Therefore, the maximum number of tensors simultaneously resident in the system memory at any moment equals the size of one local graph, $O(L \cdot D)$, plus a constant volume for the gradient-accumulation buffer. Since the temporal parameter $T$ does not enter this expression, the peak memory consumption is mathematically $O(1)$ with respect to the sequence length. $\blacksquare$

For comparative analysis, Table 2.1 presents the theoretical complexity estimates of the developed method against the existing approaches.

**Table 2.1 — Theoretical estimate of computational complexity in multi-step optimization**

| Optimization method | Memory consumption | Operation count | Gradient exactness |
|---|---|---|---|
| Standard BPTT | $O(T \cdot L \cdot D)$ | $C$ | exact |
| Gradient checkpointing | $O(\sqrt{T} \cdot L \cdot D)$ | $\approx 1.5\,C$ | $\pm 10^{-6}$ |
| Developed method (time-agnostic BP) | $O(L \cdot D) \sim O(1)$ in $T$ | $2\,C$ | $\pm 10^{-6}$ |

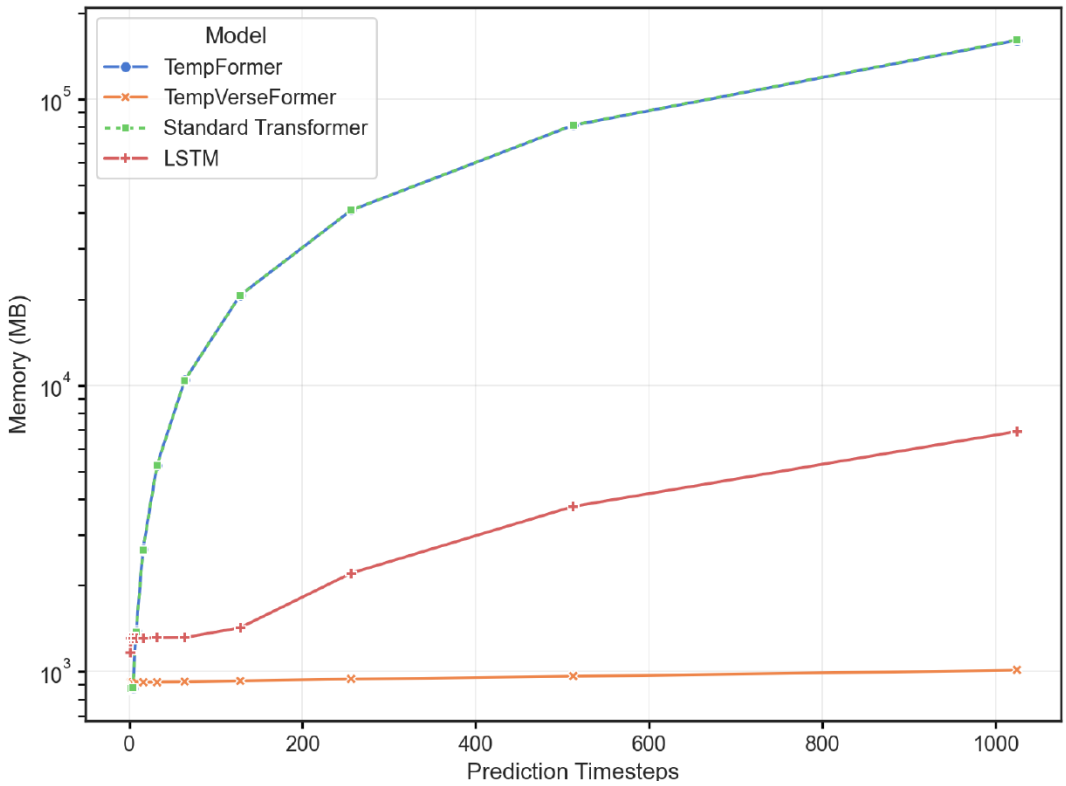



**Figure 2.12 — GPU memory consumption versus prediction horizon.** Published TempVerseFormer measurement at batch size 32. The reversible temporal model remains near 0.9 GB over 1–1,024 prediction steps, whereas the two non-reversible transformer variants grow to approximately 160 GB in the reported extrapolated/extended measurement and the LSTM grows more slowly. The logarithmic

vertical scale must be retained when typesetting. Source: Kulyk, *Temporal modeling with reversible transformers*, Fig. 13, author preprint (CC BY 4.0), DOI 10.21203/rs.3.rs-6293520/v1; version of record DOI 10.1007/s00521-025-11707-3.

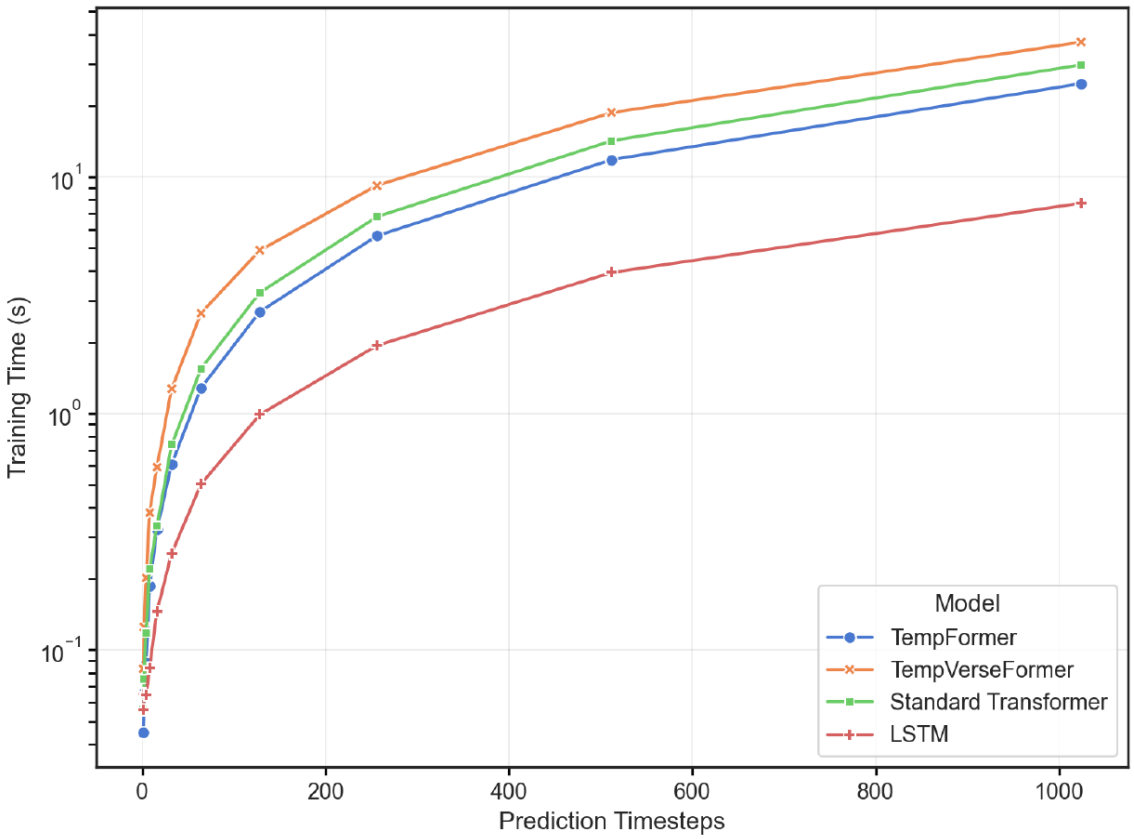



**Figure 2.13 — Training time per step versus prediction horizon.** Published comparison at batch size 32 and 1–1,024 prediction steps. Time grows with the prediction horizon for all four architectures; TempVerseFormer pays a compute premium relative to the matched non-reversible transformer while preserving the qualitatively different memory-scaling behavior in Fig. 2.12. The logarithmic vertical scale must be retained when typesetting. Source: Kulyk, *Temporal modeling with reversible transformers*, Fig. 14, author preprint (CC BY 4.0), DOI 10.21203/rs.3.rs-6293520/v1; version of record DOI 10.1007/s00521-025-11707-3.

As Fig. 2.12 shows, measured temporal-model memory remains nearly constant even at large prediction depths. Figure 2.13 also makes the trade-off explicit rather than cost-free: TempVerseFormer is slower per step than the matched TempFormer over the tested horizons, and all methods become slower as the horizon grows. Within this published configuration, the developed method is the only compared transformer whose activation-memory curve is practically independent of trajectory length. Algebraic reconstruction during the reverse pass therefore exchanges additional computation for the critically

scarce resource — GPU memory — and makes horizons of hundreds or thousands of steps operationally possible when the non-reversible activation tape would exceed the device budget.

In addition, the proposed architecture enables a unique **selective backpropagation** strategy. In many control problems the initial steps of a process are noisy, while the most informative steps are the last ones. The method allows the system to run the forward pass to full length and then reconstruct activations and compute gradients only for a narrow target segment $[T - K_{\text{opt}},\; T]$, ignoring the rest of the chain. This focuses the optimization on the relevant portions of the trajectory with zero memory cost for the skipped stages.

## Extension to Stochastic Recurrences and Limits of Reversible Computation

The time-agnostic backpropagation method was substantiated above for *deterministic* transition functors (2.13). The hierarchical video generation model developed in Chapter 3, however, is a *stochastic* dynamic system: its state transitions include an injected noise component, both as a generative modeling choice and within its iterative-refinement generation core. For the developed resource-optimization method to serve as the training foundation of that model, it must be extended to stochastic recurrences, and the boundaries of its applicability must be established. This section performs both steps; all numerical values are measured properties of the software implementation (the measurement methodology and the experimental environment are described in Chapter 4).

**(i) The exactly invertible stochastic recurrence (store-the-noise).** Consider the stochastic generalization of the temporal chain (2.13), in which the latent state $h_t$ of the modeled process evolves as

$$h_t = F(h_{t-1}) + \sigma\,\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I), \tag{2.14}$$

where $F$ is a transition operator built from reversible (additive-coupling) blocks, $\sigma$ is the noise scale, and $\epsilon_t$ is the noise sample of step $t$. Transformation (2.14), taken alone, is not invertible: the noise irreversibly injects new information. However, if the noise samples $\epsilon_t$ drawn during the forward pass are *stored* — the store-the-noise principle, extending the reversible recurrent networks of [64] — the recurrence becomes exactly invertible. Equations (2.14)–(2.15) state the principle in its simplest, *external-additive* form, in which the stored noise is subtracted before the inverse transition is applied:

$$h_{t-1} = F^{-1}\!\left(h_t - \sigma\,\epsilon_t\right). \tag{2.15}$$

The construction **implemented and measured in this dissertation** is the more general *conditioning-injected* form: within a coupled two-branch state $(s_1, s_2)$, the stored noise enters the coupling functions as a conditioning argument,

$$(y_1, y_2) \;=\; \Bigl(s_1 + F(s_2, \epsilon_t),\;\; s_2 + G\bigl(y_1\bigr)\Bigr), \tag{2.16}$$

whose closed-form inverse subtracts the couplings in reverse order with the *same* stored conditioning replayed:

$$s_2 = y_2 - G(y_1), \qquad s_1 = y_1 - F(s_2, \epsilon_t). \tag{2.16'}$$

The additive-coupling inverse is exact for an *arbitrary* conditioning argument, so (2.16') holds verbatim for any signal injected this way (noise, actions, level context); the external-additive scheme (2.14)–(2.15) is recovered as the special case in which the noise enters only as a final additive term. All numerical properties quoted below are measured on the conditioning-injected construction (2.16)–(2.16'), which is the one realized in the software complex. The memory cost of storing the noise sequence grows only as $O(T \cdot D_{\epsilon})$ — one noise tensor per step, with no layer factor $L$ and no computation-graph factor — and is negligible against the $O(T \cdot L \cdot D)$ activation graph that BPTT would store (confirmed by the direct measurement in part (iii): the total memory of the reversible method stays flat).

The following properties of this construction were verified numerically on the implemented hierarchical multi-rate recurrent core (software complex of the dissertation; measurement details in Chapter 4):

- **Exact invertibility of the deterministic coupling:** maximum state-reconstruction error $5.5 \times 10^{-15}$ (fp64); for the Lipschitz-constrained coupling in fp32, $3.1 \times 10^{-5}$.
- **Exact invertibility of the stochastic recurrence:** with the stored noise replayed, the maximum reconstruction error over the trajectory is $8.4 \times 10^{-13}$; replaying *wrong* noise destroys the reconstruction (error of order $10^{1}$) — i.e., the stored noise is exactly the information-theoretic price of inverting a stochastic step.
- **Gradient equivalence:** the gradients obtained by reverse-replay (reconstructing states backward by (2.15) and rebuilding local graphs, as in Lemma 1) coincide with the gradients of full BPTT with cosine similarity $1.000000$ — the extension preserves the exact-gradient property of the deterministic method.

Thus the time-agnostic backpropagation of §2.4 transfers to stochastic multi-level recurrences without loss of exactness, which is precisely the form required by the hierarchical generation model of Chapter 3 (Task 4 of the research in its extended formulation).

**(ii) Invertibility does not imply stability: the Lipschitz characterization.** A separate systemic question is whether the reversibility of the transition operator stabilizes the long-horizon behavior of the modeled process. The answer, established by direct measurement, is negative — and this delimits what the developed method may and may not be credited with. All recurrences in Table 2.2 are *exactly invertible* (the additive coupling has unit Jacobian determinant); they differ only in the spectral-norm bound

imposed on the coupling function $F$. The measured perturbation growth $\lVert \delta_K \rVert / \lVert \delta_0 \rVert$ of a free-running rollout over $K = 100$ steps spans four orders of magnitude:

**Table 2.2 — Perturbation growth of exactly invertible recurrences over $K = 100$ free-running steps as a function of the Lipschitz bound on the coupling**

| Coupling constraint | $\lVert \delta_{100} \rVert / \lVert \delta_0 \rVert$ | Regime |
|---|---|---|
| unconstrained | $333\times$ | explodes |
| spectral norm, $\operatorname{Lip}(F) \le 1$ | $26{,}600\times$ | **still explodes** |
| spectral norm, $\operatorname{Lip}(F) \le 0.1$ | $1.5\times$ | bounded |
| spectral norm, $\operatorname{Lip}(F) \le 0.02$ | $1.0\times$ | tightly bounded |

The mechanism is elementary: the additive-coupling step $s \mapsto s + F(s)$ has a step-Lipschitz constant

$$\operatorname{Lip}(\text{step}) \approx 1 + \operatorname{Lip}(F) \;\ge\; 1, \tag{2.17}$$

i.e., the additive structure is inherently non-contractive — constraining $F$ to $\operatorname{Lip}(F) \le 1$ is *not* sufficient (the growth factor $26{,}600\times$ in Table 2.2), and bounded long-horizon dynamics require a *small* coupling Lipschitz constant ($\operatorname{Lip}(F) \lesssim 0.1$). This is the stability analysis of invertible networks [65, 66] instantiated for temporal recurrences and confirmed by measurement. The systemic conclusion: **invertibility is a memory-and-correctness property, not a stability property.** In the developed technology, reversibility is claimed exclusively as the constant-memory enabler of long-horizon training with exact gradients; the stability of the modeled trajectory is a separate design requirement, enforced by the Lipschitz constraint on the coupling (this requirement informs the architecture of Chapter 3).

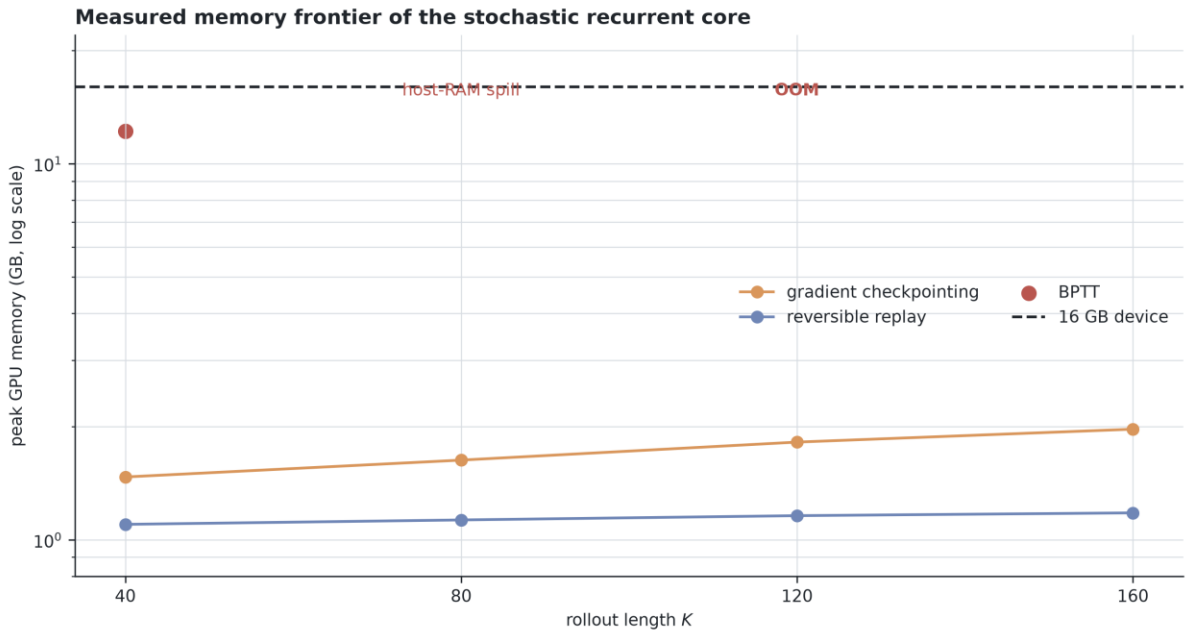



**Figure 2.14 — Measured memory frontier of the stochastic recurrent core.** Standard BPTT reaches 12.2 GB at $K=40$, spills at $K=80$ and cannot run at $K=120$ on the 16 GB device; checkpointing grows slowly, while reverse replay stays near 1.1–1.2 GB through $K=160$. Source: repository Panel-A measurements (`hvp_rev/mem_frontier.py`, `docs/SESSION_JOURNEY_AND_EBT_PLAN.md`, §2.2).

**(iii) The constant-memory autoregressive-rollout frontier (measured).** The practical payoff of the extended method is measured on the real training loop of the implemented stochastic hierarchical recurrent core (three levels, batch 32, Adam optimizer, NVIDIA RTX 4060 Ti 16 GB; the full environment is specified in Chapter 4, §4.1.4). Table 2.3 gives the peak training memory as a function of the free-running rollout length $K$:

**Table 2.3 — Measured peak training memory (MB) vs rollout length $K$: BPTT, gradient checkpointing, and the developed reversible reverse-replay on the same model**

| K | BPTT | Gradient checkpointing | Developed method (reversible) |
|---|---|---|---|
| 40 | 12,162 | 1,471 | 1,091 |
| 80 | 23,305* | 1,637 | 1,125 |
| 120 | OOM | 1,801 | 1,152 |
| 160 | OOM | 1,967 | 1,176 |

* At $K = 80$, BPTT exceeds the physical GPU memory and spills into shared host RAM, slowing a training step by a factor of $\approx 60$ (from 0.74 s at $K = 40$ to 43.8 s); at $K = 120$ it fails with a hard out-of-memory error.

The frontier confirms Lemma 1 and its stochastic extension empirically: BPTT consumes 12.2 GB already at $K = 40$ and becomes infeasible at $K = 120$, while the developed method remains essentially flat ($\approx$1.1–1.2 GB through $K = 160$), making long-horizon rollout training feasible on a consumer-grade 16 GB device. **The honest reading of Table 2.3, however, must include the checkpointing column:** gradient checkpointing also remains feasible at all measured $K$ (1.47–1.97 GB, growing slowly). The correct claim for the developed method is therefore a **constant-memory capability relative to vanilla BPTT** — with strictly exact reconstruction and gradients equal to BPTT — and an approximate **Pareto equivalence with gradient checkpointing** (comparable memory order and comparable recomputation overhead), *not* a memory victory over checkpointing. This scoping is stated deliberately: it is what the three-axis analysis below turns into a general applicability boundary.

**(iv) The three-axis taxonomy of reversible computation and its no-go result.** The reversible mechanisms developed and reviewed in this chapter can be organized along three orthogonal axes of the computation graph on which reversibility can be exploited to avoid storing states:

1. **The depth axis** — reversibility across the layers of a network at a single step (RevNet and its descendants [45]): the literature baseline on which the developed blocks build.

2. **The temporal axis** — reversibility along the physical time of the modeled process (temporal chaining of §2.3–2.4 and the stochastic extension (2.14)–(2.16)). On this axis reversibility **pays**: the stored quantity that BPTT accumulates is the $O(T \cdot L \cdot D)$ activation trajectory itself, exactly what the inverse coupling reconstructs; the measured result is the frontier of Table 2.3.

3. **The computational axis** — reversibility along the *inner optimization loop*: the $K$ pseudo-time steps of iterative refinement (gradient descent, MCMC/Langevin sampling, or a diffusion sampler) executed *inside* one generation step. The original draft of this dissertation asserted that the graph isomorphism between the temporal chain $T$ and the inner chain $K$ makes time-agnostic backpropagation the foundation for differentiating through such loops. The isomorphism of the graph *structure* is real, but the memory *accounting* is opposite, and on this axis reversibility **provably cannot pay** against checkpointing.

The accounting for the third axis is as follows. Differentiating through a $K$-step gradient-descent (or Langevin) inner loop requires, at each reverse step, a Hessian-vector product and mixed second-order terms — i.e., rebuilding the differentiable graph of $\nabla E$ (a `create_graph`-class second-order graph) from a fresh forward pass, *regardless of how the trajectory state at that step was obtained*. Reversibility

can eliminate only the storage of the trajectory states themselves. **The direct four-way measurement on the implemented system** (naive full-graph / truncated / checkpointed / reversible differentiation of the inner loop, $K \in \{4, 8, 32, 128\}$, micro-batch 12, fp32) gives: the naive full graph consumes 18.1 GB already at $K = 4$ (spilling to shared memory) and exhausts device memory at $K = 8$; the truncated, checkpointed, and reversible variants all remain flat at 5.282–5.296 GB through $K = 128$; and the reversible variant saves **14 MB = 0.26% of the peak** relative to checkpointing, at approximately twice the wall-clock time (recomputation). The per-step descent state $(y, v)$ is 0.117 MB, so checkpointing's total state storage at $K = 128$ is 15.0 MB = 0.28% of its measured peak — an upper bound on what reversibility could ever reclaim on this axis. The analytic accounting band for the same quantity is $\approx 0.6–1.0\%$, and the crossover at which trajectory states would begin to dominate the second-order graph lies at $K \approx 5{,}000–30{,}000$ inner steps, far beyond any practically meaningful refinement depth. The underlying storage arithmetic — counting the bits per step that an optimizer trajectory consumes versus the volume contraction of the update map — is due to [67], **applied** here to the video-generation inner loop, not claimed as a new theorem. Moreover, plain gradient descent is not closed-form invertible at all (inversion requires the momentum-buffer scheme of Maclaurin et al., whose kilobyte-scale state itself demonstrates that trajectory storage was never the binding constraint), and the reversible-sampler variants for diffusion inner loops are already published [68, 69, 70], with large-scale practice choosing checkpointing or truncation precisely because state $\ll$ graph [71, 72].

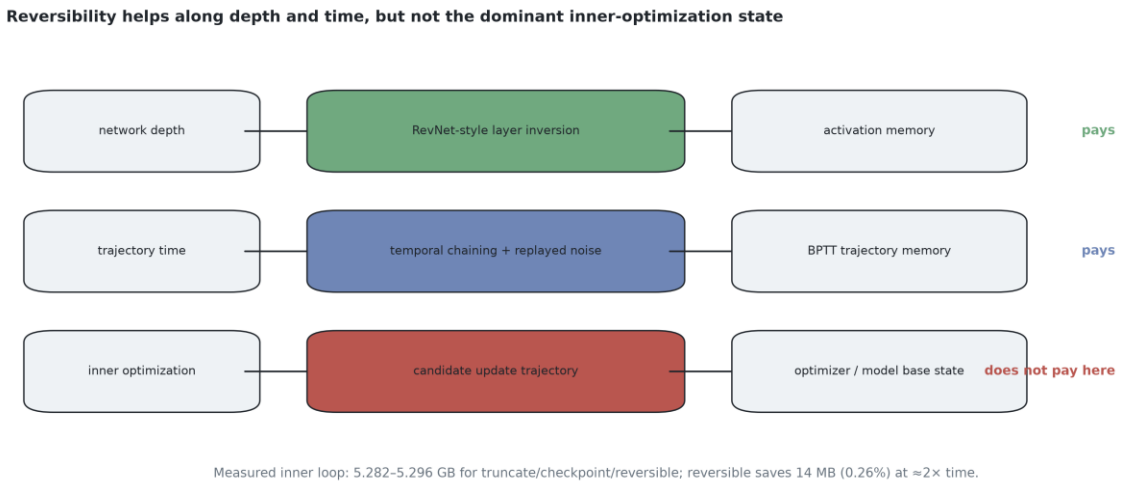



**Figure 2.15 — Three axes of reversible computation and their measured applicability.** Reconstruction pays along network depth and temporal recurrence. In the tested inner-optimization loop the model/optimizer base state dominates: reversible replay saves only 14 MB (0.26% of peak) relative to checkpointing while approximately doubling wall time. Source: author's taxonomy from `logs_rhcp/mem_panel.md`.

The resulting **applicability boundary** of the developed resource-optimization method is stated as the closing result of this chapter: *reversibility pays on recurrences whose transition operator is the invertible coupling itself — the temporal rollout of the modeled video process — and cannot pay on inner optimization loops, whose reverse step is dominated by the reconstruction of a per-step derivative graph; the latter regime belongs to gradient checkpointing.* This scoping replaces the earlier $T \leftrightarrow K$ isomorphism claim with a two-sided, measurement-backed delimitation: it simultaneously establishes where the method is the correct tool — as the **enabling instrument** for constant-memory rollout-through-time training, whose feasibility Chapter 4 demonstrates by direct memory measurement (the validated Chapter 3 model itself is trained single-window and does not presuppose it) — and closes, on the record, the axis where it is not.

## 2.5 Chapter Conclusions

In the second chapter of the dissertation, a complex of mathematical methods was developed and substantiated for resolving the structural, informational, and computational uncertainties in multi-step video generation systems. The main results are as follows.

1. In accordance with **Task 3** of the dissertation research, a method for estimating the

spatio-temporal uncertainty of video data was developed through the transition from rigid patch tokenization to adaptive entropy-based decomposition (the V-BLT architecture). It was established that computing the information entropy at the level of raw bytes yields per-pixel entropy maps that act as a reliable diagnostic indicator of complex kinematics and can be used for the optimal redistribution of the system's computational resources.

2. To ensure the structural coherence of data under spatial linearization, the mathematical

apparatus of N-dimensional rotary positional embedding (RoPE-ND) was developed. It was proven that the method algebraically guarantees the preservation of the multidimensional spatio-temporal topology inside the attention mechanisms. A coordinate-scaling mechanism was developed that provides an instrument for precise spatial localization of cross-level cross-attention between abstraction levels. This is a level-appropriate instrument for architectures that require spatial coordinate transfer; the measured Chapter 3 predictor instead uses learned level/role/frame embeddings in joint attention, so its results do not depend on RoPE-ND.

3. In accordance with **Task 4**, the fundamental problem of computational uncertainty — the

critical deficit of operating memory in BPTT optimization of long processes — was resolved. The TempVerseFormer mathematical architecture was developed on the basis of discrete reversible transformer blocks joined by the temporal-chaining mechanism.

4. The lemma on the computational complexity of the time-agnostic backpropagation algorithm was

formulated and proven: the algebraic reconstruction of local graphs during the reverse pass reduces the peak memory consumption from $O(T \cdot L \cdot D)$ to $O(L \cdot D)$ — a constant with respect to the trajectory length — while preserving gradient exactness.

5. The time-agnostic backpropagation method was **extended to stochastic multi-level

recurrences **via the exactly invertible store-the-noise recurrence: with stored noise, state reconstruction is exact (measured error $8.4 \times 10^{-13}$) and the reverse-replay gradients coincide with BPTT (cosine similarity $1.000000$). The measured memory frontier of the real training loop confirms the practical payoff — BPTT consumes 12.2 GB at rollout length $K = 40$ and fails (OOM) at $K = 120$, whereas the developed method remains at $\approx$1.1–1.2 GB through $K = 160$, with gradient checkpointing remaining a comparable alternative (1.47–1.97 GB) — i.e., the method provides a constant-memory capability relative to vanilla BPTT and approximate Pareto equivalence with checkpointing, which is stated honestly. The** applicability domain **of reversible computation was delimited by the three-axis taxonomy (depth — time — computation): the temporal-rollout axis pays and forms the resource foundation for the long-horizon autoregressive training of the hierarchical video generation model of Chapter 3; the inner-optimization-loop axis provably cannot pay against checkpointing (a saving of $\approx 0.2\%$ at $K = 128$, with the state-versus-graph accounting applied after Maclaurin et al.). It was additionally established by measurement that** invertibility does not imply stability**: exactly invertible recurrences span perturbation-growth regimes from $333\times$ to $1.0\times$ over 100 steps purely as a function of the Lipschitz bound on the coupling, so trajectory stability is a separate design requirement (a Lipschitz constraint), which is taken into account in the architecture of Chapter 3.

The results of this chapter are published in the author's works **[A1]** (§§2.1–2.2), **[A2]** (§§2.3–2.4), and the extension to stochastic recurrences with the three-axis applicability taxonomy in **[A4]**. These stable labels are bound mechanically to the final numbered publication list.

# Chapter 3 — A Hierarchical Mathematical Model for Video Generation with Long Multi-Stream Context

The preceding chapters performed the decomposition of the video-generation system and developed the mathematical instruments for uncertainty estimation (entropy-based adaptive decomposition, RoPE-ND) and for resource-efficient multi-step optimization (reversible transformations, time-agnostic backpropagation). This chapter synthesizes these results into the central contribution of the dissertation: a hierarchical mathematical model of video-sequence generation in a compact continuous hierarchical latent state space with **long-span multi-stream context**: the demonstrated system carries long-range temporal memory in a long-span context stream — the coarse (spatially downsampled) stream being its measured spatial implementation within the multi-stream family of §3.1/§3.3.5 — context signals are transferred between abstraction levels within a single joint attention mechanism, and per-frame generation is performed by an iterative-refinement (denoising diffusion) core whose selection is justified empirically by the matched-capacity and matched-NFE mechanism study of Chapter 4.

Two methodological commitments frame the chapter. First, the model is described exactly as implemented and verified in the software complex (module `hvp_rev/rhvd.py` of the developed system; the design was pre-registered before the experimental campaign): every architectural constant quoted below is the constant of the working system, not a proposal. Second, alongside the generative model itself, the chapter formalizes the **measurement instrument** — a purpose-built memory-horizon testbed with a re-entry-stratified evaluation protocol (§3.4). Under the systems-analysis methodology of specialty 124, the instrument is a deliverable of equal rank with the model: it is what operationalizes the trajectory-consistency criterion of Chapter 1 (§1.4.5) into an exactly computable, statistically disciplined measurement procedure.

## 3.1 Formalization of the Hierarchical Latent State Space

The chapter's organizing object, defined here once and used throughout (full formalization in §3.3.5): a **context-stream family**. Each context stream is a triple $s_i = (r_i,\; k_i,\; n_i)$ — spatial resolution $r_i$, temporal stride $k_i$, and slot count $n_i$, and a system configuration is a set of such streams consumed by one shared attention backbone. In this vocabulary: a **flat** configuration is a single fine-resolution stride-1 stream; a **temporal (rate) hierarchy** has two or more tick rates at any resolution; a **resolution hierarchy** has two or more spatial resolutions; a **deep hierarchy** has three or more streams; the **context budget** counts context tokens only, with total sequence length and compute reported separately (§4.1). The two-level coarse-plus-fine model developed next is the family's per-level-decodable resolution form — the dissertation's demonstrated system, not the family's definition.

### 3.1.1 Two-Level State Decomposition and Frozen Autoencoder

Chapter 1 (§1.2.2) established the hierarchical decomposition of the visual scene state into levels of abstraction as the core systems-analysis remedy for the anomalies of flat latent spaces. The developed model instantiates this decomposition with $L = 2$ levels, which is the minimal configuration that exhibits the phenomenon under study (a low-rate upper level able to carry memory that the detailed lower level cannot afford), while remaining fully trainable and evaluable on a single consumer-grade GPU — a deliberate scoping decision consistent with the resource constraints formalized in Chapter 1.

The state of the video process at frame t is represented not by the pixel frame $x_t \in \mathbb{R}^{64 \times 64}$ but by a pair of latent codes produced by a shared convolutional encoder $\mathcal{E}$:

$$z_t \;=\; \bigl(z^{\mathrm{c}}_t,\; z^{\mathrm{f}}_t\bigr) \;=\; \mathcal{E}(x_t), \qquad z^{\mathrm{c}}_t \in \mathbb{R}^{16 \times 4 \times 4}, \qquad z^{\mathrm{f}}_t \in \mathbb{R}^{16 \times 8 \times 8}, \tag{3.1}$$

where $z^{\mathrm{c}}_t$ is the **coarse** (macro-level) code — 16 channels on a $4{\times}4$ spatial grid, 256 numbers per frame — and $z^{\mathrm{f}}_t$ is the **fine** (micro-level) code — 16 channels on an $8{\times}8$ grid, 1{,}024 numbers per frame. Both codes pass through a $\tanh$ nonlinearity and are therefore bounded. The coarse code is, by construction and by training incentive, the low-rate "layout" description of the scene (global composition and camera position); the fine code carries texture and stroke detail. In the macro/meso/micro vocabulary of §1.2.2, the coarse level plays the macro role and the fine level the micro role; the meso level is absorbed — a two-level pyramid is the minimal sufficient instantiation for the claims under test.

The defining structural property of the autoencoder — adopted from the hierarchical latent video modeling line of work [32, 13, 15] in minimal form — is **per-level decodability**: each level is *independently* decodable to the full frame by its own decoder,

$$\hat{x}^{\mathrm{c}}_t = \mathcal{D}_{\mathrm{c}}\bigl(z^{\mathrm{c}}_t\bigr), \qquad \hat{x}^{\mathrm{f}}_t = \mathcal{D}_{\mathrm{f}}\bigl(z^{\mathrm{f}}_t\bigr), \tag{3.2}$$

so that neither level is a mere residual of the other: the coarse code alone suffices to reconstruct a (blurred but correctly composed) frame, and the fine code alone suffices to reconstruct a detailed frame. This property is enforced during autoencoder training by random-level dropout: at each training step exactly one of the two reconstruction losses (L1 + MSE) is supervised, with the coarse branch up-weighted ($\times 4$) so that the low-rate level is not ignored. The trained autoencoder has 1.85 M parameters and reaches a validation reconstruction error of 0.0020 (MSE, fine level) on the testbed data of §3.4.

Two systemic decisions about this autoencoder follow the unidirectional-control principle (§3.1.3) and are stated here explicitly:

1. **The autoencoder is frozen.** It is trained once, on the training clips only, and never

fine-tuned afterwards; during the training and operation of the predictive core its parameters are algorithmically excluded from the gradient graph. The latent space is therefore a *fixed coordinate system* for the entire study — every predictive mechanism compared in Chapter 4 operates in literally the same state space, which is a precondition for a valid matched-NFE comparison. This decision retains, in the new architecture, exactly the systemic function that the old design assigned to its frozen semantic encoder: invariance of the state coordinates during generation.

2. **The latent interface is standardized.** The raw $\tanh$-bounded codes are small in scale

(per-channel standard deviations average 0.121 for the coarse and 0.162 for the fine level). Every latent channel $c$ of every level $\ell \in \{\mathrm{c}, \mathrm{f}\}$ is z-scored with statistics computed **on the training set only**:

$$\tilde{z}^{(\ell)}_{t}[c] \;=\; \frac{z^{(\ell)}_{t}[c] - \mu^{(\ell)}_{c}}{\sigma^{(\ell)}_{c}}, \qquad \sigma^{(\ell)}_{c} \ge 10^{-4}, \tag{3.3}$$

and all held-out data are standardized with the *same* constants. This step is part of the model definition, not an implementation detail: the diffusion process of §3.2.3 injects noise at unit scale, and an unstandardized latent (scale $\approx 0.12$–$0.16$) silently mis-scales the entire noise schedule — the initial noise sample is then 4–7× off-scale relative to the data manifold. Chapter 4 (§4.2) documents that this single scaling error is capable of reversing mechanism-level experimental conclusions; the standardization (3.3) is therefore elevated to a protocol requirement. In the remainder of the chapter, $z$ denotes the standardized latents.

The pixel decoders $\mathcal{D}_{\mathrm{c}}, \mathcal{D}_{\mathrm{f}}$ participate in the system only downstream of the predictive core and only through the frozen interface — the old chapter's "detached pixel decoder" principle survives here in a strictly stronger form (the decoder is not merely gradient-detached but entirely frozen): pixel-space rendering can never destabilize the optimization of the dynamics core, and all pixel-space metrics of §3.4.3 are measured through one and the same fixed decoder for every compared system.

**Figure 3.1 — Model architecture and context-packaging alternatives.** *Content:* input frames → frozen two-level conv-AE (shared encoder trunk; coarse $16\times4\times4$ and fine $16\times8\times8$ taps); training-only latent standardization; the shared WindowDiT generation core consuming a selected flat, spatial or temporal package together with actions; a jointly predicted four-frame latent window; and the frozen fine decoder. The lower panel reports exact package

accounting and makes the 1,088-token temporal versus 1,152-token spatial comparison explicitly near-matched rather than exact. *Source:* **`docs/figures/fig16_context_packaging_architecture.png`** and vector PDF; generated deterministically by `scripts/make_rft_core_schematics.py` from the frozen interface.

### 3.1.2 Cross-Level Context Transfer: from Cross-Attention to Joint Attention

If the two levels of the hierarchy were optimized in complete isolation, the generated states would contradict each other — the coarse level could place scene content at one position while the fine level renders texture consistent with another. Guaranteeing the integrity of the multi-level system requires a strict information channel between the levels; Chapter 1 (§1.4.4) introduced *hierarchical consistency* as the corresponding criterion. The question is the mechanism of this **inter-level context-signal transfer**.

The classical mechanism, described in the earlier design iterations of this work, is cross-level cross-attention: queries formed from the child level, keys and values projected from the parent level, with positional coordinate scaling to align the two spatial resolutions. The developed model retains the *function* of this mechanism but generalizes its *form*: instead of a directed pair of attention maps between separate per-level networks, all tokens of both levels — context and target alike — are placed into **one joint self-attention sequence** of a single shared transformer (§3.2.1). Within every attention layer, each fine token attends to all coarse tokens (and vice versa), with the level identity, temporal position, and role of every token supplied to the network as additive embeddings and as per-token conditioning of the adaptive normalization (§3.2.2). Cross-level transfer is thus realized as a special case of joint attention: the attention sub-blocks between fine queries and coarse keys are exactly the cross-attention of the earlier design, but they are learned jointly with the within-level blocks rather than imposed as a separate module.

This generalization has three systemic consequences:

1. **Bidirectional consistency coupling at generation time.** Both levels of the target window

are denoised *jointly, in one pass, under a shared noise level* (§3.2.3). The fine level conditions on the coarse level's current (partially denoised) state and the coarse level conditions on the fine level's — at every one of the refinement steps. Hierarchical consistency is therefore enforced continuously along the generation trajectory, not once at a hand-off between stages.

2. **Asymmetric information budgets without asymmetric wiring.** The long memory enters the

sequence only as coarse tokens (48 frames × 16 tokens), while the fine level contributes only a short window (6 frames × 64 tokens). The *architecture* is symmetric; the *information design* is what makes the coarse level the carrier of long-range memory (§3.3.4). This is the precise sense in which "the coarse

level carries the memory horizon" is a designed property of the developed model rather than an emergent accident.

3. **No hand-crafted spatial masks.** The old design constrained the child-to-parent attention by a hard $2{\times}2$ receptive-field mask routed by scaled positional coordinates. In the developed model, the spatial correspondence between a $4{\times}4$ coarse cell and its $2{\times}2$ block of $8{\times}8$ fine cells is available to the network through the learned per-level spatial position embeddings, but it is not imposed as a hard constraint; at the adopted resolutions the joint attention learns the routing directly. Thus the hard $2{\times}2$ mask of the superseded 32×32 design is explicitly not part of the validated 4×4/8×8 implementation. RoPE-ND remains the scaled-coordinate instrument developed for the [A1] system in Chapter 2.

### 3.1.3 One-Way Control with Detached Gradient Paths

The old chapter postulated a strict *sequential* ordering of level generation (macro level first, then meso, then micro, each receiving the previous level's output as a gradient-detached constant) — and simultaneously, in its own algorithmic listing, executed the levels in the opposite order. The resolution adopted in this dissertation is to state what the validated implementation actually does and to re-scope the unidirectionality principle to the boundaries where it is both real and load-bearing:

**Within a prediction step, the levels are generated jointly, not sequentially.** As established in §3.1.2, one attention pass denoises both levels of the 4-frame target window simultaneously. There is no macro-first or micro-first ordering *inside* the step, and the dissertation makes no such claim. What survives of the top-down intuition is the **information-design asymmetry**: the coarse level is the only level whose history extends over the full memory horizon, so at generation time the long-range structure of the prediction can only originate from the coarse context — the fine level physically lacks the inputs to contradict it at long range. The control is unidirectional in the information-theoretic sense (long-horizon signals flow coarse → fine because only the coarse channel transports them), not in the scheduling sense.

**Gradient detachment survives at two system boundaries**, where it is exact and deliberate:

1. **Between the representation subsystem and the prediction core.** The autoencoder is frozen (§3.1.1): no gradient of any prediction objective ever reaches $\mathcal{E}$, $\mathcal{D}_{\mathrm{c}}$, or $\mathcal{D}_{\mathrm{f}}$. The state coordinate system is a fixed constant of the control problem. This is the direct descendant of the old design's Detached-KV principle — the parent representation is a constant with respect to the child's optimization — applied at the representation/dynamics boundary where it demonstrably matters.

2. **Between past and future along the rollout.** At every autoregressive step (§3.3.1), the

context buffers contain previously generated latents that enter the network as constants: during training no gradient propagates from the current window's loss into earlier windows (training is single-window by design, §3.2.3), and during generation the sampled window is appended to the buffers via an implicit detachment (sampling runs under no-grad). Destructive interference between the optimization of successive windows — the failure mode the old chapter's unidirectional control was designed to prevent — is thereby excluded structurally.

This formulation combines stop-gradient discipline, constant parent context and prevention of bidirectional gradient interference. The implemented and validated system generates both levels jointly in a single attention pass, with the coarse level distinguished by *what it remembers*, not by a separate execution phase.

## 3.2 Generative Core: Windowed Latent Diffusion

The per-frame generation mechanism of the developed model is an iterative-refinement (denoising diffusion) core. This choice is not aesthetic: Chapter 4 (§4.2) selects it by a matched-capacity, matched-capacity and matched-NFE comparison of three mechanisms sharing this chapter's backbone — direct regression, denoising diffusion, and energy-based gradient descent — in which diffusion commits to sharp, correctly located content where regression hedges toward the conditional mean and the energy-based alternative fails in a two-sided, characterized manner. This chapter presents the selected core; the selection evidence belongs to Chapter 4.

### 3.2.1 Token Sequence and Component-Wise Conditioning

The prediction problem is windowed. Let t be the first frame to predict, $W = 4$ the window length, H_{\mathrm{c}} the coarse context length and w_{\mathrm{f}} the fine context length (system configuration: $H_{\mathrm{c}} = 48$, $w_{\mathrm{f}} = 6$; §3.3.4 introduces the control configurations). One prediction step consumes:

- the coarse context $\;C^{\mathrm{c}} = \bigl(z^{\mathrm{c}}_{t-H_{\mathrm{c}}}, \ldots, z^{\mathrm{c}}_{t-1}\bigr)$ — the last 48 coarse latents;
- the fine context $\;C^{\mathrm{f}} = \bigl(z^{\mathrm{f}}_{t-w_{\mathrm{f}}}, \ldots, z^{\mathrm{f}}_{t-1}\bigr) — the last 6 fine latents (the fine context frames are by construction the *last* w_{\mathrm{f}} frames of the coarse window, so the two contexts are temporally aligned and satisfy w_{\mathrm{f}} \le H_{\mathrm{c}}$);
- the recorded control actions $a_t, \ldots, a_{t+W-1}, a \in \{-1, 0, +1\}$ (§3.4.1), where $a_{t+j}$ is the action that moves the process from frame $t+j-1$ to frame $t+j$;

- the noisy target window \;y = \bigl(y^{\mathrm{c}}_{t..t+3},\, y^{\mathrm{f}}_{t..t+3}\bigr)

at noise level \tau (during generation — the current iterate of the sampler).

Every latent map is linearized into spatial tokens (one token per spatial cell, its 16 channels linearly projected to the model width d): a coarse frame contributes $4 \times 4 = 16$ tokens, a fine frame $8 \times 8 = 64$ tokens. The full input of one prediction step is the single joint sequence

$$S \;=\; \underbrace{\bigl[\,\text{coarse ctx: } 48 \times 16\,\bigr]}_{768} \;\Vert\; \underbrace{\bigl[\,\text{fine ctx: } 6 \times 64\,\bigr]}_{384} \;\Vert\; \underbrace{\bigl[\,\text{targets: } 4 \times (16 + 64)\,\bigr]}_{320} \;=\; 1472 \ \text{tokens}, \tag{3.4}$$

of which $48 \cdot 16 + 6 \cdot 64 = 1152$ are context tokens — the context-token budget that the matched-budget flat control of §3.3.4 is required to equal exactly.

Each token carries four additive identity embeddings: a learned spatial position embedding (one table per level), a **level** embedding ($\mathrm{c}/\mathrm{f}$), a **role** embedding (context/target), and a **frame-index** embedding. Frame indices are absolute within the step: context frames occupy indices $0 \ldots 47$ (fine context frames share the indices $42 \ldots 47$ of their coarse counterparts), and the four target frames occupy indices $48 \ldots 51$ — the frame-embedding table has 52 entries. On top of the identity embeddings, every token receives a per-token conditioning vector c used by the adaptive normalization of every block:

$$c \;=\; e_{\mathrm{level}} + e_{\mathrm{frame}} + e_{\mathrm{role}} + \begin{cases} e_{\tau}(0), & \text{context tokens (clean, noise level } 0\text{)},\\[2pt] e_{\tau}(\tau) + e_{\mathrm{act}}(a_{t+j}), & \text{target tokens of frame } t{+}j, \end{cases} \tag{3.5}$$

where $e_{\tau}(\cdot)$ is a sinusoidal-MLP embedding of the continuous noise level and $e_{\mathrm{act}}$ is a learned embedding of the action alphabet. Two properties of (3.5) are essential. First, context tokens are conditioned at noise level 0 — the network is explicitly told which part of the sequence is clean evidence and which is the object of refinement; the prediction loss is computed on target tokens only. Second, the action conditioning is **per-target-frame**: frame t{+}j receives exactly its own action $a_{t+j}$, so the network never needs to disentangle a summed multi-step action code — each generated frame is bound to the control signal that produced it.

**Figure 3.2 — Token layout for one prediction step.** The 1,472-token sequence contains 768 coarse-context tokens, 384 recent fine-context tokens and four 80-token coarse-plus-fine targets. The bracket identifies the 1,152-token context budget matched by the central contiguous-fine control; the lower strips show the additive token embeddings and per-target action/noise conditioning. *Source:* `docs/figures/fig22_token_sequence_layout.png`; deterministic source

`scripts/make_dissertation_missing_figures.py`, reflecting `hvp_rev/rhvd.py::WindowDiT._ctx_tokens/forward`.

### 3.2.2 Transformer Backbone with Adaptive Normalization

The sequence (3.4) with conditioning (3.5) is processed by a stack of $N_{\mathrm{blk}} = 8$ pre-norm transformer blocks of width $d = 384$ with 6 attention heads and MLP expansion ratio 4 (22.18–22.19 M trainable parameters for the two-level arms; the 0.01 M difference is due to the package-specific input/output interface). Attention is full (every token attends to every token): the causality of the overall system is enforced by sequence construction and by the autoregressive recursion (§3.3.2), not by an attention mask inside the step.

Each block applies **adaLN-Zero conditioning** [18]: the per-token conditioning vector c is mapped (SiLU → linear) to six modulation signals ($\gamma_1, \beta_1, g_1, \gamma_2, \beta_2, g_2$), and the block computes

$$\operatorname{mod}(h; \beta, \gamma) = h\,(1 + \gamma) + \beta, \tag{3.6}$$

$$h \leftarrow h + g_1 \cdot \operatorname{Attn}\bigl(\operatorname{mod}(\operatorname{LN}(h); \beta_1, \gamma_1)\bigr), \qquad h \leftarrow h + g_2 \cdot \operatorname{MLP}\bigl(\operatorname{mod}(\operatorname{LN}(h); \beta_2, \gamma_2)\bigr), \tag{3.7}$$

with the modulation projection zero-initialized, so every block starts as the identity — the standard stable initialization for a denoiser trained as a residual refiner. Queries and keys are RMS-normalized per head. After the final block, only the 320 target tokens are read out: each level has its own final adaptive layer-norm (again zero-initialized) and its own zero-initialized linear head projecting $d \to 16$ channels, after which the tokens are reshaped back into the latent geometry — $(W, 16, 4, 4)$ for the coarse and $(W, 16, 8, 8)$ for the fine level. The backbone, tokenizer, and conditioning are shared verbatim with the alternative mechanisms of the Chapter 4 mechanism study; only the output interpretation and the training objective differ — this is what "matched capacity" means operationally in this work.

### 3.2.3 v-Parameterization

Why a *window* of $W = 4$ frames rather than one frame at a time? The reason is rollout-consistency of the training distribution. The autoregressive rollout of §3.3 appends *all* frames it generates to the context buffers; if training predicted only a single frame at offset $k = 4$ (the design of the preliminary single-step harness), the rollout context would consist of stride-4 samples while the training context consisted of consecutive frames — a train/rollout distribution mismatch built into the protocol itself. Predicting a window of W *consecutive* frames per step and advancing the recursion by exactly W keeps

the buffers consecutive at all times: the layout of every context the model sees at rollout is identically distributed with the contexts it saw in training.

Let $z_0 = \bigl(z^{\mathrm{c}}_{t..t+3},\, z^{\mathrm{f}}_{t..t+3}\bigr)$ denote the clean (standardized) target-window latents. The forward diffusion process corrupts them with Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$ at a noise index $\tau \sim \mathrm{U}\{1, \ldots, T\}$, $T = 1000$, *shared across the window's frames and both levels within a training sample* (one $\tau$ per sample — the coupling that ties the two levels and four frames to a common refinement trajectory):

$$y \;=\; \sqrt{\bar\alpha_\tau}\, z_0 \;+\; \sqrt{1 - \bar\alpha_\tau}\,\varepsilon, \tag{3.8}$$

with the cosine noise schedule [4]

$$\bar\alpha_\tau \;=\; \frac{f(\tau)}{f(0)}, \qquad f(\tau) \;=\; \cos^{2}\!\Bigl(\frac{\tau/T + s}{1 + s} \cdot \frac{\pi}{2}\Bigr), \qquad s = 0.008, \qquad \bar\alpha_\tau \in [10^{-4},\, 0.9999]. \tag{3.9}$$

The network is trained in the **v-parameterization** [73]: the regression target is the velocity

$$v \;=\; \sqrt{\bar\alpha_\tau}\,\varepsilon \;-\; \sqrt{1 - \bar\alpha_\tau}\, z_0, \tag{3.10}$$

which interpolates between noise prediction (at low noise) and sign-flipped data prediction (at high noise) and is well-conditioned across the whole schedule. Denoting by $v_\theta\bigl(y,\, C^{\mathrm{c}},\, C^{\mathrm{f}},\, \tau,\, a\bigr)$ the two-level output of the backbone (§3.2.2), the training objective is the unweighted sum of the two levels' mean-squared v-errors:

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{\,z_0,\,\tau,\,\varepsilon} \Bigl[ \bigl\lVert v^{\mathrm{c}}_\theta - v^{\mathrm{c}} \bigr\rVert^2_2 \;+\; \bigl\lVert v^{\mathrm{f}}_\theta - v^{\mathrm{f}} \bigr\rVert^2_2 \Bigr]. \tag{3.11}$$

Equal weighting is meaningful only because of the standardization (3.3), which puts both levels' latents at unit per-channel variance; without it the loss would be silently dominated by whichever level's raw scale happened to be larger.

Optimization protocol (constants of the validated system): AdamW, learning rate $3 \cdot 10^{-4}$, weight decay $10^{-4}$; linear warmup over the first 2{,}000 steps to a constant learning rate (restart-safe by design — training is resumable from atomic checkpoints); gradient-norm clipping at 1.0; batch size 16–32; 30{,}000 training steps per system arm; an exponential moving average (EMA) of the parameters with decay 0.999 maintained throughout, with *all* evaluation performed on the EMA weights. Training uses clean (ground-truth) contexts only — no self-generated context injection in the base protocol; the pre-registered design reserves context-noise augmentation as a single budgeted fallback. It was not

activated: the observed deep-horizon sharpness-retention failure is reported as a boundary result rather than repaired by an adaptive retraining branch.

Training-noise/teacher-forcing scope note for the defense: the long-horizon capability claimed in Chapter 4 is achieved *without* backpropagation through the rollout — the model is trained on single windows with clean contexts, and its long memory is an inference-time property of the information design (§3.1.2), not of a long-horizon training objective. Where a rollout-through- time objective would be required, the constant-memory machinery of Chapter 2 is the enabling instrument; the two contributions compose but do not presuppose each other.

### 3.2.4 DDIM

Generation inverts the diffusion with a deterministic DDIM sampler [17] over $N_{\mathrm{DDIM}} = 16$ steps on a rounded-linear sub-grid $\tau_1 = T > \tau_2 > \cdots > \tau_{16} \ge 1$ of the training schedule. Starting from pure noise $y^{(1)} \sim \mathcal{N}(0, I)$ over the whole two-level window, each step evaluates the network once and converts its v-output into the implied clean-data and noise estimates:

$$\hat{z}_0 \;=\; \sqrt{\bar\alpha_{\tau_i}}\, y^{(i)} \;-\; \sqrt{1 - \bar\alpha_{\tau_i}}\, v_\theta\bigl(y^{(i)}, \cdot\bigr), \qquad \hat{\varepsilon} \;=\; \sqrt{1 - \bar\alpha_{\tau_i}}\, y^{(i)} \;+\; \sqrt{\bar\alpha_{\tau_i}}\, v_\theta\bigl(y^{(i)}, \cdot\bigr), \tag{3.12}$$

$$y^{(i+1)} \;=\; \sqrt{\bar\alpha_{\tau_{i+1}}}\; \hat{z}_0 \;+\; \sqrt{1 - \bar\alpha_{\tau_{i+1}}}\; \hat{\varepsilon}, \qquad y^{(N_{\mathrm{DDIM}}+1)} \;:=\; \hat{z}_0 \ \ (\text{final step}), \tag{3.13}$$

i.e. the deterministic ($\eta = 0$) DDIM update; the sampler's only stochasticity is the initial noise draw, which the evaluation protocol controls with common random numbers (§3.4.4). The computational price of generation is thus fixed: 16 network evaluations per 4-frame window — 4 network function evaluations per generated frame.

### 3.2.5 Regression Control Head

For the mechanism claims of Chapter 4 (hypothesis H2: the generative mechanism earns its keep at rollout), the system admits a drop-in **regression** variant on the identical backbone: the noisy target inputs are replaced by zeros, the noise level is fixed at $\tau \equiv 0$, the output is interpreted directly as $\hat{z}_0$, and the objective is the smooth-L1 error on both levels. Its "sampling" is a single forward pass. Architecture, tokenization, conditioning, context design, parameter count, optimizer, and evaluation protocol are all shared with the diffusion core — the two systems differ in the training objective and nothing else. The chapter records this variant as part of the model family because the

honest form of the dissertation's mechanism claim is comparative, and the comparison is meaningful only under this strict parity.

## 3.3 Autoregressive Rollout with Ring-Buffer Memory

The developed technology operates as a fully **autonomous** autoregressive predictor: its inputs are past video frames (through the latent interface) and the recorded low-dimensional control track; no exogenous modality (in particular, no text) steers the generation. The control track is not a side channel of convenience: in the systems framing of Chapter 1 the actions *are* part of the observed state of the control system, and Chapter 4 (§4.2) documents that withholding them forces every mechanism to hedge over the control-marginal future and manufactures a spurious blur verdict. The salvaged autonomy principle of the old chapter therefore survives in a sharpened form: autonomous with respect to semantic side information, conditioned on the recorded control signal.

### 3.3.1 Rollout Recurrence

The state of the rollout between prediction steps is the pair of **ring buffers**

$$B^{\mathrm{c}}_n = \bigl(\hat z^{\mathrm{c}}_{t_n - H_{\mathrm{c}}}, \ldots, \hat z^{\mathrm{c}}_{t_n - 1}\bigr), \qquad B^{\mathrm{f}}_n = \bigl(\hat z^{\mathrm{f}}_{t_n - w_{\mathrm{f}}}, \ldots, \hat z^{\mathrm{f}}_{t_n - 1}\bigr), \tag{3.14}$$

holding the last $H_{\mathrm{c}} = 48$ coarse and the last $w_{\mathrm{f}} = 6$ fine latents, where $\hat z$ denotes ground-truth latents during warm-up and self-generated latents thereafter. The recursion is:

**Algorithm 3.1 — Autoregressive rollout with ring-buffer memory** *(implemented as `hvp_rev/rhvd.py::rollout_eval`)*

**Input:** ground-truth frames $x_{0..H_{\mathrm{w}}-1}$ for warm-up ($H_{\mathrm{w}} = 48$), recorded actions $a_{1..T}$, frozen AE ($\mathcal{E}$, $\mathcal{D}$), trained EMA weights $\theta$, horizon $T$.

**Output:** predicted frames $\hat x_{H_{\mathrm{w}}..T-1}$.

1. Encode and standardize the warm-up frames; initialize $B^{\mathrm{c}}$, $B^{\mathrm{f}}$ with their last $H_{\mathrm{c}}$ and $w_{\mathrm{f}}$ latents; $t_0 \leftarrow H_{\mathrm{w}}$.

2. **for** $n = 0, 1, \ldots, \lfloor (T - H_{\mathrm{w}})/W \rfloor - 1$ **do**

3.    read the actions $a_{t_n}, \ldots, a_{t_n+W-1}$ of the current window;

4.    sample the window by DDIM (§3.2.4):

$(\hat z^{\mathrm{c}}_{t_n..t_n+3},\, \hat z^{\mathrm{f}}_{t_n..t_n+3}) \leftarrow \mathrm{DDIM}_{16}\bigl(v_\theta;\, B^{\mathrm{c}}_n,\, B^{\mathrm{f}}_n,\, a_{t_n..t_n+3}\bigr)$;

5.    append all W predicted latent pairs to the buffers; drop the oldest W entries of each (ring update); $t_{n+1} \leftarrow t_n + W$;

6.    decode predictions for evaluation:

$\hat x_{t_n+j} = \mathcal{D}_{\mathrm{f}}\bigl(\hat z^{\mathrm{f}}_{t_n+j}\bigr)$, $j = 0..3$ (through un-standardization and the frozen fine decoder).

7. **end for**

On the 200-frame evaluation clips of §3.4: warm-up 48 frames, then $(200 - 48)/4 = 38$ rollout steps, i.e. 152 self-generated frames per clip — more than three full loop periods of the testbed world.

Because the training windows are consecutive and the recursion advances by exactly W, the buffer contents at every rollout step are distributionally identical to training contexts (§3.2.3) — the property that makes clean-context training sufficient. The warm-up length is **universal across all compared systems** (48 frames even for arms whose context is shorter, e.g. the flat control warms its 18-frame buffer from ground-truth frames 30–47): every arm starts generating at the same frame of the same clip, so the per-(clip, horizon) records of §3.4 pair exactly across arms.

**Figure 3.3 — Ring buffers for autoregressive rollout.** After the common 48-frame ground-truth warm-up, each step combines the fixed 48-slot coarse and 6-slot fine buffers with four actions, samples four latent pairs with DDIM-16, and appends/evicts four positions. The stored state is 18,432 latent numbers and does not grow with rollout length. *Source:* `docs/figures/fig23_rollout_ring_buffer.png`; deterministic source `scripts/make_dissertation_missing_figures.py`, reflecting `hvp_rev/rhvd.py::rollout_eval`.

### 3.3.2 Temporal Routing and Block-Causal Semantics

The old chapter's prediction skeleton promised two routing mechanisms: a block-causal attention mask (full attention within a frame, causal across frames) over a jointly optimized past+future token tensor, and 3D-RoPE relative spatio-temporal positional coding. The developed system realizes the *functions* of both, by different and simpler means, and the dissertation states the mapping explicitly rather than letting the old vocabulary stand:

- **Block-causality holds at window granularity and is enforced by construction, not by a

mask.* *Within one prediction step, attention over the sequence (3.4) is full — and legitimately so: every token in the sequence belongs either to the observed past (context) or to the window currently being generated, whose frames are refined* jointly* as one block under a shared noise level. Across steps, the recursion of Algorithm 3.1 makes information flow strictly causal: frames of window $n\{+\}1$ are generated conditioned on (at most) windows $\le n$ through the buffers, and no token of any future window exists at step $n$. Viewed over an entire rollout, the effective attention pattern is exactly block-causal with block size $W = 4$ — realized structurally, which eliminates both the mask bookkeeping and any possibility of future-information leakage (there is no future present to leak).

- **Temporal position is encoded by learned absolute frame-index embeddings** (52-entry table,

§3.2.1) rather than rotary relative coding. Because every prediction step re-indexes its context to the fixed range $0 \ldots 47$ and its targets to $48 \ldots 51$, the absolute indices *are* relative to the window, so the scheme is translation-invariant across the rollout by construction. RoPE-ND and these learned indices are therefore presented as level-appropriate instruments: RoPE-ND for the variable-patch, cross-resolution [A1] system in Chapter 2, and fixed learned indices for the re-indexed window used by the validated hierarchical predictor. No unmeasured RoPE-ND retrofit is attributed to the final model.

### 3.3.3 Memory and Computational Complexity

**Rollout memory is constant in the horizon.** The complete inter-step state of Algorithm 3.1 is the buffer pair (3.14):

$$M_{\mathrm{roll}} \;=\; H_{\mathrm{c}} \cdot \bigl|z^{\mathrm{c}}\bigr| \;+\; w_{\mathrm{f}} \cdot \bigl|z^{\mathrm{f}}\bigr| \;=\; 48 \cdot 256 \;+\; 6 \cdot 1024 \;=\; 18{,}432 \ \text{numbers} \;=\; O(1) \ \text{in the rollout length } T, \tag{3.15}$$

plus the (constant) transient activations of one network evaluation. Generation compute is linear in the horizon with a fixed per-frame constant — $N_{\mathrm{DDIM}} / W = 4$ network evaluations per generated frame — and, crucially, the per-step attention cost never grows with how long the system has been rolling, because the sequence length (3.4) is fixed. This replaces the old chapter's inconsistent claims (its §3.4 derived a per-image memory reduction $O(K \cdot T \cdot L \cdot D) \to O(T \cdot L$

$\cdot D$) while its §3.5 asserted $O(K \cdot L \cdot D) \to O(D)$ for the same algorithm) with a single derivation that is true of the implemented system: there is no inner optimization chain $K$ to reverse at all — sampling runs gradient-free — and the memory object that matters for long-horizon operation, the rollout state (3.15), is constant by design.

**Training memory is single-window and horizon-independent.** The training graph spans one network evaluation over 1,472 tokens and $N_{\mathrm{blk}} = 8$ blocks; no gradient crosses window boundaries (§3.1.3). The stored residual-state term is $O(N_{\mathrm{blk}} S d)$, while a naïve attention implementation can additionally materialize $O(N_{\mathrm{blk}}S^2)$ attention-score storage. Memory-efficient or fused attention changes that kernel-dependent term but not the model's $O(N_{\mathrm{blk}}S^2d)$ attention compute. Accordingly, peak VRAM is treated as an empirical implementation property. The justified invariance is to the number of rollout windows; memory still depends on token count $S$ and hence on the selected package.

**Token-budget economics of the hierarchy.** The systemic argument for the two-level design is a counting argument. Let $H$ be the number of past frames a model can consult ("effective memory frames"). A flat single-level model pays $64$ tokens per remembered frame; the hierarchical model pays $16$ tokens per remembered frame beyond its short fine window:

$$S^{\mathrm{flat}}_{\mathrm{ctx}}(H) \;=\; 64\,H, \qquad S^{\mathrm{hier}}_{\mathrm{ctx}}(H) \;=\; 16\,H \;+\; 64\,w_{\mathrm{f}} \quad (w_{\mathrm{f}} \ \text{fixed}), \tag{3.16}$$

an asymptotic 4× token advantage per memory frame, hence (attention being quadratic in sequence length within each of the $N_{\mathrm{blk}}$ layers, $O(S^2 d)$) an asymptotic ≈16× advantage in attention cost for the same memory horizon. At the system's operating point the concrete numbers are: hier-long consults 48 frames for 1152 context tokens; a flat model consulting the same 48 frames pays $48 \cdot 64 = 3072$ context tokens — 2.67× the tokens and, at full sequence lengths (1472 vs 3392 tokens), a ≈5.3× ratio in per-layer attention cost. Conversely, a flat model *constrained to hier-long's token budget* can afford only $1152 / 64 = 18$ frames of memory — which is precisely the matched-budget control of §3.3.4. The claim is about **hierarchical context packaging whose benefit is mediated by reach**, not an intrinsic superiority of coarse features and not context length treated as a free variable: at equal budget, the flat alternative's horizon (18 frames) falls short of the testbed's loop period (≈48 frames, §3.4.1), while the hierarchical allocation covers it.

**Disposition of the reversible inner loop (salvage of old §3.3).** The earlier design stage of this work proposed integrating the reversible-transformation apparatus of Chapter 2 *inside* the per-frame generation loop ("Reversible MCMC": reconstructing the activations of a $K$-step gradient-descent sampling chain instead of storing them). The final system contains no such component, for two reasons established in this dissertation. First, the three-axis reversibility taxonomy of Chapter 2 (§2.4) proves

that the inner-optimization-loop axis cannot outperform gradient checkpointing there: the chain's states must be retained for exact reversal, which *is* checkpointing under another name — the axis is closed by analysis, not abandoned by neglect. Second, the mechanism study of Chapter 4 removed the premise: the energy-descent core whose $K$-step chains motivated the machinery was itself the characterized-negative mechanism, and the selected diffusion core requires no gradient flow through its sampling recursion (3.12)–(3.13) at all. Reversibility remains a contribution of this dissertation on the axis where it pays — the temporal-rollout axis of Chapter 2 — and this paragraph is the honest record of where it does not.

### 3.3.4 System Configurations and Matched-Budget Controls

The information design of §3.1.2 turns context allocation into an experimental variable. The model family is parameterized by $(H_{\mathrm{c}}, w_{\mathrm{f}}, \text{core})$, and the dissertation's claims are comparative statements over the following pre-registered configurations (all sharing the backbone of §3.2.2, the training protocol of §3.2.3, and the evaluation protocol of §3.4):

**Table 3.1 — System configurations (arms) and context-token accounting.**

| Configuration | Coarse ctx $H_{\mathrm{c}}$ | Fine ctx $w_{\mathrm{f}}$ | Context tokens | Core | Systemic role |
|---|---|---|---|---|---|
| **hier-long** | 48 | 6 | $48 \cdot 16 + 6 \cdot 64 = 1152$ | diffusion | the developed system: coarse memory ≥ loop period |
| hier-short | 6 | 6 | $6 \cdot 16 + 6 \cdot 64 = 480$ | diffusion | memory ablation (hierarchy without the horizon) |
| flat-budget | 0 | 18 | $18 \cdot 64 = 1152$ | diffusion | **token-budget-matched flat control** (equal cost, 18-frame horizon) |
| flat-long | 0 | 48 | $48 \cdot 64 = 3072$ | diffusion | window-matched flat exhibit (equal horizon, 2.67× budget) |
| reg-long | 48 | 6 | 1152 | regression | mechanism control (H2), §3.2.5 |
| persistence / AE round-trip | — | — | — | — | floor / ceiling references (§3.4.4) |

The flat configurations set $H_{\mathrm{c}} = 0$: coarse *context* tokens are absent, while the coarse *target* tokens are retained, so the latent interface, the output heads, and the loss (3.11) remain identical across all arms — the flat arm differs from the hierarchical one in exactly one respect, the absence of coarse memory. The decisive comparison is hier-long versus flat-budget: both pay 1152 context tokens per step, but the hierarchical allocation stretches those tokens over a 48-frame horizon while the flat

allocation reaches only 18 frames. The pre-registered prediction — validated in Chapter 4 — is that the two systems separate exactly where the testbed's re-entry events lie beyond the 18-frame flat horizon, and are otherwise comparable; if they had *not* separated, the honest conclusion recorded in the design memo was "context length, not hierarchy," and the dissertation would have reported it as such.

### 3.3.5 Generalization to a Family of Hierarchical Configurations

The token-stream design of §3.2.1 generalizes beyond the two-level, stride-1 family of Table 3.1 without any change to the backbone. A configuration is a set of context streams, each parameterized by the triple **(spatial resolution, temporal stride, number of frames)**: the coarse stream may tick at stride $k_t > 1$ (one slot per $k_t$ frames, block-end anchors, $W/k_t$ target slots per prediction window so the slot buffer keeps phase for any $k_t$ dividing $W$); a full-resolution stream may itself be strided (temporal abstraction without a resolution change); and a third, **top** stream at $2{\times}2$ (4 tokens per frame) extends the pyramid to depth 3 at factor 2 per level ($2{\times}2$ / $4{\times}4$ / $8{\times}8$). The top latent is a third tap of the frozen autoencoder trained on the **frozen trunk** (§3.1.1), so the two lower levels remain bit-identical and every measurement of Chapter 4 stays valid across the extension. Two systemic properties distinguish this family from recurrent hierarchies of the CW-VAE class: (i) **depth is backbone-capacity neutral** — all streams share one attention backbone; a third level adds only per-level input/output parameters (+0.30 M on the 22.19 M predictor, +1.4%) and the frozen autoencoder tap (1.25 M), measured and disclosed; and (ii) every configuration's context cost is an exact token count, so families are comparable at **matched budget** — the property that licenses the shape comparisons of §4.4 (simplified configurations: `hier-s2-min` = 24 slots at stride 2 + 2 fine = 512 tokens; `strided-flat` = 11 full-resolution slots at stride 4 + 6 fine = 1,088; `hier3` = 48 top + 24 coarse + 9 fine = 1,152).

## 3.4 Memory-Horizon Testbed and Re-Entry-Stratified Evaluation

A model whose defining property is a *memory horizon* cannot be validated on benchmarks that do not exercise one. On Moving-MNIST-class data nothing ever leaves the frame and returns: every future frame is either fully determined by a short context or permanently unknowable, so there exists no *hard-but-computable-with-memory* regime in which a long-memory mechanism could demonstrate its value — this project's own null results on such data (Chapter 1, §1.3; Chapter 4) trace to exactly this degeneracy. Existing memory-oriented benchmarks (Memory Maze and TECO-Minecraft [20, 15], and the recent long-term spatial-memory evaluation protocols for world models [21, 23, 22]) pose genuine long-horizon problems but derive their re-entry events from 3-D egomotion in rendered environments, where determining *which* pixels of a target frame were knowable from a given context requires geometry processing at best — and their model scales exceed a single-GPU budget. The instrument constructed here is the minimal fully analytic member of this family: hardness labels are computed

**exactly** from the recorded control track, and the complete benchmark trains and evaluates on one 16 GB consumer GPU.

### 3.4.1 Cyclic World with Inertial Panning

Each clip is generated (module `scripts/gen_looping.py`) from a static **world**: a $192 \times 64$ texture on a horizontal torus, composed of a smooth background (a sum of four sinusoids with integer wavenumbers, hence seamlessly wrapping; intensities in $[0.15,\ 0.5]$) and three MNIST digits max-composited at fixed world positions. A $64 \times 64$ **viewport** pans horizontally; the observed clip is the sequence of viewport crops, so two thirds of the world (128 of 192 columns, 67%) is off-screen at every instant.

The pan is a **momentum action walk**: at each frame the camera holds with probability $p_{\mathrm{hold}}$, otherwise continues its heading with probability $p_{\mathrm{cont}}$ and reverses otherwise (measured on the shipped data: $p_{\mathrm{hold}} \approx 0.10$, $p_{\mathrm{cont}} \approx 0.90$). The executed action $a_t \in \{-1,\ 0,\ +1\}$ is recorded, and the viewport advances by $4\,a_t$ pixels:

$$x^{\mathrm{vp}}_t \;=\; \bigl(x^{\mathrm{vp}}_{t-1} + 4\,a_t\bigr) \bmod 192 \,. \tag{3.17}$$

The momentum is the essential design element. A symmetric random walk diffuses too slowly to circumnavigate the torus within a clip, so off-screen content would be either still inside a short context window (determined) or never seen again (unknowable) — reproducing the degenerate Moving-MNIST regime. Directed motion cycles the torus with loop period $\approx 192/4 = 48$ frames: a digit that exits the viewport re-enters roughly 48 frames later at a world position fixed since frame 0. A model with memory spanning the loop period can predict the re-entry exactly; a short-context model cannot — and 48 frames is precisely the coarse horizon $H_{\mathrm{c}}$ of the developed system (§3.3.4), while the token-matched flat control's 18-frame horizon falls short of it. The testbed thereby *instantiates* the memory-horizon regime as a controlled experimental variable.

Datasets (fixed, generated once, shared by every experiment): `loopmom_train` — 3{,}000 clips × 100 frames (split by clip into 2{,}700 train / 300 validation); `loopmom_test` — 500 clips × 200 frames, providing a held-out pool of 250 evaluation clips whose 200-frame duration supports the multi-loop rollouts of Algorithm 3.1 (rollout evaluation uses 100 of them per arm). Test latents are standardized with the training statistics (3.3).

**Figure 3.4 — LoopMom geometry and analytic reach band.** The 64-pixel viewport traverses a 192-pixel toroidal static world. Saved positions make geometric observability exact; the highlighted 14-to-48-frame band contains evidence available to the long-reach package but outside the compared short flat window. The separate pose-mosaic inset is marked as a privileged reference because it receives exact

pose and raw stored pixels. *Source:* `docs/figures/fig15_loopmom_observability_schematic.png`; deterministic source `scripts/make_rft_core_schematics.py`.

### 3.4.2 Exact Re-Entry Difficulty Labels

Because the viewport trajectory (3.17) is a deterministic function of the recorded actions, the set of world columns covered by any context window is computable in closed form — hardness is an analytic label, not a heuristic. For a target frame $t$ and a reference window of the $m$ preceding frames, let $\mathrm{cov}_m(t) \subset \{0, \ldots, 191\}$ be the union of world columns visible in frames $t-m, \ldots, t-1$, and let $F(t)$ be the set of world columns occupied by the target frame's sprite pixels (detected by the foreground threshold of §3.4.3). The **novel-foreground fraction** is

$$\nu_m(t) \;=\; \frac{\bigl|\, F(t) \setminus \mathrm{cov}_m(t) \,\bigr|}{\bigl|\, F(t) \,\bigr|} \;\in\; [0, 1], \tag{3.18}$$

the fraction of the target sprite that is geometrically unobserved in that context. It is an exact observability label for the generator, not information-theoretic necessity: a shared digit prior can still support a statistical guess. The protocol fixes the reference window at $m = 6$ frames for *all* systems regardless of their actual context length ("novel-fg-at-6"), so every arm is stratified on the same hardness axis, and defines three strata: **easy** ($\nu_6 = 0$ — the sprite was fully visible in the short window), **mid** ($0 < \nu_6 \le 0.1$), and **hard re-entry** ($\nu_6 > 0.1$ — a substantial part of the sprite is not geometrically visible in the short window). The single-step mechanism study of Chapter 4 uses the analogous *novel-strip* label (the same construction restricted to the newly revealed leading strip of the pan — the only genuinely generative region of a single predicted frame). This stratification is what turns the trajectory-consistency criterion of §1.4.5 into a measurement: the dissertation's headline quantities are error deltas *within the hard re-entry stratum* as a function of rollout horizon.

### 3.4.3 Decoded Frame, Foreground and Object Metrics

Aggregate pixel metrics are excluded by protocol. On the archived CW-VAE Moving-MNIST population (100 clips, 36 context frames, 964 open-loop targets), a deterministic all-black prediction scores 0.7681 SSIM and 13.941 dB, while the trained CW-VAE's best-of-5-by-SSIM curve scores 0.5708 and 12.918 dB. The source dataset, archived curve and evaluator are hash-pinned in `artifacts/cwvae_black_baseline/results.json`; record-level CW-VAE values were not retained, so this is a same-population metric diagnostic without paired inference, not a budget-matched model comparison. A benchmark on which the empty image wins its full-frame metrics is not evidence about dynamics. All metrics below are therefore computed on decoded pixels through the frozen fine decoder and reported with foreground, background, union, full-frame and component-sensitive views.

**Table 3.4-A — Same-population Moving-MNIST metric sanity diagnostic.**

| Open-loop horizon | CW-VAE SSIM | All-black SSIM | CW-VAE PSNR, dB | All-black PSNR, dB |
|---|---|---|---|---|
| +1–48 | 0.6338 | 0.7679 | 13.944 | 13.943 |
| +49–192 | 0.5665 | 0.7679 | 12.625 | 13.942 |
| +193–964 | 0.5677 | 0.7682 | 12.909 | 13.941 |

- **Foreground MSE (fg-MSE).** Predictions and ground truth are decoded to $[0,1]$; the error is

averaged over pixels where the ground-truth frame exceeds the threshold $0.5$. The threshold is testbed-calibrated and verified: the background texture occupies ≈0.18–0.41 while sprite strokes reach ≈0.99, so 0.5 isolates the digits (a conventional 0.1 threshold would classify the entire panning background as foreground).

- **Foreground sharpness (fg-sharp).** Per-image Laplacian energy on that image's foreground

mask: $\sum (\nabla^2 \hat{x})^2$ weighted by the mask, normalized by mask area, averaged over images. Deliberately *not* a batch-variance statistic: batch variance rewards inter-sample diversity, whereas per-image Laplacian energy rewards per-frame detail — which is what distinguishes a committed prediction from a mean-hedged blur.

- **Rollout records.** During Algorithm 3.1, every generated frame contributes a record

(clip, horizon $h = t - H_{\mathrm{w}} + 1 \in \{1, \ldots, 152\}$, hardness $\nu_6$, fg-MSE, fg-sharp, and the persistence reference for the same frame). Analyses bin horizons as 1–8, 9–24, 25–60, 61+ — the 9–24 bin brackets the flat control's 18-frame horizon boundary, and 25–60 brackets the loop period.

- **Decoded-sample review** is a first-class protocol element: every evaluation saves panels of

decoded frames (last context | prediction | ground truth) from the same samples that produced the scores. Twice during this research, scalar metrics concealed a qualitative failure (a slow fade to the mean; a blank-output collapse) that one glance at the panels exposed; the protocol therefore mandates the panels alongside the numbers.

### 3.4.4 Statistical Protocol, References and Boundaries

All system-versus-system comparisons are **paired**: evaluation draws use fixed seeds so every arm sees identical (clip, target) pairs; stochastic samplers are re-seeded per batch with common random numbers; and rollout records pair exactly across arms by (clip, horizon) thanks to the universal 48-frame warm-up (§3.3.1). Headline deltas are reported with 95% confidence intervals from a paired **clip-cluster** bootstrap (10{,}000 resamples), so all horizons belonging to one clip are resampled together. The untouched frontier and protected robustness families use a studentized max-statistic clip-cluster procedure for simultaneous coverage across their registered contrasts. Intervals are computed within

each stated hardness×horizon population; single-step evaluation pools are sized at ≈768 samples so the hard stratum retains power, and rollout evaluation covers 100 clips × 152 generated frames = 15{,}200 records per arm. The headline arms (hier-long, flat-budget) are replicated over 2 seeds. Where a best-of-$N$ statistic appears it is an oracle *coverage* diagnostic (selection by pixel-space fg-MSE against ground truth) and is always reported alongside the mean over $N$; headline tables use best-of-1.

Every table is bracketed by references measured under the identical pipeline: the **persistence floor** (copy the last context frame's fine latent, decode) scores fg-MSE 0.2367 / fg-sharp 0.0515 at the 4-frame offset — any model below that error has learned at least the action-conditioned warp; and the **sharpness ceiling set by the frozen decoder**: raw ground-truth frames measure fg-sharp 0.4785, but their autoencoder round-trip measures 0.2412, so all model sharpness is read against ≈0.24 (roughly half the raw sharpness is spent in the autoencoder before any prediction model acts), and a decoded output below ≈0.08 contains no digit structure at all (the empirically calibrated collapse threshold used throughout the mechanism study).

**Table 3.2 — Testbed and evaluation protocol summary.**

| Component | Value | Notes |
|---|---|---|
| World / viewport | 192×64 torus / 64×64 crop | 128 columns (67%) off-screen at any time |
| Content | background texture + 3 MNIST digits | fixed world positions; max-composited |
| Pan process | momentum walk, 4 px/frame | $p_{\mathrm{hold}} \approx 0.10$, $p_{\mathrm{cont}} \approx 0.90$ (measured); actions $\{-1,0,+1\}$ recorded |
| Loop period | ≈ 48 frames | $192/4$; sets the memory horizon; $= H_{\mathrm{c}}$ of hier-long |
| Train / test | 3,000×100 / 500×200 frames | clip-disjoint; test standardized with train statistics |
| Latent interface | frozen 2-level conv-AE, 1.85 M | coarse 16 ch @ 4×4, fine 16 ch @ 8×8, per-level decodable; fine val recon MSE 0.0020 |
| Standardization | per-channel z-score, per level | training-set statistics only; eq. (3.3) |
| Error metric | fg-MSE, threshold 0.5 | decoded pixels; background ≈0.18–0.41, sprite ≈0.99 |
| Sharpness metric | per-image masked Laplacian energy | GT 0.4785; AE round-trip ceiling 0.2412; collapse < ≈0.08 |
| Hardness axis | $\nu_6$ — novel-fg vs 6-frame window, exact | easy $=0$; mid $(0, 0.1]$; hard $> 0.1$; eq. (3.18) |

| Component | Value | Notes |
|---|---|---|
| Rollout | 48-frame GT warm-up; 38 steps × 4 frames | 152 generated frames/clip; 100 clips; horizons binned 1–8 / 9–24 / 25–60 / 61+ |
| Statistics | paired clip-cluster 95% CI; CRN for same target shapes | 10,000 resamples; simultaneous max-statistic families where registered; ≈768-sample single-step pools; 15,200 rollout records/arm; training seeds reported separately |
| Floors / ceilings | persistence fg-MSE 0.2367 / fg-sharp 0.0515 | k = 4 offset; identical pipeline for every arm |

Scope and limitations of the instrument are stated plainly, as required of a measurement deliverable: motion is one-dimensional and piecewise-deterministic given the action, so the generative burden is concentrated in the re-entry events by design; world content is static in world coordinates, so the testbed does not probe memory of *dynamic* off-screen state; all conclusions are at 64×64 grayscale and single-digit complexity; and the frozen decoder's sharpness ceiling caps every downstream comparison. The analytic hardness labels that justify the design would not transfer unchanged to 3-D egomotion benchmarks, which are positioned as the natural next rung (Chapter 4 and General Conclusions).

## 3.5 Chapter Conclusions

1. In accordance with **Task 5** of the dissertation research, a mathematical model of

multi-stream video-data generation in a compact continuous hierarchical latent state space has been developed. It supports streams with different temporal rates and spatial resolutions over one shared attention backbone. Its measured spatial implementation decomposes the state (3.1) into a coarse (macro) level — 16 channels at 4×4, independently decodable, standardized per channel — and a fine (micro) level — 16 channels at 8×8 — over a frozen two-level autoencoder. In that implementation the coarse stream economically packages long-range history: it is the stream whose context extends over the full 48-frame memory horizon, at a cost of 16 tokens per remembered frame against the fine level's 64 (eq. 3.16, a 4× per-frame and asymptotically ≈16× attention-cost advantage).

2. The inter-level transfer of context signals has been formalized as joint self-attention over

a single token sequence of both levels (eq. 3.4–3.5), generalizing the cross-level cross-attention of the earlier design and enforcing hierarchical consistency continuously along the refinement trajectory (both levels denoised jointly under a shared noise level). The unidirectional-control principle with gradient detachment has been retained at the two boundaries where it is exact and load-bearing — between the frozen representation subsystem and the prediction core, and between past and future along the autoregressive recursion — while the internally contradictory strict level-ordering claim of the earlier formulation has been withdrawn in favor of the validated joint-generation formulation (§3.1.3).

3. The per-frame generation core has been formalized as window diffusion in the standardized latent space: v-parameterized (3.10) denoising of 4-frame two-level windows under a cosine schedule (3.9) with per-target-frame action conditioning, trained on single windows (3.11) and sampled by 16-step deterministic DDIM (3.12)–(3.13) at a fixed cost of 4 network evaluations per generated frame. The window length equals the recursion advance, which makes the training and rollout context distributions identical by construction — the property that permits clean-context training without rollout-time distribution shift of the context layout. A capacity-matched regression variant on the identical backbone (§3.2.5) completes the model family for the mechanism claims whose evidence is presented in Chapter 4.

4. The autoregressive rollout algorithm with ring-buffer memory (Algorithm 3.1) has been developed and its resource profile derived: the inter-step state is constant at $48 \cdot 256 + 6 \cdot 1024 = 18{,}432$ numbers (3.15) independently of the rollout horizon, per-step cost is fixed by the constant 1472-token sequence, and block-causality holds at window granularity by construction rather than by masking (§3.3.2). The re-derivation (§3.3.3) removes the complexity-claim inconsistency of the earlier chapter version, and the earlier reversible-inner-loop proposal is disposed of honestly: by the three-axis taxonomy of Chapter 2 that axis cannot outperform checkpointing, and the selected diffusion core requires no gradient through its sampler — reversibility remains a contribution on the temporal-rollout axis where it pays.

5. In accordance with **Task 6**, the measurement instrument of the intelligent information technology has been formalized: the momentum looping memory-horizon testbed (3.17) with exactly computable re-entry hardness labels (3.18), foreground-masked per-image metrics with verified thresholds and decoder-set ceilings, mandatory decoded-sample review, and a paired bootstrap protocol with common random numbers and a universal warm-up that pairs every record across systems. The instrument operationalizes the trajectory-consistency criterion of Chapter 1 into an analytic, statistically disciplined measurement procedure, and its scope limits are stated as part of the deliverable.

6. The complete pre-registered experimental design has been fixed within the model chapter (Table 3.1): the developed system (hier-long), the memory ablation (hier-short), the token-budget-matched flat control (flat-budget, 1152 tokens = 18-frame horizon), the window-matched flat exhibit (flat-long, 2.67× budget), and the regression control (reg-long) — with the decisive comparison, hier-long versus flat-budget at equal token cost, constructed so that the hierarchy's memory horizon (48 frames ≥ loop period) and the flat horizon (18 frames < loop period) straddle the testbed's re-entry structure. The experimental verification of the pre-registered hypotheses on this design — including the honest fallback conclusions specified in advance — constitutes §§4.2–4.4 of Chapter 4.

The results of this chapter form the methods and model section of [A4]. Its final numbered bibliographic position and any related conference theses are publication-administration fields, not scientific dependencies of the chapter.

# Chapter 4 — Experimental Evaluation and Practical Application

This chapter performs the comprehensive experimental verification required by **Task 7** of the dissertation research: the developed methods and the intelligent information technology are evaluated on the testbed family formalized in §3.4, under the pre-registered protocol of Table 3.1/Table 3.2, with every quoted comparison paired, stratified by the exact re-entry hardness label, and interval-estimated by the clip-level cluster bootstrap. The order of the sections is deliberate. The generation-head study (§4.2) is settled first, because every subsequent table depends on knowing how much of a system arm's score is owed to its head rather than to its memory design. The matched-token-budget memory study (§4.3) is the headline result of the dissertation. The spatial-abstraction factorial (§4.4) isolates the factor that every hierarchical arm otherwise confounds. The temporal-capability boundary study at the second operating point (§4.5) closes the capability line of the experimental program with an honestly bounded negative and the methodology lessons it produced. The generalization study (§4.6) then carries the configuration selected by §§4.2–4.4 onto the standard and synthetic benchmark suites — its measured components include the standard Moving-MNIST negative-control bridge and the corrected persistent-object long Moving-MNIST external boundary. §4.7 documents the implementation of the results, and §4.8 concludes the chapter.

Two standing conventions of the chapter, fixed by the validity audit of the experimental campaign and stated once here: (i) the long-horizon evaluation protocol is a **48-frame ground-truth warm-up followed by 152 autoregressive frames** (38 scheduler steps × 4 frames) — all text says "152 AR frames", never "200-frame rollout"; (ii) all confidence intervals quoted in this chapter are 95% paired **clip-level cluster bootstrap** intervals unless a disclosure states otherwise (§4.1.5).

## 4.1 Data, Hardware and Measurement Protocol

### 4.1.1 Testbed Family: Two Operating Points

All experiments of §§4.2–4.5 run on the memory-horizon testbed family of §3.4 at two operating points.

**Operating point A (primary)** is the momentum looping world *loopmom* formalized in §3.4.1: a $192 \times 64$ toroidal world observed through a panning $64 \times 64$ viewport at 4 px/frame ($x^{\mathrm{vp}}_t = (x^{\mathrm{vp}}_{t-1} + 4a_t) \bmod 192$; measured $p_{\mathrm{hold}} \approx 0.10$, $p_{\mathrm{cont}} \approx 0.90$), so ~67% of the world is off-screen at any instant and off-screen content re-enters with a loop period of ≈48 frames. Datasets: 3,000 training clips × 100 frames and 500 test clips × 200 frames, clip-disjoint (§3.4.1); the frozen two-level autoencoder interface, standardization, hardness labels, metrics, and floors are exactly those of §§3.1.1 and 3.4.2–3.4.4.

**Operating point B (capability regime)** is *hgrid3*: the same generator scaled to a 768-px world (12 digits, speed 4 px/frame, near-ballistic circulation), whose **effective loop** is $L_{\mathrm{eff}} \approx 213$ frames (eq. 4.1, §4.5.2) — deliberately beyond any affordable flat context window, which is what makes it the *capability* rather than *efficiency* operating point. Operating point B carries its own frozen autoencoder (fine reconstruction MSE 0.0024, coarse 0.0059) and its own re-measured references (Table 4.1); its regime properties are verified before any model run: median re-entry gap 192 frames, 204 beyond-48 re-entry frames per clip, warm-up world-coverage 88.5%, and viewport-crop uniqueness 0.004–0.010 (the positional-landmark property whose destruction is diagnosed in §4.5.1).

**Table 4.1 — Reference floors and ceilings, measured under the identical pipeline per operating point (never mixed across operating points).**

| Reference | Op. point A (loopmom) | Op. point B (hgrid3) |
|---|---|---|
| Persistence floor, fg-MSE (single-step, k = 4) | 0.2367 | 0.2381 |
| Persistence floor, fg-MSE (rollout protocol) | 0.2552 | — |
| Persistence fg-sharp (k = 4) | 0.0515 | — |
| AE round-trip fg-MSE ceiling | — | 0.0108 |
| AE round-trip sharpness ceiling | 0.2412 | 0.2084 |
| Raw ground-truth sharpness | 0.4785 | 0.4218 |
| Collapse threshold (no digit structure below) | ≈0.08 | ≈0.08 |
| Frozen AE reconstruction (fine / coarse MSE) | 0.0020 / — | 0.0024 / 0.0059 |

**Figure 4.A — LoopMom geometry and the registered reach band.** *Content:* toroidal world, moving viewport, persistent digits and the pair-derived `14 < dt <= 48` population. The privileged pose-indexed raw-memory reference is drawn separately from token-budget-matched neural packages. *Source:* `docs/figures/fig15_loopmom_observability_schematic.png` and vector PDF; generated deterministically from the frozen E2 geometry by `scripts/make_rft_core_schematics.py`.

### 4.1.2 [A2]

The experimental validation of the reversible-transformer method and time-agnostic backpropagation (§§2.3–2.4) was published in [A2] on procedurally synthesized **2-D rotating shapes** at $64 \times 64$ resolution. The suite controls the temporal law—constant, changing, oscillating, interrupted, and combined rotation—and therefore isolates the effect of prediction depth and temporal discontinuity. The published regime uses a 12-frame context, grows the prediction depth from one to five frames during 30,000 training steps, evaluates 2,000 independently generated sequences, and trains all

compared models with Adam at learning rate $10^{-5}$ on one NVIDIA RTX 4060 Ti 16 GB. The memory-scaling measurements use batch 32. These values replace the unsupported old-draft claims of AdamW with cosine decay, 100–300 epochs, and A100 80 GB hardware.

The **3-D polycube** and entropy-tokenization material belongs to [A1], not [A2]. Moving MNIST, MMNIST-CIFAR, TaxiBJ, and KTH are likewise not reported as [A2] experiments: Moving MNIST and TaxiBJ are separate dissertation bridges (§§4.6.2 and 4.6.5), whereas MMNIST-CIFAR and KTH remain explicitly unexecuted extensions. This separation preserves the content and scope of the existing publications instead of retroactively enlarging them.

### 4.1.3 Bridge to the OpenSTL/Moving-MNIST Benchmark

**[MEASURED 2026-07-06 — full results, table, and reading in §4.6.2 (Table 4.8).]** For continuity with the literature, the developed stack is additionally evaluated on the standard OpenSTL Moving-MNIST suite [74]. The pre-registered expectation, recorded before the runs: mid-pack aggregate MSE/SSIM against the OpenSTL leaderboard — aggregate metrics on a ≈93%-black-background benchmark reward mean prediction (this project's own measurement: a constant all-black prediction outperforms a trained CW-VAE [13] on aggregate SSIM/PSNR, §3.4.3) — together with competitive perceptual (LPIPS) behavior and an autoencoder round-trip ceiling row that decomposes decoder error from dynamics error. The purpose of the bridge is connectivity to the standard suite, not a leaderboard claim; the memory-horizon claims of this chapter cannot be tested on Moving MNIST at all, for the degeneracy reasons established in §3.4. The measured outcome (§4.6.2) confirmed the connectivity, ceiling-row, and perceptual-versus-persistence clauses, **refuted the "mid-pack aggregate MSE" clause** (180.91 against the deterministic specialists' 22.06–29.80 — reported as a refutation, per the standing convention of §4.2.3), and returned the predicted flat ≈ hierarchical null as the negative control of the memory-horizon claim.

### 4.1.4 Hardware and Software Environment

All experiments of the [A4] campaign (§§4.2–4.5) were performed on a single consumer-grade GPU (NVIDIA GeForce RTX 4060 Ti, 16 GB) — a deliberate constraint of the research design: the resource-efficiency claims of Chapter 2 are made *for* this class of hardware, and the complete benchmark (data generation, autoencoder training, all system arms, all evaluations) reproduces on one such device. The software complex is implemented in Python/PyTorch (§4.7; the module map is given in the appendix **[AUTHOR: appendix letter]**). The memory-consumption measurements quoted in Chapter 2 (Table 2.3: BPTT 12.2 GB at rollout length $K = 40$ and OOM at $K = 120$ versus the developed method's ≈1.1–1.2 GB through $K = 160$; the inner-loop panel behind the 0.26%-measured no-go result) were performed in this same environment, on the implemented stochastic hierarchical recurrent core, and constitute part of the experimental program of this chapter even though their tables live with the theory they verify.

Training protocol of the rollout arms (operating point A): 30,000 optimizer steps, batch 16, bf16, EMA decay 0.999, evaluation on EMA weights (§3.2.3); the flat-long arm is the disclosed exception (§4.3.4). The DiT backbone of every arm is the §3.2.2 configuration (d = 384, depth 8, 6 heads; 22.19 M parameters).

### 4.1.5 Statistical Protocol and Threats to Validity

The evaluation protocol is the §3.4.4 instrument, hardened by a six-surface adversarial validity audit of the campaign (data/splits, rollout protocol, gate protocol, energy-model protocol, autoencoder provenance, metrics/statistics; 24 findings, 2 confirmed major, both fixed and recomputed the same day). The audit's material corrections and standing disclosures are part of the protocol definition and are stated here once:

1. **Cluster-robust intervals.** Rollout records are 100 clips × 152 consecutive AR frames with lag-1 autocorrelation +0.86; frame-iid bootstrap intervals are 2–7× too narrow. All quoted CIs are therefore paired clip-level cluster bootstrap intervals. Under the corrected intervals, three secondary bin verdicts of the campaign flipped to null; **every headline verdict survived**. Records whose ground-truth frame contains no foreground (3,263 of 15,200 = 21.5%, where fg-MSE would silently fall back to full-frame MSE) are excluded from all stratified bins and reported as their own background-only row.

2. **Corrected hardness anchoring.** An audit-confirmed bug in the mechanism-study hardness labels (the target-anchored coverage window counted the k − 1 = 3 bridge frames as covered, mislabeling 11–14% of hard samples as easy; hard fraction 0.19 buggy → 0.27 correct) was fixed and Table 4.2 recomputed from the saved checkpoints with a single raw-ground-truth metric reference for all rows. The memory-gate and rollout evaluations always used the correct anchor. All hardness labels are computed from ground-truth geometry, independently of any model and identically across arms.

3. **Metric scope disclosures.** fg-MSE scores only ground-truth-sprite pixels; hallucinated off-sprite content is invisible to it — it is therefore never interpreted alone but jointly with foreground sharpness, decoded-sample review (§3.4.3), and the complete-metric evidence artifacts, which include full-frame, union-mask and false-positive-sensitive endpoints. Per-horizon *sharpness* series are chunk-batch means duplicated per record — valid for overall means and trend curves, never for per-bin or per-clip readings; every sharpness-versus-horizon statement in this chapter is so scoped.

4. **Pairing and replication conventions.** All arm-versus-arm comparisons share evaluation clips and horizons; sampling noise is common only when target tensor shapes match. Cross-shape cells retain record pairing but not elementwise CRN (unbiased, with less variance reduction). "2 seeds" for the single-step memory gate means two *training* seeds sharing one CRN evaluation set (by design, for exact pairing). At operating point B the evaluation adds the pre-registered **dt-since-last-visible** stratifier

(per-record time since the target region was last in view, computed from the recorded trajectory; bins dt ∈ (48, 96], (96, 192], > 192, crossed with hardness) and a warm-up-SEEN restriction, so that gates are read only on re-entries of content that was answerable by construction.

5. **External validity.** The testbed is purpose-built and synthetic; digit assets are shared

across splits (texture familiarity only — re-entry memory requires clip-specific trajectory content, which does not transfer across clips), while trajectories and world layouts are split-disjoint. All claims are scoped to the memory-horizon regime the testbed operationalizes; the OpenSTL bridge (§4.1.3, measured in §4.6.2) anchors short-horizon behavior on a standard benchmark. Generation seeds of the shipped dataset files were not recorded (measured properties are stated as measured; a frame-hash duplication check found no cross-file clip duplication).

## 4.2 Matched-Compute Mechanism Study: Selecting the Generative Core

The claim defended in this section is deliberately conditional: **which prediction head wins depends on the inference regime; there is no one best head.** Three results establish this — the single-step coverage study (§4.2.2), its rollout reversal (§4.2.3), and the two-sided study of the energy-based head (§4.2.4). The section thereby supplies the empirical selection evidence promised in §3.2 (the diffusion core) *and* the honest boundary of that selection.

**Figure 4.1 — Three prediction heads in the shared context testbed.** Direct regression predicts one target latent, diffusion predicts a denoising velocity and commits through DDIM sampling, while the energy head learns a scalar landscape traversed by candidate descent. The diagram separates objective and inference from the shared context question. *Source:* `docs/figures/fig24_generation_objectives.png`; deterministic source `scripts/make_dissertation_missing_figures.py`.

### 4.2.1 Shared Mechanism Testbed

All Part-A predictors are transformers of one shape (width 256, 6 blocks, 8 heads) over one token layout — 16 coarse plus 64 fine tokens per frame with joint attention over both levels' tokens (the §3.1.2 conditioning mechanism) — with parameter counts 7.27 M (regression), 7.53 M (diffusion), and 7.28 M (energy-based): the differences are the mechanism-specific heads and embeddings only. Context is 6 frames of both levels; the target is the frame k = 4 steps after the last context frame; the four bridging actions are embedded through the same conditioning path in every arm. Training is single-step and leakage-free (ground-truth context and target; no prediction fed back); batch 48, fp32; the regressor trains 6,000 steps (its loss plateaus at 0.096), the diffusion arm 8,000 steps. Latents are standardized per channel and per level with training-set statistics — the §3.1.1 protocol requirement, and a finding in its own right: with raw latents (per-channel scale ≈0.12 coarse / ≈0.16 fine) the diffusion arm

*underperformed* the regressor, because unit-scale noise buried the signal mid-schedule and sampling initialized 4–7× off-scale; the one-line standardization fix reversed the mechanism ranking. The failure mode is silent — losses fall, samples look plausible, nothing errors — which is why standardization is elevated to a protocol requirement rather than an implementation detail.

Inference budgets are matched in network function evaluations (NFE), not step counts: one DDIM step = 1 forward pass; one energy-descent step = forward + gradient ≈ 2 forward passes; one regression sample = 1 forward pass (evaluation keeps dropout 0.1 active, so N stochastic passes yield N samples — MC-dropout, making best-of-N comparable across arms). Best-of-N selection is performed in decoded pixel space on foreground MSE against ground truth — an **oracle coverage diagnostic** (§3.4.4), always reported alongside the mean over N.

### 4.2.2 Single-Step Oracle-Coverage Regime: Diffusion Wins Every Column

Table 4.2 reports the corrected mechanism comparison (single-step, k = 4, best-of-8, NFE 16; six evaluation batches of 48; hard split = novel-strip > 0, covering 27% of samples; references per Table 4.1).

**Table 4.2 — Matched-capacity head comparison, single-step (corrected hardness anchor, single raw-GT reference).**

| Head | fg-MSE (all) | easy | HARD | mean-over-8 | fg-sharp |
|---|---|---|---|---|---|
| Regression (MC-dropout) | 0.0664 | 0.0232 | 0.1808 | 0.0714 | 0.1591 |
| Diffusion (v-pred, DDIM) | **0.0512** | **0.0120** | **0.1549** | **0.0640** | **0.1886** |
| Persistence floor | 0.2367 | — | — | — | 0.0515 |

**Table 4.3 — Matched-NFE control (best-of-8 fg-MSE).**

| NFE per sample | Regression | Diffusion |
|---|---|---|
| 4 | 0.0680 | 0.0546 |
| 8 | 0.0680 | 0.0514 |
| 16 | 0.0680 | 0.0503 |

Three observations. First, both heads are far below the persistence floor on every column: the action-conditioned warp is learned by everything, as expected where the easy region is deterministic given the action. Second, the separation concentrates exactly where the hardness stratification says it should: on the easy bin diffusion's advantage is 0.011 absolute (0.0232 → 0.0120), on the genuinely generative HARD bin it is 0.026 (0.1808 → 0.1549), and diffusion is sharper (0.1886 vs 0.1591 against the 0.2412

ceiling). Third, the advantage is not an inference-budget artefact (Table 4.3): the regressor is budget-flat at 0.0680 while diffusion already leads at NFE 4 and improves monotonically. The regressor's mean-over-8 ≈ best-of-8 gap is near zero — it has *averaged over* the possible futures rather than distributing probability mass across them; diffusion's best-of-N gap (0.0640 → 0.0512) is the coverage signature of a head that commits to distinct hypotheses per sample. This is a controlled replication of the "regression averages, sampling-based generation commits" finding of the stochastic video-prediction literature [75, 76, 77] under matched capacity, matched conditioning, matched inference budget, and exact hardness stratification — and its protocol scope is the point: best-of-8 selection is an oracle coverage regime.

**Figure 4.2 — Single-step regression and diffusion predictions.** Each row shows last context, prediction and ground truth on identical inputs. The panels are descriptive examples of the registered single-step regime; the aggregate and paired results, not a selected row, determine the comparison. *Source:* `docs/figures/fig29_mechanism_single_step_samples.png`, composed without pixel editing from `logs_rhcp/step1_mech_clean_{reg,diff}.png` by `scripts/make_dissertation_missing_figures.py`.

### 4.2.3 Rollout Reversal: Regression Wins Best-of-1

The pre-registered expectation for the rollout campaign was that the regressor, rolled out autoregressively, would fade toward the mean while diffusion held sharpness. The measured outcome **refutes** this expectation, and the reversal is reported as such, not reframed: under the deterministic best-of-1 rollout protocol (§4.1), **the regression arm beats the hierarchical diffusion arm at every horizon band and every hard bin at equal sharpness** — reg-long vs hier-long: fg-sharp 0.0741 vs 0.0729; ALL fg-MSE 0.1682 vs 0.1848; direct paired cluster CI on the ALL delta [+0.0110, +0.0227]. The same holds at the matched flat context: reg-flat 0.2015 vs flat-budget (diffusion) 0.2038.

The two results reconcile without contradiction once the protocols are stated side by side: the Table 4.2 diffusion win is single-step **best-of-8** — oracle coverage switched on; the rollout result is **best-of-1** — coverage off, and every sampled hypothesis is fed back and compounded over 152 frames. Diffusion wins the coverage regime; regression wins the deterministic best-of-1 rollout regime. Any claim of the form "head X is better" that omits the regime is, on this evidence, wrong.

### 4.2.4 Two-Sided Energy-Head Study: A Characterized Negative Result

The third candidate head is energy-based descent: Energy-Based Transformers [39] train a scalar energy $E(\text{context}, \text{candidate})$ — reviewed in §1.3.3 as a global consistency criterion — and generate by gradient descent on the candidate. The mechanism is attractive for this dissertation's problem precisely because descent steps are a natural "think-longer" knob for hard re-entry frames. The study below is *two-sided* in that it decomposes the observed failure into two separable capabilities: whether the trained energy landscape **orders** candidates correctly (target below mean below noise), and

whether descent from the prescribed initialization can **reach** a structured output. The headline result is that these dissociate.

**Setup: the official implementation, imported unchanged.** This dissertation's own reimplementation of the mechanism, trained in the §4.2.1 harness, collapsed to a blank near-mean output; a reimplementation failure is uninformative, so the study removes the author's code from the loop entirely. The control module of the software complex imports the reference repository's model classes unmodified (RMSNorm blocks, rotary positions, causal autoregressive attention, the prepended descent-step time-embedding token, the bias-free linear energy head, the causal replay buffer), and transcribes the reference MCMC descent control flow line-for-line (random-noise initialization, per-step step-size randomization, Langevin noise, gradients with respect to the pre-projection frame embeddings, single-graph and truncated variants per recipe). Two adaptations were unavoidable and are flagged: (i) *frame-as-token* — the reference flattens each frame's latent into one token via a learned down-projection; the identical scheme is applied to the concatenated standardized latents of §3.1.1 ($16 \cdot 4 \cdot 4 + 16 \cdot 8 \cdot 8 = 1{,}280$ dimensions), with descent performed in the original 1,280-d space exactly as in the reference; (ii) an additive learned *action embedding* on the candidate tokens (the reference video task is action-free; this testbed is action-conditioned). One recipe number is re-tuned: the descent step size $\alpha$, whose published values are calibrated to the reference latent space's gradient scale, is recalibrated to this latent scale by the reference's own movement principle ($\alpha \cdot |\nabla E| \cdot n_{\text{steps}} \approx$ the initialization-to-target distance).

Both published training recipes are run at the paper's "xxs" video shape — width 384, 6 layers, 6 heads, FFN multiplier 4, context 16 frames; 14.66 M parameters in the frame-token instantiation — with the paper's scheduler shape, peak learning rate $4 \times 10^{-4}$, weight decay 0.01, gradient clip 1.0, effective batch 128 (micro-batch 32 × 4 accumulation), 20,000 optimizer steps, fp32. The **S2 recipe** uses a single training landscape with randomized extra final-landscape steps, truncated second-order graph, Langevin noise, a replay buffer, and fixed recalibrated $\alpha = 15.97$; the **S1 recipe** (the configuration behind the paper's published video generations) uses two descent steps with detach between them, a second-order graph on every step, no replay, no training-time Langevin noise, and a learnable $\alpha$ (recalibrated initialization 23.95). Every 1,000 steps, held-out probes record: the energies $E(\text{target})$, $E(\text{mean latent})$, $E(\text{random noise})$ on the final landscape; the cold-descent output's latent distance to target $d_{\mathrm{tgt}}$ (the mean-to-target distance is 0.964 in the same units); and the decoded output's foreground sharpness (collapse threshold ≈0.08; ceiling ≈0.24). For S1, the paper's own inference recipe is additionally run at each probe (best-of-4 by minimum energy, 4 descent steps, inference $\alpha/4$, Langevin noise, probed at both the current-$\alpha$ and initial-$\alpha$ readings).

**Figure 4.3 — Energy head and gradient-descent generation loop.** The tested head scores a 1,280-dimensional per-frame candidate together with context and actions. Generation repeatedly updates the candidate using its energy gradient and the registered noise rule. This makes landscape ordering and reachability by descent separately testable. *Source:* `docs/figures/fig25_ebt_generation_loop.png`; deterministic source `scripts/make_dissertation_missing_figures.py`.

**The S2 recipe: a stably mis-ordered landscape.** The S2 run converges (training loss settles at ≈0.597 from roughly step 7,000 to the end) — and its landscape is stably wrong. At **every one of the 21 probes** from step 0 to step 20,000, $E(\text{mean}) < E(\text{target})$: the batch-mean latent has strictly lower energy than the true target, with a stable gap of ≈0.6–1.2 energy units from step 3,000 onward. Cold descent behaves accordingly: $d_{\mathrm{tgt}}$ stays flat at 1.26–1.29 (against the mean's own 0.964 — the output is no closer to the target than the mean is), and decoded foreground sharpness never leaves the collapse regime (0.008–0.014 across all probes, versus the 0.08 structure threshold). Twenty thousand converged steps at the paper's shape, batch semantics, and scheduler do not repair the ordering; the objection that the earlier negative was an under-training artefact is thereby closed for this recipe. One disclosure: the batch-mean latent has roughly eight times lower norm than an individual target, so a norm preference could in principle carry part of the ordering; two probe features rebut a pure norm artefact — $E(\text{random noise})$ sits far above both throughout, and the S1 recipe flips the *same* probe to the correct ordering, which a norm effect alone cannot do.

**The S1 recipe: ordering repaired, generation still out of reach.** The S1 run is the informative one. Its loss falls further (0.361 at 20,000 steps, still improving), the learnable $\alpha$ drifts upward (23.95 → 33.27), and — unlike S2 — the landscape *becomes correctly ordered*: $E(\text{target})$ is strictly lowest at **all fifteen consecutive probes from step 6,000 to 20,000**, with a target-below-mean gap ranging from +0.22 to +3.09 energy units across those probes (the exact flip-onset step rests on one fixed 64-sample probe batch; the fifteen-consecutive-probe persistence is the claim, not the onset). By the ordering criterion that indicted S2, S1's energy is *right*. Generation still never materializes: cold descent from the prescribed random-noise initialization stays at $d_{\mathrm{tgt}} \approx 1.12$–$1.17$, with foreground sharpness creeping from 0.011 to a maximum of 0.035 — a real but small trend that remains a factor of ≈2.3 below the structure threshold and ≈7 below the decoder ceiling. The paper's own inference recipe does no better: $d_{\mathrm{tgt}} \approx 1.19$–$1.24$ and fg-sharp 0.010–0.015 at every probe and both $\alpha$ readings (final probe: 1.207 / 0.0127 at the current-$\alpha$ reading; 1.233 / 0.0120 at the init-$\alpha$ reading). A correctly ordered landscape, probed under the publication's own sampler, produces no structured output at this scale.

**Decomposition: ordering × reachability.** Combining the two runs with the positive-control probes gives a clean two-factor decomposition. *(a) Landscape ordering is recipe-dependent:* S2 never orders the landscape correctly (21/21 mean-lowest probes at converged loss); S1 orders it correctly from step

6,000 onward (15/15 target-lowest). The choice between the two published training recipes decides whether the energy function even prefers the truth. *(b) Descent reachability from noise fails under both, and is not bought back by inference-time compute.* In positive-control experiments on the same official model (copy and shift tasks), the trained landscape is correct by every probe — on copy, $E(\text{target}) = -65.6$ vs mean -57.2 vs noise -14.8, and an energy-conditioning matrix identifies the matching context for 8/8 targets with an off-diagonal gap of +26.4 — yet cold descent still lands blank (fg-sharp 0.030 on copy, 0.016 on shift). Sweeping cold-descent steps 2 → 128 leaves $d_{\mathrm{tgt}}$ flat (1.131–1.149); warm-starting *at the target* and descending does not hold it (at the operating step size the output drifts to $d_{\mathrm{tgt}}\ 0.41 \to \approx 0.55$, settling in a shallow basin between target and mean — the target is at best a weak attractor along the descent dynamics); the minimum-energy sample among 16 cold descents remains collapsed (fg-sharp 0.0145); and a persistence-initialized descent also fails to move toward the target ($d_{\mathrm{tgt}}$ 0.835, fg-sharp 0.046, versus 0.348 and 0.088 for a plain regressor on the same task). The two factors are separable because S1 fixes (a) while leaving (b) intact: what is observed at 16 GB-reachable scale is not "the energy model learns the wrong criterion" simpliciter — with the generation recipe it learns a usefully ordered criterion whose minima are nonetheless unreachable by the prescribed sampler from the prescribed initialization. This is consistent with theoretical accounts of entropy collapse in gradient-descent sampling [78], cited as context being instantiated, not as a discovered mechanism.

**Scope and non-claims.** This study tests a capability the EBT publication does not claim at a scale it did not run: the paper's video-section claims concern validation-loss scaling and energy-as-uncertainty, not sharp generation, and its published video generations come from a larger model class trained on natural video through a large pretrained autoencoder on multi-GPU hardware. Nothing here is a reproduction failure, and nothing here contradicts the published claims; published small-model positives in other domains [79] are likewise untouched. What the study establishes is a **capability boundary with a mechanism attached**: at the scales reachable in this work (Table 4.4), the energy-descent head does not convert even a correctly ordered landscape into committed video predictions — while the diffusion control at roughly half the parameter count (7.53 M vs 14.66 M), on the same frozen latents and the same standardization, shows recognizable digit structure by 1,500 optimizer steps and at convergence commits to sharp, correctly located digits (Table 4.2). Whatever resource the energy sampler is missing at this scale, diffusion does not miss it. Residual limitations are stated: the scaled runs remain a factor of 3–5 below the reference step budgets and 2 below its batch size; the S1 sharpness trend (0.011 → 0.035), while far below structure, is non-zero, and a longer run could in principle continue it — the claim is scoped accordingly.

**Table 4.4 — Configuration gap between this study's fair-scale runs and the reference training configurations (the extrapolation-risk record).**

| | This study (S2 / S1 arms) | Reference S2 | Reference S1 |
|---|---|---|---|
| Model shape | xxs: d 384, 6L, 6H, FFN 4× (14.66 M with 1,280-d frame tokens) | xxs | xxs |
| Context | 16 frames | 16 frames | 16 frames |
| Steps (warm-up) | 20,000 (5,000) | 60,000 (10,000) | 100,000 (10,000) |
| Effective batch | 128 fresh + 64 replay / 128 fresh | 256 fresh + replay | 256 fresh |
| Step size \alpha | fixed 15.97 / learnable, init 23.95 (recalibrated to latent scale) | fixed 3,000 | learnable 30,000 |
| Data | synthetic 64×64 grayscale; 2,700 clips × 100 frames | natural video, 224×224, pretrained-AE latents | same |
| Hardware | 1× RTX 4060 Ti 16 GB, fp32 | 4 GPUs/node | 4 GPUs/node |

**Figure 4.4 — Energy-landscape ordering under two recipes.** S1 changes to the desired target-below-mean ordering after approximately 6,000 updates; S2 keeps the batch mean below the target through 20,000 updates. Both series include the random-candidate reference. *Source:* `docs/figures/fig26_ebt_landscape_ordering.png`, parsed deterministically from `logs_rhcp/ebt_scaled_{s1,s2}.log` by `scripts/make_dissertation_missing_figures.py`.

**Figure 4.5 — Foreground sharpness after cold energy descent.** Both recipes remain below the preregistered 0.08 structure threshold and far below the approximately 0.24 decoder reference, even after S1 learns the desired energy ordering. *Source:* `docs/figures/fig27_ebt_descent_sharpness.png`, parsed deterministically from `logs_rhcp/ebt_scaled_{s1,s2}.log` by `scripts/make_dissertation_missing_figures.py`.

### 4.2.5 Generative-Core Conclusion

The head ranking is **regime-dependent**: diffusion for the single-step best-of-N coverage regime, regression for the deterministic best-of-1 rollout regime, energy-based descent for neither at 16 GB-reachable scale (a two-sided characterized negative: ordering recipe-dependent, reachability failing under both recipes and under the paper's own sampler). The memory studies that follow therefore use **both** viable heads for the central matched-budget memory comparison (§4.3.5), while §4.4 estimates and reports package preference separately by head and training seed. The central reach result is not tied

to one head; the broader ordering of temporal and spatial packages is explicitly head- and seed-conditional.

## 4.3 Single-Step Memory Gate and Matched-Budget Rollout Study

This section tests the central pre-registered prediction of Table 3.1: at an equal context-token budget, does a hierarchical allocation (long coarse memory + recent fine window) beat a flat allocation when only the former reaches memory-relevant history? The measured studies then bracket the tested budgets where that prediction holds and where full-resolution fidelity instead wins; they do not assume a continuous or unique transition.

### 4.3.1 Preliminary Gate: Recall Emerges Only after Sufficient Optimization

Before any rollout machinery, the single-step memory gate asked whether a long *coarse* context helps at all: coarse context 48 versus 6 at fixed fine context 6 (1,232 vs 560 tokens including the target slots), best-of-8, stratified by the exact hardness label. The result is an emergence-with-budget pattern: at 6,000 steps the HARD-bin delta is null on both training seeds (+0.0014 [−0.0004, +0.0033] and +0.0008 [−0.0009, +0.0025]); at 20,000 steps it is a decisive win on both (+0.0135 [+0.0080, +0.0196] and +0.0143 [+0.0083, +0.0208]; hard-bin absolute error 0.1053 → 0.0918 on seed 0 — a ≈13% relative reduction), with sharpness tied (≈0.188 both arms): the win is retrieval, not sharpness. Four disclosures attach: "2 seeds" means two training seeds sharing one CRN evaluation set (§4.1.5); these evaluation windows are near-independent rather than fully clustered, so the audit's worst-case design-effect correction widens the 20k hard-bin CIs to ≈[+0.005, +0.022] (seed 0) and ≈[+0.006, +0.023] (seed 1) — the win survives; **each budget is its own complete, fully annealed training schedule** (a 6,000-step run and a 20,000-step run, budget-matched within each contrast), so the compared quantity is the optimization budget with schedule shape a bundled covariate — and the 6,000-step arms were the better-converged endpoints of their own schedules, which works against a mid-anneal objection; and the 20,000-step budget was a **disclosed post-null follow-up** — the originally registered read at 6,000 steps returned its null, which is on the record, and the extension was then run and replicated on both seeds, so the claim rests on the twice-replicated 20k result, not on a selected read. The methodological moral carries through the whole chapter: **retrieval-from-memory requires a large optimization budget, and short-budget ablations (6,000 steps here) would have spuriously killed it.**

**Figure 4.6 — Emergence of single-step memory benefit with optimization budget.** The HARD-bin effect is unresolved at 6,000 updates for both seeds and positive with intervals excluding zero at 20,000 updates for both seeds. This is an optimization-budget result, not a claim about asymptotic convergence. *Source:* `docs/figures/fig28_memory_gate_optimization.png`, parsed deterministically from the four `logs_rhcp/mem_gate_full_s{0,1}{,_t20k}.log` files by `scripts/make_dissertation_missing_figures.py`.

### 4.3.2 Rollout Matrix

Table 4.5 gives the full arm ordering under the rollout protocol (ALL-row fg-MSE over 152 AR frames after the 48-frame ground-truth warm-up, seed 0; arms per Table 3.1).

**Table 4.5 — Rollout arm ordering, operating point A.**

| Arm | Context | Ctx tokens | ALL fg-MSE | fg-sharp | Note |
|---|---|---|---|---|---|
| flat-long (diff) | 48 fine frames | 3072 (2.7×) | **0.1564** | **0.0996** | B = 8 × 60k sample-matched (caveat, §4.3.4) |
| reg-long | 48c + 6f, regression | 1152 | 0.1682 | 0.0741 | |
| hier-long (diff) | 48c + 6f | 1152 | 0.1848 | 0.0729 | THE system arm |
| reg-flat | 18 fine frames, regression | 1152 | 0.2015 | 0.0512 | |
| flat-budget (diff) | 18 fine frames | 1152 | 0.2038 | 0.0582 | token-matched flat control |
| hier-short (diff) | 6c + 6f | 480 | 0.2260 | 0.0392 | memory ablation |
| persistence | — | — | 0.2552 | — | floor |

Three seed-0 regularities are visible in the ordering, each defended separately below: (i) the 48-frame coarse-plus-fine arm beats the shorter contiguous-fine control under both tested heads at this seed; (ii) regression wins the deterministic best-of-1 rollout (§4.2.3); and (iii) the full-resolution 48-frame window wins outright at 2.7× the tokens (§4.3.4). The first observation replicates at seed 1 for this particular long-versus-short comparison, but it must not be promoted to a head-independent law. The complete five-seed temporal-versus-spatial factorial (§4.4.3) instead finds stable package direction under diffusion and sign-changing direction under regression. The defensible claim is therefore configuration- and head-scoped: reach helps where the shorter control cannot address the relevant past, while the preferred way to package that reach is not universally mechanism-agnostic.

**Figure 4.7 — Error and sharpness by rollout horizon.** *Content:* foreground MSE on all frames, foreground MSE on HARD re-entry frames, and foreground-sharpness retention, with persistence and frozen-decoder references. *Source:* `docs/figures/fig1_claim1_horizon_curves.png`; deterministic source `scripts/make_figures.py`.

### 4.3.3 Main Two-Seed Pair: Advantage beyond the Flat Memory Horizon

Table 4.6 is the headline of the dissertation: hier-long versus flat-budget at equal 1,152 context tokens, the full horizon × hardness matrix (Δ = flat − hier; positive = hierarchy better).

**Table 4.6 — hier-long vs flat-budget (seed 0), 152 AR frames after the 48-frame GT warm-up, paired per-frame fg-MSE, clip-level cluster bootstrap; empty-GT-foreground records (21.5%) excluded from stratified bins.** Sharpness 0.0729 vs 0.0582; persistence floor 0.2552.

| Horizon | Hardness | n | flat-budget | hier-long | Δ | 95% CI | Verdict |
|---|---|---|---|---|---|---|---|
| h 1–8 | easy | 434 | 0.0440 | 0.0419 | +0.0022 | [−0.0040, +0.0089] | null |
| h 1–8 | HARD | 150 | 0.1435 | 0.1359 | +0.0075 | [−0.0164, +0.0327] | null |
| h 9–24 | easy | 919 | 0.1216 | 0.1189 | +0.0028 | [−0.0075, +0.0133] | null |
| h 9–24 | mid | 45 | 0.1661 | 0.1624 | +0.0038 | [−0.0121, +0.0182] | null |
| h 9–24 | HARD | 282 | 0.1800 | 0.1728 | +0.0071 | [−0.0110, +0.0260] | null |
| h 25–60 | easy | 2286 | 0.2210 | 0.1989 | +0.0221 | [+0.0080, +0.0368] | **WIN** |
| h 25–60 | mid | 79 | 0.2218 | 0.2036 | +0.0182 | [−0.0049, +0.0437] | null |
| h 25–60 | HARD | 550 | 0.2554 | 0.2209 | **+0.0344** | [+0.0162, +0.0531] | **WIN** |
| h 61+ | easy | 5610 | 0.3060 | 0.2788 | +0.0272 | [+0.0166, +0.0382] | **WIN** |
| h 61+ | mid | 209 | 0.3224 | 0.2842 | +0.0382 | [+0.0178, +0.0616] | **WIN** |
| h 61+ | HARD | 1353 | 0.3073 | 0.2759 | **+0.0314** | [+0.0202, +0.0431] | **WIN** |
| bg-only | | 3263 | 0.0067 | 0.0047 | +0.0020 | [+0.0013, +0.0026] | WIN |
| **ALL** | | 15200 | 0.2038 | 0.1848 | **+0.0189** | [+0.0121, +0.0264] | **WIN** |

On seed 0 the matrix reads as the claim in one table: **null in every bin below horizon 25; a CI-excluding-zero win in every easy/HARD bin from h 25–60 onward** (the sparse mid cell, n = 79, is null at h 25–60 and a win at h 61+). The advantage appears exactly where the memory horizon demands it — h ≈ 24 is half a loop period, past which the required horizon exceeds what 18 flat frames carry — and nowhere else.

**Seed-1 replication (pre-registered before running).** The same pairing retrained at seed 1 wins **in every cell**: ALL +0.0354 [+0.0267, +0.0447]; HARD h 25–60 +0.0622; sharpness 0.088 vs 0.057. Two-seed status of the headline: **CONFIRMED** (seed 0: ALL +0.0189 [+0.0121, +0.0264]). The honest nuance is the onset boundary: it is seed-variable — seed 0 is null below h 25 while seed 1 wins already from h 1–8, and the hier-long arm itself moves 0.1848 → 0.1690 across seeds over a stable flat baseline. The robust two-seed claim is therefore **the win and its mid-horizon concentration, not the exact onset frame**; the "no advantage before h 25" phrasing is a seed-0 observation, retained only as such.

**Per-horizon behavior and the deep-horizon disclosure.** hier-long stays below the persistence floor at *every* horizon out to 152 frames (more than three loop periods); its error saturates at 0.221–0.222 over the last 70 frames rather than diverging — bounded drift. Sharpness, however, fades monotonically for **all** arms (hier-long 0.2023 at h 1–4 → 0.0323 at h 105–152; batch-mean series, §4.1.5): the pre-registered sharpness-retention clause of gate G2b (> 0.5× single-step sharpness at depth) is **not met** at h ≥ 81, and this is stated plainly as the deep-horizon limitation rather than tuned away. The blur ceiling is owned by the frozen decoder (AE round-trip 0.2412), not by the dynamics.

**Figure 4.8 — Effect map by horizon and difficulty.** *Content:* horizon × hardness map of Δ(flat − hierarchy), seeds 0 and 1 side by side. The horizon-gated onset is seed-0-specific; the common feature is the positive effect and mid-horizon peak rather than one exact onset. *Source:* `docs/figures/fig2_claim1_winmap.png`; deterministic source `scripts/make_figures.py`.

**Figure 4.9 — Qualitative slices of autoregressive rollout.** *Content:* identical ground truth and seed-1 `hier-long`/`flat-budget` predictions at registered horizons. Both models become blurred at long horizon; the figure illustrates a modest average advantage, not a universal per-frame recovery or a claim that one named arm uniquely hallucinates. *Source:* `docs/figures/fig7_samples_claim1.png`, from the saved rollout checkpoints.

### 4.3.4 flat-long: the 2.7×-Token Pareto Reference

The flat-long arm (48 full-resolution frames, 3,072 tokens) beats every other arm on both error and sharpness: ALL 0.1564, fg-sharp 0.0996; against flat-budget its ALL delta is +0.0473 [+0.0362, +0.0592]. This is reported with the same prominence as the headline, because it is what keeps the claim honest: **at operating point A, hierarchy is an efficiency point, not a capability necessity.** The hierarchical arm recovers most of the long-memory benefit at ≈37% of the context tokens; a party willing to pay 2.7× buys the frontier outright. (Protocol caveat, disclosed wherever flat-long is quoted: it trained at B = 8 × 60k steps — sample-matched to B = 16 × 30k, but with a different EMA horizon and gradient-noise scale than the other arms.) The regime where no affordable flat window can cover the loop at all is exactly what operating point B exists to test (§4.5).

### 4.3.5 Prediction-Head Robustness and Its Boundary

The preregistered two-seed gate asked whether the specific `reg-long` versus token-matched `reg-flat` comparison reproduced the diffusion reach result. It did: `reg-long` beats `reg-flat` in every bin at both tested seeds (ALL +0.0334 [+0.0248, +0.0423] at seed 0 and +0.0501 [+0.0396, +0.0612] at seed 1; hard-bin wins at every horizon, including h 1–8 HARD +0.0447; sharpness 0.0741 versus 0.0512 at seed 0). The regression pairing has no null prelude, so the seed-0 diffusion onset near h = 25 is not a general property of the memory geometry.

This is evidence that one long-coarse-versus-short-fine reach advantage is not solely an artifact of the diffusion head. It is **not** evidence that package preference is mechanism-agnostic. The complete five-seed temporal-versus-spatial factorial in §4.4.3 gives two temporal and three spatial regression winners and a sign-changing head × package interaction. Accordingly, the dissertation reports the two-seed replication as a scoped robustness result and rejects the broader head-independent wording.

### 4.3.6 Completing the Token-Budget Frontier with Matched Upper and Lower Partners

The arm matrix of Table 4.5 leaves the two extreme token budgets without same-budget partners: flat-long (3,072 tokens) has no hierarchical arm at its budget, and hier-short (480 tokens) has no flat control at its. Two frontier-completion arms close the grid, pre-registered before launch (the FRONTIER COMPLETION block of the factorial pre-registration): **hier-rich** (48 coarse + 36 fine frames = 3,072 tokens, trained B = 8 × 60,000 steps — sample-matched to flat-long's disclosed regime, so the pairing inherits flat-long's §4.3.4 protocol caveat *symmetrically*) and **flat-short** (8 fine frames = 512 tokens ≈ hier-short's 480, standard B = 16 × 30,000). Diffusion heads first; the regression twins are optional.

The reading was fixed in advance and is quoted verbatim from the pre-registration: "at 3072 tokens the window is SATURATED at 48 frames by the T=100 train clips — hierarchy cannot buy more window, so per the factorial's abstraction cost the honest expectation is hier-rich ~<= flat-long (CONVERGENCE at the frontier top). The efficiency claim's correct form is 'the hierarchy curve reaches a given quality at fewer tokens and converges with flat where the window saturates' — a frontier-from-below, not dominance at every budget. At the low end, hier-short vs flat-short tests the frontier's bottom."

**Table 4.6a — Token-budget frontier of operating point A (letter-suffixed to avoid renumbering the audited table register; final numbering at typeset). Measured anchor rows from Tables 4.5/4.6; frontier-completion cells measured 2026-07-07 (`logs_rhcp/frontier_top.log`, `logs_rhcp/frontier_bottom.log`); lower-edge pair measured 2026-07-07 (`logs_rhcp/frontier_low.log`, pre-registered the same morning before the runs).**

| Token budget | Flat arm | ALL fg-MSE | Hierarchical arm | ALL fg-MSE | Paired Δ (flat − hier), cluster CI |
|---|---|---|---|---|---|
| ≈3,072 | flat-long | 0.1564 | hier-rich | 0.1695 | **−0.0131 [−0.0201, −0.0069]** (flat wins; seed 1 REPLICATED: −0.0107 [−0.0163, −0.0058]) |
| 1,152 | flat-budget | 0.2038 | hier-long | 0.1848 | +0.0189 [+0.0121, +0.0264] (Table 4.6) |
| **896** | flat-min (14 f) | 0.2106 | hier-min (48c+2f) | **0.1706** | **+0.0399 [+0.0299, +0.0501]** (hier wins EVERY cell; seed 1 REPLICATED: +0.0358 [+0.0267, +0.0454], every cell) |
| ≈480–512 | flat-short | 0.2224 | hier-short | 0.2260 | **−0.0036 [−0.0063, −0.0010]** (flat wins) |

**Figure 4.10 — Confirmed token-budget frontier.** *Content:* ALL fg-MSE versus context-token budget at operating point A, one 6-point curve per family (flat: flat-short → flat-min → flat-budget → flat-c1 → flat-c2 → flat-long; hierarchical: hier-short → hier-min → hier-long → hier-c1 → hier-c2 → hier-rich). Mark the tested family-wise-confirmed hierarchy points at 896/1,152, the unresolved 1,792 point, and the confirmed flat points at 2,432/3,072. Draw the geometric span threshold as an explanatory annotation and the persistence floor (0.2552); do not shade an interpolated win band. *Source:* `docs/figures/article_confirmed_frontier.png`, generated from the untouched E2 simultaneous family in `artifacts/confirmation_e2/results.json`. The historical development `fig9_frontier.png` is retained for provenance but is not used for confirmation.

The measurement resolved the pre-registered dual branch to a **third outcome, stricter than the pre-registered convergence expectation, and reported as measured under the no-soft-reads rule** (the default-to-FAIL discipline of §4.5 applied to wording). At the top budget hier-rich does not converge with flat-long — it loses CI-clean at equal tokens and equal training regime (−0.0131 [−0.0201, −0.0069], a small uniform loss across essentially every horizon × hardness bin); at the bottom budget flat-short beats hier-short (−0.0036 [−0.0063, −0.0010], concentrated at h ≤ 60, null beyond). The frontier therefore *inverts at both ends*: this lands on the first fallback branch at the top (the abstraction cost does not vanish at the top budget — the −0.0131 loss is of the same kind as the diffusion-head abstraction cost of §4.4) and refutes the matched-budget law at the bottom, where the 6–8-frame window is below the memory-relevance scale and coarse slots are redundancy rather than memory. The mechanism reading: hierarchy pays where its coarse tokens buy a memory window that (a) exceeds what fine tokens afford and (b) matters; at window saturation (48 frames, clip-bounded) coarse slots merely displace full-resolution frames, and below the memory-relevance threshold they duplicate it. Sharpness follows the window, not the hierarchy: hier-rich 0.0898 versus flat-long 0.0996, hier-short 0.0392 versus flat-short 0.0432 — at matched saturation the hierarchy does **not** retain sharpness better.

This mechanism reading makes a quantitative prediction, which was pre-registered and then measured the same day (the lower-edge row of Table 4.6a): the win must switch on at the cheapest budget whose coarse span covers the ≈48-frame loop — 48 × 16 coarse tokens plus a minimal fine tail, ≈896 tokens, a budget at which a flat model affords only a 14-frame window. The measured pair confirms the prediction, and more strongly than expected: hier-min beats flat-min in **every horizon × hardness cell** (overall +0.0399 [+0.0299, +0.0501] — the largest hierarchical advantage measured in the campaign), with the visibility-stratified gates showing the reach signature exactly where the mechanism operates (dt ≤ 48 HARD +0.0750; dt 49–96 HARD +0.0282; dt 97–192, beyond the 48-frame span, null). Two structural conclusions follow. First, the win smallest tested dense-family hierarchy win coincides with

the geometric span-threshold arithmetic written before the run: the next smaller dense-family point favors flat, while hierarchy wins at

896. This is a tested ordering rather than a continuous switch. Second, the **largest tested dense-family advantage occurs at 896 tokens**: reducing the budget removes the flat model's window from the memory horizon faster than the hierarchical abstraction cost grows — the efficiency argument is strongest exactly where budgets are tightest, and hier-min at 896 tokens (0.1706) reaches a quality the flat curve attains only at substantially larger budgets in the development curve. The untouched family leaves 1,792 unresolved and confirms flat at 2,432, so no equal-quality multiplier or interpolated crossing is claimed. The pre-registered seed-1 replication confirmed the point: every cell WIN again (overall +0.0358 [+0.0267, +0.0454]) with the same visibility-stratified reach signature — the 896-token 896-token point is therefore a two-seed result (seed-mean Δ ≈ +0.0379, the largest two-seed effect of the campaign), alongside the two-seed 1,152-token pairing of §4.3.3. The §4.3 efficiency claim takes its final, tested-bracket form:** the simultaneous confirmation family supports hierarchy at 896/1,152, leaves 1,792 unresolved, and supports flat fidelity at 2,432/3,072; no interval, exact edge or unique crossing is inferred.**

## 4.4 Forms of Hierarchical Abstraction: Spatial, Temporal and Depth

A hierarchy in a latent video predictor is not one construction but a **family of forms**. A context-token budget can be packaged along four axes: **spatial abstraction** (every frame, at lower resolution), **temporal abstraction** (full resolution, at a coarser tick), **depth** (how many nested levels), and **allocation** (the split between long-span and recent-detail streams). Each shape buys memory span at a different token price and pays a different information cost. Throughout, hierarchy means multi-rate and/or multi-RESOLUTION context — independent axes: a configuration can be hierarchical in tick rate while single-resolution (the §4.4.3 strided control — a two-rate temporal hierarchy in the clockwork/CW-VAE sense, whose levels are likewise temporal) or hierarchical in resolution while single-rate (dense coarse arms); where the two depth counts differ, both are reported (resolution depth / temporal-rate depth). The systems-analysis question is therefore not whether hierarchy helps universally, but **which form pays, where, and why**. Three cells of this grid are measured in earlier sections (allocation — §4.3.6; the spatial factorial — §4.4.1 below; the slow-clock boundary — §4.5); the remaining cells (temporal family §4.4.2, span control §4.4.3, depth §4.4.5) follow the pre-registered program of `SPATIAL_FACTORIAL.md` (final block), and §4.4.5 assembles the grid into one comparison table in the manner of the Clockwork VAE benchmark [13] — the format adopted, the controls tightened (Chapter 1, §1.3; fourth article): matched token budgets, backbone-capacity-neutral depth (levels add tokens to a shared backbone; per-level I/O heads +0.30 M / +1.4% on the 22.19 M predictor + the frozen 1.25 M AE tap, disclosed), foreground-masked metrics.

**Table 4.7-A — What each packaging axis buys and costs.**

| Axis | Form | What it buys | Information or causal cost | Evidence |
|---|---|---|---|---|
| spatial | dense 4×4 coarse stream, stride 1 | 4× reach per token; per-level decodability; dense temporal coverage | −0.008 under diffusion and approximately null under regression at fixed window | §4.4.1, measured |
| temporal | full-resolution anchors, stride 4 | 4× reach per token while retaining detail at anchors | blind intervals between anchors; no cheap dense decodable level | §4.4.3: temporal package wins under diffusion on all five seeds; regression is heterogeneous |
| spatial × temporal | coarse stream, stride 2 | 8× nominal reach per token; cheapest tested loop coverage | couples both information costs and the legacy target-interface change | §4.4.2: +0.0330/+0.0338 at 512 tokens under diffusion |
| depth | additional 2×2 stream | a third representational scale | target-loss allocation, optimization load and top-stream floor are inseparable | §4.4.5: −0.0196 versus two levels at fixed budget, seed 0; appendix-grade |
| allocation | spend on reach before recency/detail | confirmed lower/interior hierarchy operating points | sacrifices full-resolution fidelity after reach is saturated | E2: hierarchy at 896/1,152; 1,792 unresolved; flat at 2,432/3,072 |

### 4.4.1 Spatial Factorial: an Economical, Head-Dependent Verdict

Every hierarchy arm of §4.3 confounds two factors: the coarse spatial resolution of the history, and the longer window its cheaper tokens buy (an accepted external-review finding). The factorial of this section isolates the resolution factor: a window × history-resolution grid at operating point A (shared frozen AE, seed 0, protocol identical to the rollout campaign so all existing records pair exactly), holding recency equal — every coarse-history arm keeps the 6 most recent fine frames, exactly the hier-long design. The hypothesis under test, in its original form: coarse history is a *more abstract, less distracted* memory representation.

**Table 4.7 — Window × history-resolution factorial (fg-MSE ALL over 152 AR frames, cluster CIs; context tokens in parentheses).**

| Window | Full-res history | Coarse history | Paired Δ (full − coarse) |
|---|---|---|---|
| 18 f, diffusion | flat-budget 0.2038 (1152) | spat18 0.2120 (672) | **−0.0082 [−0.0115, −0.0050] LOSS** (short-horizon-concentrated, null by h > 120; sharp 0.058 vs 0.050) |
| 18 f, regression | reg-flat 0.2015 (1152) | spat18-reg 0.2008 (672) | **+0.0008 [−0.0013, +0.0029] null** (h 61–120 HARD +0.0047 WIN; tiny h 1–8 loss) |

| Window | Full-res history | Coarse history | Paired Δ (full − coarse) |
|---|---|---|---|
| 48 f, diffusion | flat-long 0.1564 (3072) | hier-long 0.1848 (1152) | −0.0284 (from Tables 4.5/4.6; 2.7× token gap) |

**Abstraction sufficiency at the fixed 18-frame window (read G-S1) — head-dependent.** For the diffusion head, coarse history is a small, real, short-horizon-concentrated cost: −0.0082 ALL at 42% token savings. For the regression head — the better rollout head (§4.2.3) — coarse history is **statistically free**: +0.0008 [−0.0013, +0.0029], null ALL at the same 42% savings, with the long-horizon HARD bin slightly *better* (h 61–120 HARD +0.0047). The "intrinsically better abstraction" hypothesis is therefore **not supported**; the supported form of the claim is **economic and head-conditional**: spatial abstraction is lossy compression whose token savings buy the memory window, and under the regression head the loss term is ≈0.

**Interaction (read G-S2).** The abstraction cost per saved token is similar at both windows for the diffusion head (≈0.007–0.008 per 480 tokens); there is no evidence the cost shrinks as the window grows.

**Window at fixed representation (read G-S3).** hier-long 0.1848 (W48, 1,152 tokens) versus spat18 0.2120 (W18, 672 tokens): extending the window via coarse tokens buys −0.027 — an order that dwarfs the abstraction cost, the same window-ordering law as Table 4.5.

**Within-family readout (measured arm ordering, not a factorial claim).** spat18-reg (672 tokens) at 0.2008 matches or beats the diffusion flat control flat-budget (1,152 tokens) at 0.2038 — regression + coarse history at 58% of the tokens. At the long window, the complete five-seed factorial below separates a stable diffusion package effect from regression and cross-head training-seed heterogeneity.

**Honest link to the prior spatial-gate null.** This project's earlier per-pixel spatial-gate study on Moving MNIST found that an upper spatial level engaged unstably and added no quality — a clean null for spatial abstraction as an *intrinsic* representational win. The factorial is consistent with that history rather than overturning it: isolated from the window confound, coarse history is not intrinsically better (it costs under diffusion and is at best free under regression); its value is what its cheapness buys. The temporal-only corner (strided full-resolution history), declared an omission in the factorial's pre-registration, is filled at operating point A by the span control of §4.4.3.

**Figure 4.11 — Spatial factorial by prediction head.** *Content:* the 2×2 factorial as a diagram: axes window (18/48) × history resolution (full/ coarse), each cell annotated with arm name, token count, ALL fg-MSE; paired deltas drawn on the edges (resolution edges: −0.0082 diff / +0.0008 reg; window edge: −0.027 at fixed coarse representation), with the combined-optimum path highlighted. *Source:* `docs/figures/fig4_claim3_factorial.png`; deterministic source `scripts/make_figures.py`. The figure does not promote the combined arm ordering to a training-seed-robust head interaction.

### 4.4.2 Temporal Family and the Moving Tested Boundary

Span-per-token provides an explanatory calculation, not an exact edge law. The dense coarse configuration first favors hierarchy at the tested 896-token point. The stride-2 coarse configuration (`hier-s2-min`, 24 slots + 2 fine) purchases the same 48-frame reach at **512 tokens** and wins on both development seeds (+0.0330/+0.0338) and on the untouched E2 holdout (+0.0340/+0.0381). Its reach-band effects are also positive on both E2 seeds. This is strong evidence for the complete stride-2 configuration and is consistent with the span calculation.

It is not a pure context-stride intervention: the legacy arm also changes target slots, target-token count, auxiliary supervision and total sequence length. Therefore the result is not worded as an exactly moved lower edge. A future stride dose-response, including any 320-token point, would require a fixed target interface and a new preregistration; it is not part of the current closure portfolio. The measured regression seed-0 B1 result (+0.0639 [+0.0516,+0.0763]) supports the configuration but does not remove this confound. **Figure 4.13:** `docs/figures/fig13_family_edges.png`; the caption treats B1 as a configuration comparison and does not draw a continuous edge or unmeasured point.

### 4.4.3 Coarse Level or Reach Alone? Sparse Full-Resolution Control

The control `strided-flat` keeps NO coarse level: 11 full-resolution slots at stride 4 (span 48)

- 6 recent fine = 1,088 tokens — 64 short of the 1,152 pairing, a deficit that *favors the

control*. The complete five-seed factorial returns the following package effects (positive means the temporal package wins; negative interaction means regression shifts toward spatial coarsening):

| Seed | Diffusion temporal advantage | Regression temporal advantage | Head × package interaction |
|---|---|---|---|
| 0 | +0.0197 [+0.0143, +0.0252] | +0.0107 [+0.0062, +0.0158] | **−0.0089 [−0.0151, −0.0025]** |
| 1 | +0.0046 [+0.0007, +0.0084] | −0.0104 [−0.0146, −0.0062] | **−0.0150 [−0.0199, −0.0104]** |
| 2 | +0.0111 [+0.0070, +0.0151] | −0.0208 [−0.0270, −0.0150] | **−0.0319 [−0.0401, −0.0245]** |
| 3 | +0.0075 [+0.0037, +0.0114] | +0.0248 [+0.0181, +0.0323] | **+0.0173 [+0.0106, +0.0245]** |
| 4 | +0.0060 [+0.0027, +0.0092] | −0.0166 [−0.0229, −0.0108] | **−0.0226 [−0.0297, −0.0160]** |

Diffusion favors temporal subsampling on all five seeds (mean +0.0098, SD 0.0060, t95 [+0.0023, +0.0173]). Regression has two temporal and three spatial material winners (mean −0.0024, SD 0.0195, t95 [−0.0266, +0.0217]). The head interaction changes sign and has mean −0.0122 with t95 [−0.0353, +0.0109], so the preregistered robust negative-shift rule fails. The clip-cluster intervals condition on each trained model; the t intervals use training seeds as units. The measured statement is that, at this operating point, the diffusion head consistently uses temporally sparse full-resolution packaging,

whereas the regression head is training-seed dependent—not that one package is universally preferred or that a directional head law holds.

All cells in this factorial use the existing adaptively used evaluation pool; E4a/E4b is seed-resolved development evidence, not a second untouched confirmation.

**Figure 4.12 — Head × package heterogeneity across training seeds.** *Content:* five seed-resolved diffusion, regression and interaction effects. Clip-bootstrap intervals condition on each trained model; the shaded t intervals use the five training seeds as the inferential units. *Source:* `docs/figures/fig17_head_package_seed_heterogeneity.png`, generated directly from `artifacts/e4b_head_package/results.json`.

Three binding scope qualifiers (mirror of the article §4.4.3): (1) **static world** — nothing changes between stride-4 anchors, so the testbed is maximally kind to temporal subsampling; on data with inter-anchor dynamics the stride aliases what a dense stream tracks—this is an operating-point result, not a general law; (2) the abstraction cost is a property of this autoencoder's 4×4 layout code (frozen-AE discipline fixes the choice); (3) the winning control is itself a hierarchy in the RATE sense (two tick rates, one resolution — the clockwork quadrant), so the measured statement is that the multi-rate form beats the multi-resolution form at this operating point under diffusion, not that hierarchy is unnecessary.

### 4.4.4 What Each Level Preserves: Blinding Probes

Rerun the trained arm's full 152-frame rollout with one context stream **blinded** (zeroed in standardized latent space) at every step. **MEASURED (2026-07-10): asymmetric essentiality.** Blind-coarse = the predicted retrieval signature (hard re-entry +0.073 vs h1–8 +0.022; ALL 0.2236 past flat-budget) ✓; blind-fine = renderer collapse, expectation partially wrong (h1–8 0.2427; sharp 0.0086 below the structural threshold; ALL below persistence — brittleness-dominated per the binding OOD caveat); blind-top (hier3) = +0.0035 ALL only — the top stream carries almost nothing usable, independent convergent evidence for the §4.4.5 mechanism reading. Report-only. Binding interpretation caveat (per the pre-registered amendment): a zeroed stream is an out-of-distribution input, so the collapse magnitude bounds the stream's contribution from below confounded with OOD brittleness from above — the probe licenses “stream removed → capability lost,” not a precise information accounting. **Figure 4.14:** `docs/figures/fig14_level_blinding.png`; measured eval-only blinding curves with the out-of-distribution qualification retained.

### 4.4.5 Consolidated Comparison: Depth, Heads and the Assembled Grid

**Table 4.7-B — Consolidated context-shape grid (152-step autoregressive rollout; `ALL` foreground MSE, lower is better).**

| Context budget | Resolution / temporal depth | Temporal factor | Spatial factor | Head | Configuration | Reach | ALL fg-MSE |
|---|---|---|---|---|---|---|---|
| 512 | 1 / 1 | 1 | 1 | diffusion | `flat-short` | 8 | 0.2215 (2 seeds) |
| 512 | 2 / 2 | 2 | 2 | diffusion | `hier-s2-min` | 48 | 0.1881 (2 seeds) |
| 512 | 2 / 2 | 2 | 2 | regression | `hier-s2-min-reg` | 48 | 0.1566 (1 seed) |
| 896 | 1 / 1 | 1 | 1 | diffusion | `flat-min` | 14 | 0.2104 (2 seeds) |
| 896 | 2 / 1 | 1 | 2 | diffusion | `hier-min` | 48 | 0.1725 (2 seeds) |
| 1,088 | 1 / 2 | 4 | 1 | diffusion | `strided-flat` | 48 | 0.1644 (2 seeds) |
| 1,088 | 1 / 2 | 4 | 1 | regression | `strided-flat-reg` | 48 | 0.1590 (5 seeds) |
| 1,152 | 1 / 1 | 1 | 1 | diffusion | `flat-budget` | 18 | 0.2042 (3 seeds) |
| 1,152 | 2 / 1 | 1 | 2 | diffusion | `hier-long` | 48 | 0.1762 (3 seeds) |
| 1,152 | 1 / 1 | 1 | 1 | regression | `reg-flat` | 18 | 0.2009 (5 seeds) |
| 1,152 | 2 / 1 | 1 | 2 | regression | `reg-long` | 48 | 0.1592 (5 seeds) |
| 1,152 | 3 / 1 | 1 | 2×2 | diffusion | `hier3` | 48 | 0.2044 (1 seed) |
| 1,792 | 1 / 1 | 1 | 1 | diffusion | `flat-c1` | 28 | 0.1876 (1 development seed) |
| 1,792 | 2 / 1 | 1 | 2 | diffusion | `hier-c1` | 48 | 0.1799 (1 development seed; E2 simultaneous result unresolved) |
| 2,432 | 1 / 1 | 1 | 1 | diffusion | `flat-c2` | 38 | 0.1703 (B16×30k) |
| 2,432 | 2 / 1 | 1 | 2 | diffusion | `hier-c2` | 48 | 0.1820 (B16×30k) |
| 3,072 | 1 / 1 | 1 | 1 | diffusio | `flat-` | 48 | 0.1572 (B8×60k; 2 seeds) |

| Context budget | Resolution / temporal depth | Temporal factor | Spatial factor | Head | Configuration | Reach | ALL fg-MSE |
|---|---|---|---|---|---|---|---|
| | | | | n | long | | |
| 3,072 | 2 / 1 | 1 | 2 | diffusion | hier-rich | 48 | 0.1691 (B8×60k; 2 seeds) |

*Notes.* Multi-seed values are means; exact seed effects and intervals appear in the corresponding subsections. `hier-s2-min` also changes the target interface and is therefore a whole-configuration effect. `strided-flat` has 1,088 context tokens and `hier-long` 1,152: the comparison is near matched, not exact. The machine-generated grid with sequence length, throughput, VRAM and sharpness is stored in `docs/CWSTYLE_TABLE.md`.

The depth row `hier3` (2×2/4×4/8×8 factor-2 pyramid, 48 top + 24 coarse + 9 fine = the same 1,152 tokens) tests a task with a single memory scale and a protocol-capped 48-frame history. **MEASURED (2026-07-10, s0) — the pre-registered ALTERNATIVE (loss) branch:** hier3 ALL **0.2044** / sharp 0.0564 (class-lowest): CI-clean loss to hier-long (**−0.0196 [−0.0248, −0.0146]**), null vs flat-budget (−0.0007) — the third stream erases the two-stream win at fixed budget. Three inseparable disclosed mechanisms: loss-weight dilution (per the amendment; the class-lowest sharpness is its predicted signature), fixed-step optimization load, and the top-stream information floor (2×2 = 32-px cells vs ~20-px digits, with the coarse stream cut 48→24 to pay for the top). It is reported as a depth-plus-loss-allocation probe that costs approximately 0.02 at fixed budget, not as evidence that depth is intrinsically harmful; the depth hypothesis stands (single-scale data cannot reward a third scale — consistent with the per-pixel-gate null, the §4.5 boundary, CW-VAE's two-levels-sufficed observation). Disposition: appendix-grade; a weight-matched follow-up would be required for any causal depth statement. **Figure 4.15:** `docs/figures/fig12_cwstyle_curves.png`, degradation curves over the 152-frame rollout for the measured configuration grid.

## 4.5 Operating Point B

Operating point A proves *efficiency at matched budget*: the loop (≈48 frames) sits inside the affordable flat window (48 frames), so flat-long can always buy the result with tokens. The boundary study asks the capability question on the pre-registered design frozen before any launch (the `docs/HGRID_DESIGN.md` preregistration trail is included in the reproducibility appendix): in a regime where **no affordable flat window covers the memory horizon** (hgrid3, effective loop ≈196–213 frames), does temporal abstraction — a strided, slow-rate coarse level — deliver what flat context cannot? The seed-0 study closed with a **negative rollout outcome under a binding pre-committed decision rule**. The registered

E5a repetition then retained the genuine single-step positive and the early self-fed benefit, but did not reproduce the late inversion at either new training seed. This section therefore reports a negative replication and three diagnosed methodology lessons, not a trust boundary or capability law. The §4.3 matched-budget result is unaffected under every branch of this study, and remains the headline of the dissertation.

### 4.5.1 Aliasing Episode: Diagnosed Failure, Not a Discarded Run

The first dataset iteration at the wide operating point (*hgrid*: world 768 px, with background wavenumbers scaled ×4 so that the frozen loopmom autoencoder could be reused) produced a dead-flat null at the 20,000-step single-step pre-check (HARD −0.0008 [−0.0027, +0.0012]; all dt bands null) and a CI-clean **LOSS** at the budget-matched 40,000-step extension (HARD −0.0025 [−0.0042, −0.0008]; ALL −0.0012; dt > 192 −0.0026): more optimization made the unaddressable memory slots a pure attention cost. The cause is established, not speculative: the ×4 wavenumber scaling made the background **exactly 192-px periodic**, so the four aliased viewport positions are pixel-identical — the crop test measures pairwise MSE **0.000000** between them. Retrieval lost the absolute-position landmark it plausibly used at operating point A (where unit-wavenumber components made positions unique). The autoencoder-reuse constraint *caused* the failure and was dropped (rule adopted: a fresh autoencoder per testbed variant; within-grid comparability needs only one shared autoencoder per campaign). The second iteration (hgrid2, a DC brightness cue) was aborted mid-run in favor of the mechanistically stronger variant; two position-probe attempts from that period failed their own positive controls and are **retracted on the record**. The third iteration, **hgrid3** (unscaled background with unit wavenumber forced), restores position uniqueness (crop-uniqueness 0.004–0.010, versus hgrid2's 0.0006–0.003) and passes the full regime verification of §4.1.1.

### 4.5.2 Geometry: Memory Reach versus Effective Loop Period

The pre-check arm carries 48 coarse slots at stride 4 — a nominal memory span of 192 frames. The relevant horizon, however, is not $\mathrm{world}/\mathrm{speed} = 192$ but the **effective loop**

$$L_{\mathrm{eff}} \;=\; \frac{\mathrm{world}}{\mathrm{speed} \cdot (1 - p_{\mathrm{hold}})} \;\approx\; \frac{768}{4 \cdot 0.9} \;\approx\; 213 \ \text{frames}, \tag{4.1}$$

because hold frames advance time without advancing the pan. The 48-slot arm's 192-frame span therefore falls *short* of its own target loop (≈196–213 measured) — a geometry error inherited from the operating-point-A intuition, and the motivation for the span-224 arm of §4.5.4.

### 4.5.3 Recall Exists: Extended Evaluation and Internal Control

The 4×-widened paired re-evaluation of the same hgrid3 models (evaluation batches 24 → 96; pre-registered before running; both reads disclosed) flips the underpowered nb = 24 nulls to CI-clean wins

for the strided arm — **48-slot stride-4 (reach 195 frames): HARD +0.0016 [+0.0007, +0.0025] WIN; REACHABLE-HARD +0.0021 [+0.0008, +0.0036] WIN; dt 49–96 +0.0032 WIN; ALL +0.0005 WIN** — while the contiguous-48 positive-control arm (operating point A's exact winning design, but with reach only 51 frames here) is **null in every cell**. The effect appears exactly where the memory reach covers the re-entry interval and vanishes where it cannot: a reach dose-response, serving as the internal control. Two readings attach. First, beyond-flat-window retrieval on the position-unique wide world **exists** but is ≈7× smaller than at operating point A (+0.002 vs +0.0135 absolute; ≈2% vs ≈13% relative) — consistent with the span shortfall of §4.5.2 diluting it and with the harder addressing problem at this scale. Second, the nb = 24 pre-checks were underpowered for effects of this magnitude, not wrong about the aliased hgrid (whose 40k read was a CI-clean LOSS; the aliasing verdict stands).

### 4.5.4 Reach-Threshold Hypothesis and Preregistered Single-Step Test

The pre-approved span-fix shot — 56 slots at stride 4, span 224 > loop, 40,000 steps, evaluation width 96 — returned, on a single seed: **HARD +0.0273; REACHABLE-HARD +0.0363; beyond-227 null**, with the effect confined to the hard-reachable cells (easy, mid, and beyond-reach all null). Taken at face value this is the threshold signature — an order-of-magnitude jump once the span exceeds the effective loop, localized exactly where reach predicts — but it was explicitly recorded as *"threshold HYPOTHESIS — systematic-or-lucky, undetermined"* (single seed, no same-pool 48-vs-56 comparison, measured against the weakest baseline). The resolution rule was written before the disambiguation runs, converged with two external reviewers, and was binding; it is quoted verbatim in the pre-registration appendix: R1 (replication on an independent seed, CI > 0), R2 (reach causality: the strided arm must beat a contiguous arm with *identical* token and slot counts, reach 227 vs 59, CI > 0), R3 (informative, non-gating: the 48-slot arm at seed 1), with PASS = R1 AND R2 gating the decisive rollout pair, any uncovered outcome defaulting to FAIL, and FAIL closing the study with no further capability spend.

**Single-step outcomes (the characterized positive).** R1 passed: the span-224 arm beats its baseline on an independent seed — REACHABLE-HARD +0.0134 [+0.0107, +0.0161], HARD +0.0089 [+0.0069, +0.0109] — at 2.7× smaller magnitude than seed 0 (+0.0363 / +0.0273; seed mean ≈ +0.025 reachable): replicable but seed-noisy. R2 passed: the strided arm beats the contiguous same-token control at both seeds (differential-reach band +0.0374 [+0.0334, +0.0416] seed 0; +0.0144 [+0.0118, +0.0171] seed 1). R3: the span-192 arm stays ≈flat at both seeds (+0.0004; +0.0011), and the contiguous arms never help at either seed. The mechanism *attribution* is seed-dependent and is not over-read: at seed 1 the win concentrates in the reach-differential bands exactly as a per-sample reach-causal account predicts (shared +0.0053 vs differential +0.0181; versus the contiguous control, shared null −0.0005 vs differential +0.0144), but at seed 0 it is uniform across shared and differential bands (+0.0343 vs +0.0366) — a circuit-level rather than per-sample-reach signature. The reliable two-seed single-step statement is therefore: **a memory span exceeding the effective loop reliably enables retrieval learning**

**(2/2 span > loop runs CI-positive versus 0/4 span ≤ loop runs at ≈0), while the magnitude and per-sample usage pattern of the learned circuit vary by seed.** Disclosure: the single-step gate CIs in this subsection are sample-iid, not clip-clustered; the load-bearing cells survive a conservative 2–3× design-effect widening, and a clip-clustered recomputation is reported alongside. The teacher-forced caveat is stated where it belongs: the single-step study conditions on ground-truth context, and the rollout pair of §4.5.5 is the real test.

### 4.5.5 Decisive Rollout Pair and Registered Negative Replication

The pre-registered decisive pair (the span-224 strided arm versus the long-context diffusion baseline on hgrid3; warm-up 224 frames; uniform B = 8 × 40k training) returned **RED**: over 256 self-fed rollout frames the pre-registered gate cells are null — dt 49–96 HARD +0.0020 [−0.0071, +0.0111]; dt 97–192 −0.0028 [−0.0088, +0.0035]; dt > 192 −0.0018 [−0.0065, +0.0027] — so the seed-0 programme did not establish rollout capability on its registered gate. Its horizon profile nevertheless suggested an early hierarchy benefit followed by a late inversion.

E5a preregistered this early-positive/late-negative pattern and repeated the same pair at training seeds 1 and 2. Positive effects favor the hierarchy: seed 0 is +0.0204 [+0.0133,+0.0279] at h 1–24 and −0.0064 [−0.0095,−0.0032] at h 61–256; seed 1 is +0.0082 [+0.0035,+0.0133] and +0.0074 [+0.0038,+0.0112]; seed 2 is +0.0230 [+0.0150,+0.0315] and +0.0087 [+0.0035,+0.0139]. Thus the early benefit is CI-clean at all three seeds, but neither new seed reproduces the late-negative direction. The registered verdict is **NOT_REPLICATED**, and the gated dose-response is not authorized. Addressability and self-fed reliability remain distinct constructs, but the present study does not establish a stable approximately-24-frame trust boundary. The §4.3 matched-budget result is unaffected.

**Figure 4.16 — Registered negative replication of the self-fed boundary.** *Content:* seed-resolved paired Δ (strided − baseline) versus rollout horizon with cluster-CI bands: early hierarchy benefit at seeds 0–2, late inversion at seed 0 only, and positive late effects at seeds 1–2; the registered primary windows are highlighted. *Source:* `docs/figures/fig18_trust_negative_replication.png`, generated from `artifacts/e5a_trust_replication/results.json` and independently validated E5a manifests.

### 4.5.6 Methodological Lessons from the Boundary Study

1. **The effective-loop formula.** A memory mechanism's span must be sized against $L_{\mathrm{eff}} = \mathrm{world}/(\mathrm{speed} \cdot (1 - p_{\mathrm{hold}}))$ (eq. 4.1), not $\mathrm{world}/\mathrm{speed}$: hold frames consume slots without advancing the loop. The 48-slot arm's 192-frame span silently fell ≈2–10% short of its own target (≈196–213), plausibly diluting the effect it was built to detect.

2. **The autoencoder-reuse trap.** Reusing a frozen autoencoder across testbed variants forced a

background transformation (×4 wavenumber scaling) that made the world exactly viewport-periodic and destroyed the positional landmarks retrieval depends on — a dataset-side confound that manifested as a *mechanism* failure (memory slots scoring as pure attention cost). Rule adopted: a fresh autoencoder per testbed variant; share one autoencoder only within a campaign.

3. **Evaluation power.** The nb = 24 pre-checks were adequately powered for operating-point-A magnitudes (+0.0135) but not for the ≈+0.002 effects of the harder regime; the pre-registered 4× widening flipped nulls to CI-clean wins on the *same* checkpoints. Reading a null requires a power argument against the plausible effect size, not just a confidence interval spanning zero — and both the underpowered and the widened reads are disclosed.

## 4.6 Generalization across Standard and Synthetic Datasets

The studies of §§4.2–4.5 select a configuration and delimit it on the purpose-built testbed family; this section carries that configuration outward — to the standard benchmark of the field and to the synthetic suites on which the reversible-transformation method was published ([A2], §4.1.2) — so that the dissertation's two experimental programs meet on shared ground. Scope discipline, stated once: **§§4.6.2 and 4.6.5 are measured**; the 2D-rotation, 3D-polycube, MMNIST-CIFAR and KTH extensions in §§4.6.3, 4.6.4 and 4.6.6 are future protocols and contribute no projected number or performance claim.

### 4.6.1 Selected Configuration by Inference Regime

The recipe transferred below is not a preference but the measured selection of §§4.2–4.4, and it is regime-dependent, exactly as §4.2.5 concluded. For the **deterministic best-of-1 autoregressive rollout** — the deployment regime of every long-horizon claim — the selected head is **regression** (it wins the rollout reversal, §4.2.3, and the memory win replicates under it, §4.3.5), combined with a context package selected to cover the relevant span with the least measured information loss. The coarse-plus-fine package (long coarse memory + recent fine window) is the replicated spatial option, whose coarse history is statistically free under this head (+0.0008 [−0.0013, +0.0029], Table 4.7), at the **largest affordable window** — window extension at fixed representation dwarfs the abstraction cost (−0.027 versus −0.008, read G-S3). The strongest replicated spatial recipe is reg-long (seed mean 0.1593 at 1,152 tokens, §4.4), while the cross-family temporal option remains seed/head-gated. For the **single-step best-of-N coverage regime**, the selected head is **diffusion** (it wins every column at matched capacity and NFE, Table 4.2), with the same hierarchical context rule. Transfers below use the rollout recipe as primary and report the diffusion head where sampling diversity is part of the question; where a measured transfer predates this formulation, its actual arm configuration is disclosed in place (§4.6.2 used the diffusion head, best-of-1).

### 4.6.2 OpenSTL Moving-MNIST Bridge: Measured Negative Control

The bridge announced in §4.1.3 is measured. Protocol: the canonical OpenSTL 10→10 suite on the canonical test set (n = 10,000 sequences), with bit-verified reimplementations of the OpenSTL metric conventions (pixel-sum MSE/MAE) and a faithful port of the standard sequence generator; a fresh two-level autoencoder trained for this benchmark (per the §4.5.6 rule: a fresh autoencoder per testbed variant); the unmodified hierarchical DiT backbone of §3.2.2 (22.19 M parameters), trained B = 16 × 15,000 steps, bf16, evaluated best-of-1 with three autoregressive window steps (4 + 4 + 2 frames). Two arms: **mm-flat** (10 fine frames of context, 640 tokens) and **mm-hier** (10 coarse + 6 fine, 544 tokens).

**Table 4.8 — OpenSTL Moving-MNIST bridge (10→10, canonical test set, n = 10,000; pixel-sum MSE/MAE per the OpenSTL convention).**

| Arm | MSE | MAE | SSIM | PSNR | LPIPS |
|---|---|---|---|---|---|
| mm-flat (10 fine frames, 640 tokens) | 180.91 | 229.94 | 0.7625 | 13.55 | **0.1320** |
| mm-hier (10 coarse + 6 fine frames, 544 tokens) | 181.29 | 230.45 | 0.7607 | 13.54 | 0.1389 |
| AE round-trip ceiling | 3.30 | 20.79 | 0.9917 | 30.93 | 0.0062 |
| Persistence | 286.28 | 337.89 | 0.6528 | 11.56 | 0.2533 |

Leaderboard reference (deterministic specialist regressors at ≈200 epochs, OpenSTL documentation): ConvLSTM-S 29.80, PredRNN 23.97, PredRNN++ 22.06, SimVP+gSTA-S 26.69, TAU 24.60 aggregate MSE [74] (range parity with article Table 5.1; the strongest specialist, PredRNN++ 22.06, sets the quoted floor).

Three readings, in the order of their epistemic weight.

1. **The protocol anchor is achieved.** The harness reproduces the standard benchmark protocol exactly (canonical test sequences, bit-verified metric conventions), and the autoencoder ceiling row (3.30 against the arms' ≈181) shows the tokenizer is not the limiter: the gap to the specialists is owned by the dynamics model and objective, not the representation.

2. **The absolute aggregate numbers are modest, with disclosed reasons — and the "mid-pack aggregate MSE" clause of the §4.1.3 pre-registered expectation is refuted and reported as such. **The measured 180.91 stands against the specialists' 22.06–29.80: the evaluated arm is a 22 M stochastic latent-diffusion *generalist* at ≈15,000 optimizer steps, evaluated best-of-1, against deterministic pixel-regressors trained roughly an order of magnitude longer and optimized exactly for the pixel-sum-MSE aggregate — an aggregate that structurally punishes sampling variance (§3.4.3: on this ≈93%-black benchmark, aggregate metrics reward mean prediction). Accordingly,** no statement of the

form "competitive with, or beating, the OpenSTL leaderboard on aggregate MSE" appears anywhere in this dissertation** — the measurement refutes it. What does hold: the model halves persistence's LPIPS (0.1320 vs 0.2533) and beats persistence on every metric of the table.

3. **The flat ≈ hier null (Δ ≈ 0.4 aggregate MSE ≈ 0) is the measured short-protocol boundary,

not a failure.** Moving MNIST has no off-screen re-entry at 10→10, so this protocol does not expose the re-entry-defined demand isolated at operating point A. It does not prove that persistent identity/velocity is irrelevant over long rollout; a valid persistent-object study would test that separately. Thus the bridge supplies a short/no-re-entry boundary observation, while the matched-budget hierarchy win is two-seed CI-clean (§4.3.3). The question "when does hierarchy pay" receives its negative control from the standard benchmark itself.

**Historical Phase C preregistration (2026-07-10; before any result).** The original long-horizon design registered `mmL-flat` and `mmL-hier` at seeds 0/1/2, a 48-frame warm-up and approximately 300 self-fed frames. Its continuity primary was paired foreground-MSE: replicated practical equivalence required every seed's complete clip-cluster interval to lie inside ±0.005, with horizon bins secondary. The design also proposed digit identity, trajectory and count endpoints with preliminary margins ±0.02 accuracy, ±1 pixel and ±0.05 digit/frame. A sound positive hierarchical effect was to be examined and reported. **The study did not launch.**

**Validity amendment (2026-07-11; before launch).** The registered foreground-MSE question is retained narrowly. The legacy factor evaluator inferred identity from horizontal component order, merged overlaps, used many-to-one trajectory matching and intersected frames scorable in both arms; those choices cannot support identity, trajectory, coverage or equivalence conclusions. The corrected hybrid protocol stores persistent object/glyph IDs, official train/test glyph splits, boxes and per-object trajectories; uses one-to-one Hungarian assignment; penalizes misses and extras; reports coverage; validates the digit classifier on rendered, overlapping and autoencoder-reconstructed crops; and justifies native-unit margins. The original registration and this amendment are both retained in the scientific record. The conditional protocol is documented in `docs/PHASE_C_V2_PROTOCOL_2026-07-11.md`; no result or figure enters the dissertation until its CPU, classifier and protocol-freeze gates pass.

### 4.6.2a Persistent-object long Moving-MNIST: a measured external boundary

The corrected E6 protocol passed every CPU, codec, classifier, production-crop, data-integrity and protocol-freeze gate before protected evaluation. The frozen comparison used `mmL-flat` and `mmL-hier` at training seeds 0/1/2, a 48-frame ground-truth warm-up, 300 self-fed target frames and 100 protected test clips. Both arms used the same 1,152 context-token budget, predictor regime, codec, sampler and evaluation geometry. The generator saved persistent object and source-glyph identity, boxes and trajectories; the evaluator used one-to-one Hungarian assignment, explicit missed/extra-object

penalties and intention-to-score coverage. The official train/test glyph split and classifier-domain floors were verified before the holdout was opened.

**Table 4.8a — E6 persistent-object long Moving-MNIST.** Positive effects favor hierarchy; all intervals belong to one simultaneous 95% max-statistic clip-cluster family over the 15 co-primary effects.

| Seed | Foreground MSE | Union MSE | Trajectory error, px | Identity accuracy | Object coverage |
|---|---|---|---|---|---|
| 0 | +0.0036 [−0.0074,+0.0146] | +0.0102 [+0.0019,+0.0185] | +2.4786 [+0.2130,+4.7443] | −0.0762 [−0.1106,−0.0418] | +0.0304 [+0.0016,+0.0591] |
| 1 | −0.0031 [−0.0127,+0.0065] | +0.0122 [+0.0052,+0.0192] | +1.4223 [−1.0646,+3.9092] | −0.0561 [−0.0934,−0.0188] | +0.0159 [−0.0163,+0.0482] |
| 2 | −0.0024 [−0.0131,+0.0083] | +0.0169 [+0.0093,+0.0245] | +4.1325 [+1.5271,+6.7380] | −0.0647 [−0.0955,−0.0338] | +0.0498 [+0.0161,+0.0834] |

The preregistered verdict is **MIXED_BOUNDARY**, not equivalence and not a null result. Union-mask error favors hierarchy on all three seeds. Trajectory and coverage favor hierarchy cleanly on seeds 0 and 2 and are directionally positive but unresolved on seed 1. Missed-object rate, full-frame error, background error and foreground F1 also improve for hierarchy at all seeds as secondary descriptive endpoints. Digit identity, however, favors flat context with simultaneous intervals wholly below zero at all three seeds; foreground-only MSE is unresolved and changes sign.

This division is consistent with the dissertation's systems interpretation. The coarse stream retains spatial support economically—where an object is, whether it persists and how its motion continues—but discards fine glyph information needed to preserve what the object is. Flat full-resolution storage retains that identity fidelity better. Thus E6 extends, but does not universally generalize, the LoopMom reach–fidelity frontier. It is a controlled synthetic external boundary, not evidence of natural-video transfer.

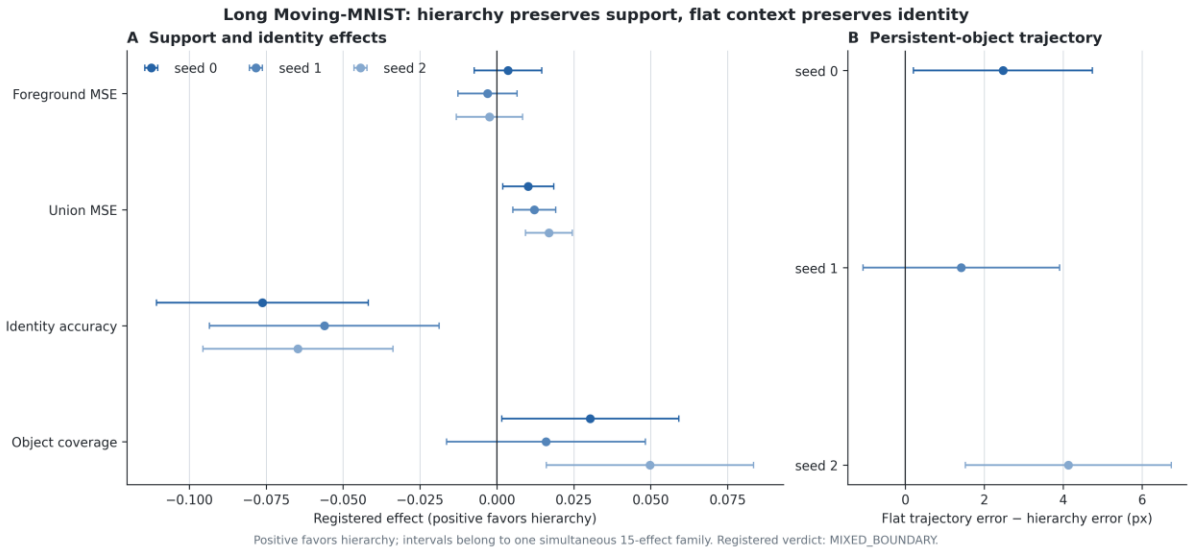



**Figure 4.22 —** Persistent-object long Moving-MNIST simultaneous effects. Positive favors hierarchy; support/continuity and fine identity fidelity separate.

### 4.6.3 Synthetic [A1]/[A2] Testbeds: 2D Rotations and 3D Polycubes

Two distinct published sources motivate this future bridge. [A2] supplies the $64 \times 64$ procedural 2-D rotation suite and its controlled temporal laws; [A1] supplies the V-BLT 2-D-polygon and 3-D-polycube sequences with entropy maps. Neither publication evaluates the Chapter 3 reach–fidelity model. A future transfer study would therefore ask whether the selected hierarchical configuration retains accuracy and consistency on these source-faithful suites, using each publication's exact split, metrics, horizon and baselines. No such transfer has been executed, so no result table or projected advantage is inserted here. The published examples remain documented in Chapter 2, Figures 2.3–2.4.

### 4.6.4 Moving-MNIST-CIFAR

The MMNIST-CIFAR variant replaces the black background with natural-image texture, removing the background degeneracy that makes aggregate metrics mean-seeking on standard Moving MNIST (§3.4.3) while keeping the same no-re-entry kinematics — so the pre-registered expectation of the §4.6.2 negative control (flat ≈ hier) carries over, now with more informative aggregate metrics. This extension is not required to establish the present reach–fidelity result and has not been executed; no numerical outcome is implied.

### 4.6.5 TaxiBJ: Measured Real Spatio-Temporal Forecasting Data

TaxiBJ is the one *real-data* suite of the transfer set: Beijing taxi in/out-flow grids, $32 \times 32 \times 2$ (two flow channels), 4 → 4 frame prediction under the standard OpenSTL protocol, scored by the benchmark's own metric implementation invoked verbatim (the normalization convention — data

rescaled to $[0,1]$ by the benchmark's loader — was verified in the OpenSTL source before any model number was read). Instead of the anticipated dedicated 32-px model variant, the transfer used the **exact-upsample equivalent** (disclosed, registered before the runs): frames nearest-upsampled ×2 through the *unchanged* 64-px two-level autoencoder (2-channel input) and WindowDiT stack of Chapter 3, predictions downsampled back by the exactly inverse 2×2 mean before scoring — an information-identical formulation that keeps the architectural constants of §3.2 literally unchanged across every dataset of the study. Regime: fresh autoencoder (12,000 steps), four arms at batch 64 × 15,000 steps (≈48 epochs, comparable to the leaderboard's 50-epoch protocol), best-of-1, unconditional, seed 0.

**Table 4.9 — TaxiBJ 4→4, MEASURED (n = 500 canonical test pairs; OpenSTL convention; leaderboard rows = OpenSTL's own 50-epoch retraining, locally verified).**

| Arm | MSE | MAE | SSIM | PSNR |
|---|---|---|---|---|
| **Flat, regression (256 ctx tokens)** | **0.3642** | **14.96** | 0.9781 | **39.83** |
| Hierarchical, regression (128 ctx tokens) | 0.4170 | 15.44 | 0.9772 | 39.64 |
| Hierarchical, diffusion (128 ctx tokens) | 0.4986 | 18.37 | 0.9692 | 37.87 |
| Flat, diffusion (256 ctx tokens) | 0.5152 | 18.44 | 0.9686 | 37.85 |
| AE round-trip ceiling (2-ch) | 0.0192 | 4.48 | 0.9987 | 50.92 |
| Persistence | 3.8268 | 48.63 | 0.7853 | 30.20 |
| TAU (OpenSTL, 50 ep) | 0.3108 | 14.93 | 0.9848 | 39.74 |
| MIM (OpenSTL, 50 ep) | 0.3110 | 14.96 | 0.9847 | 39.65 |
| PredRNN (OpenSTL, 50 ep) | 0.3194 | 15.31 | 0.9838 | 39.51 |
| SimVP+gSTA-S (OpenSTL, 50 ep) | 0.3246 | 15.03 | 0.9844 | 39.71 |
| PhyDNet (OpenSTL, 50 ep) | 0.3622 | 15.53 | 0.9828 | 39.46 |

The measured readings, stated per metric and without a leaderboard-superiority headline (the §4.2.3 rule): the regression arm of the *unchanged generalist stack* lands **inside the leaderboard band on MAE (14.96 — tying MIM for second-best of the quoted rows) and on PSNR (39.83 — above every quoted row)**, at the band's lower edge on MSE (0.3642, 0.55% above PhyDNet), and below the band on SSIM (0.9781 vs 0.9828–0.9848). Persistence is beaten ≈10.5×; the autoencoder ceiling (0.0192, 16× below the best specialist row) shows the frozen tokenizer is not the limiter — the pre-registered decoder-capacity fallback branch is measured-moot. The regression-over-diffusion best-of-1 ordering replicates on real data (0.364/0.417 vs 0.515/0.499), and the no-memory-horizon boundary replicates as pre-registered: at half the context tokens the hierarchical regression arm pays −0.053 MSE with faster per-

horizon degradation (4-frame protocol → coarse history is pure token economy). Disclosure: single seed (a transfer row, not a claim-bearing contrast, per the campaign's two-seed rule).

### 4.6.6 KTH-64 — Future Extension

KTH (human actions, grayscale, evaluated at 64 × 64) is a natural-video extension named in the research prospects; it is not an [A2] result. It is retained as a future external-validity test, not as an optional empty result shell.

### 4.6.7 Boundaries of the Measured Generalization Claim

The measured generalization claim is deliberately narrow. On OpenSTL Moving MNIST the package comparison is a no-re-entry negative control and does not match specialist aggregate MSE. On TaxiBJ the unchanged generalist stack is competitive on some reported metrics, but the hierarchical package loses to the larger flat regression context and only one training seed was tested. The remaining synthetic and natural-video suites are future work. Accordingly, **a claim of beating (or matching) the OpenSTL leaderboard on aggregate MSE is never made** — Table 4.8 refutes it (180.91 versus SimVP's 26.69), and the honest statement of that comparison is the disclosed-generalist reading of §4.6.2.

### 4.6.8 Independent Confirmation on an Untouched LoopMom Holdout

The original test population informed qualitative monitoring and configuration selection. Accordingly, the central reach claims were evaluated once on the newly generated, hash-pinned holdout `loopmom-confirmation_holdout-20260712-e033abee6e04`. Before first access, the protocol froze 26 checkpoint evaluations, two primary pairs, their geometric reach bands, the complete metric suite and one studentized max-statistic clip-cluster family for 12 correlated frontier contrasts. No model was trained or selected on this holdout.

For `hier-min` versus `flat-min`, the registered `14 < dt <= 48` effects are +0.0980 [0.0788, 0.1179] at seed 0 and +0.0914 [0.0736, 0.1098] at seed 1. For the B1 configuration versus `flat-short`, the `8 < dt <= 48` effects are +0.0692 [0.0543, 0.0845] and +0.0743 [0.0588, 0.0899]. Whole-population, union-mask, full-frame, recall/F1, missed-component and coverage endpoints corroborate both directions. B1 still changes its target interface and is therefore a whole-configuration result.

The simultaneous frontier confirms hierarchy at 896 and 1,152 context tokens, leaves 1,792 unresolved, and confirms flat full-resolution context at 2,432 and 3,072. Thus 1,152 is the largest tested family-wise-confirmed hierarchy point and 2,432 the smallest tested confirmed flat point. No monotonic curve, unique crossing or interpolation is inferred. In exact memory arithmetic, 48 fine frames cost 3,072 context tokens, while 48 coarse frames plus two or six recent fine frames cost 896 or 1,152. The hierarchy therefore provides the same nominal 48-frame reach with 2.7–3.4 times fewer context tokens.

This is a reach-efficiency statement, not a claim of equal retained information or equal performance at every budget.

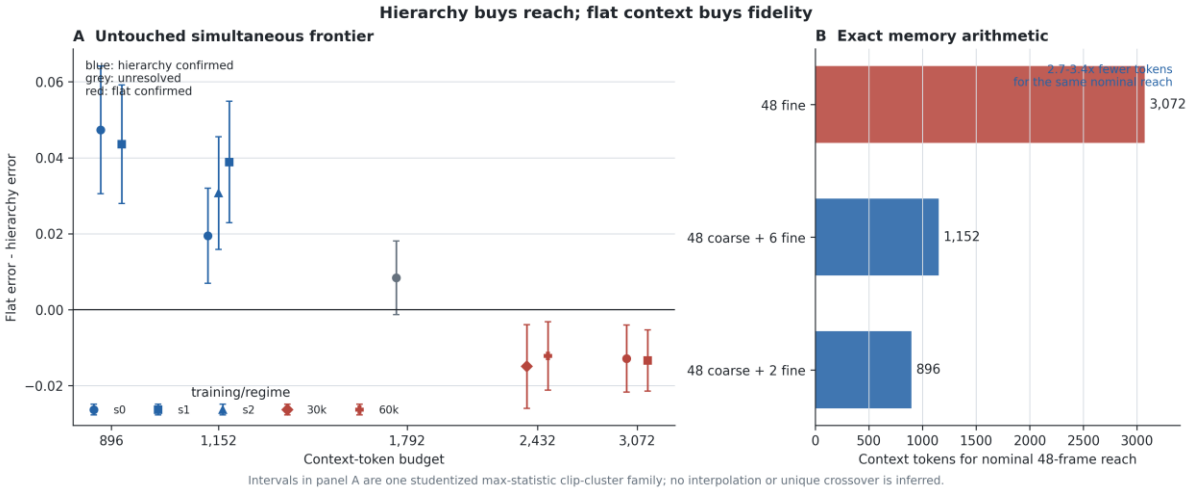



**Figure 4.18 — Untouched reach–fidelity frontier and memory arithmetic.** Positive effects favor hierarchy. All intervals in the left panel belong to one simultaneous family; the right panel shows exact token counts for nominal 48-frame reach. *Source:* `artifacts/confirmation_e2/results.json`; generated by `scripts/make_rft_result_figures.py`.

### 4.6.9 Robustness to Sampling Budget and DDIM Randomness

E7c separates a property of the trained checkpoints from dependence on one default diffusion allocation. Four frozen seed-0 contrasts—896-token reach, the 1,152-token interior point, the near-matched 1,088-token temporal versus 1,152-token spatial package shape, and the 2,432-token flat-fidelity inversion—were crossed with NFE 8/16/32 at sampler seed 0 and NFE 16 at sampler seeds 1/2. All 20 effects were covered by one preregistered simultaneous family.

All 20 point estimates retain the registered direction and 18 simultaneous lower bounds exceed zero. The reach, interior and package-shape contrasts are family-wise clean for every variant. The upper inversion remains positive but is unresolved at sampler seed 1 (+0.0098 [−0.0001,+0.0196]) and seed 2 (+0.0077 [−0.0013,+0.0167]). The registered verdict is therefore `DIRECTIONALLY_ROBUST`, not universal sampler invariance. Non-empty foreground, full-frame, background and union-mask error, recall and F1 agree in all 20 variants; component counts are not uniformly favorable.

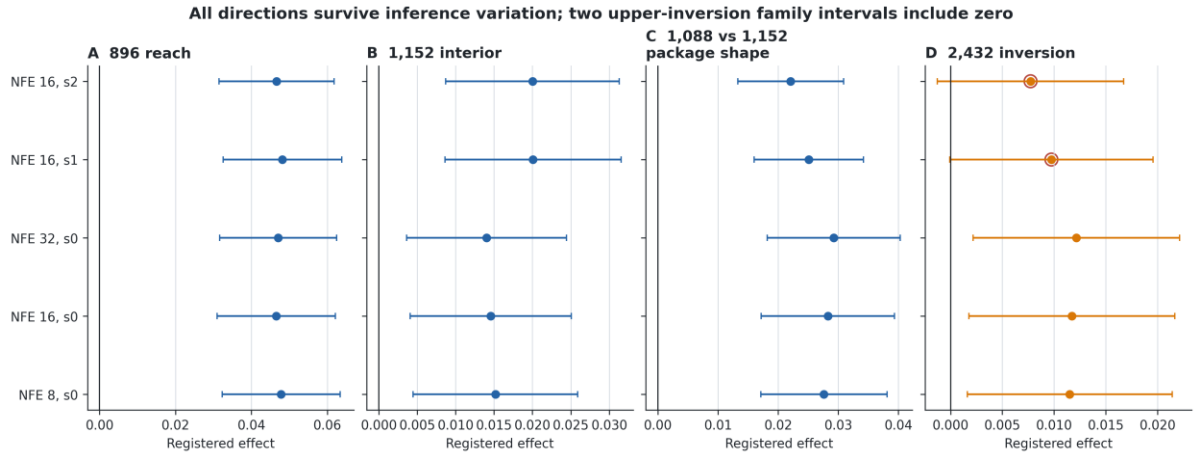



**Figure 4.19 — Robustness to NFE and sampler randomness.** The two open upper-inversion points have simultaneous intervals that include zero. *Source:* `artifacts/e7c_sampler_nfe/results.json`.

### 4.6.10 Robustness to an Independently Trained Latent Representation

E7a is a whole-codec intervention: a second codec seed is trained independently, normalization is recomputed, and the same seven seed-0 diffusion predictors are retrained in the new latent space. It does not isolate codec weights, reconstruction error, latent scale or normalization. Four original-codec and four alternate-codec effects share one new purpose-locked holdout and one simultaneous family.

**Table 4.10 — Robustness of the four registered directions to an independent codec.**

| Codec | 896 reach | 1,152 interior | 1,088 vs 1,152 shape | 2,432 inversion |
|---|---|---|---|---|
| Original | +0.0494 [0.0339, 0.0649] | +0.0283 [0.0169, 0.0397] | +0.0253 [0.0163, 0.0343] | +0.0150 [0.0065, 0.0235] |
| Alternate | +0.0521 [0.0374, 0.0669] | +0.0439 [0.0310, 0.0569] | +0.0112 [0.0039, 0.0184] | +0.0079 [−0.0006, 0.0165] |

All eight point estimates retain direction and seven simultaneous lower bounds exceed zero. The sole unresolved interval is the alternate-codec 2,432-token inversion. The verdict is `CODEC_DIRECTIONALLY_ROBUST`. Codec seed is not an inferential replicate, so descriptive effect changes do not identify a component-level latent-space cause.

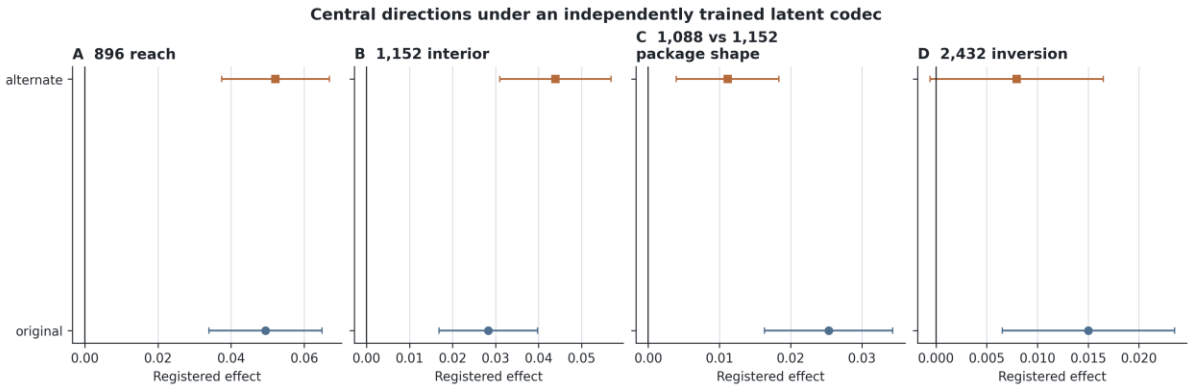



**Figure 4.20 — Whole-codec robustness.** The open alternate-codec upper-inversion interval includes zero. *Source:* `artifacts/e7a_codec_robustness/results.json`.

### 4.6.11 Privileged Pose-Indexed Explicit-Access Reference

E8a separates having relevant history from addressing and reusing it. A fixed world-coordinate mosaic is built from raw warm-up pixels and queried at the exact saved future viewport positions. Unobserved columns use a row prior computed only from training data. The mosaic receives raw pixels and future poses, has no token-matched budget and is not learned; it is therefore a privileged access reference, not an architecture or efficiency competitor.

**Table 4.11 — Simultaneous E8a effects for pose-indexed raw memory.**

| Registered contrast | Estimate | Simultaneous 95% CI |
|---|---|---|
| Raw K14 minus raw K48 | +0.0723 | [+0.0548, +0.0898] |
| Raw K38 minus raw K48 | +0.0210 | [+0.0096, +0.0325] |
| `hier-min` minus raw K48 | +0.1219 | [+0.1031, +0.1407] |
| `hier-long` minus raw K48 | +0.1368 | [+0.1190, +0.1545] |
| `strided-flat` minus raw K48 | +0.1134 | [+0.0944, +0.1324] |
| `flat-c2` minus raw K48 | +0.1310 | [+0.1122, +0.1498] |

K48 reaches foreground MSE 0.0494, full-frame MSE 0.0027 and foreground recall 0.8438, versus foreground MSE 0.1628–0.1862 for the four neural references. All neural-minus-K48 lower bounds are positive, yielding `RAW_ACCESS_CEILING_CI_CLEAN`. Median stored world columns rise from 80 at K6 to 154 at K48, and mean observed future-crop coverage from 0.454 to 0.809. The result shows that reach alone does not guarantee reliable access; it does not rank the privileged mosaic against deployable neural models.

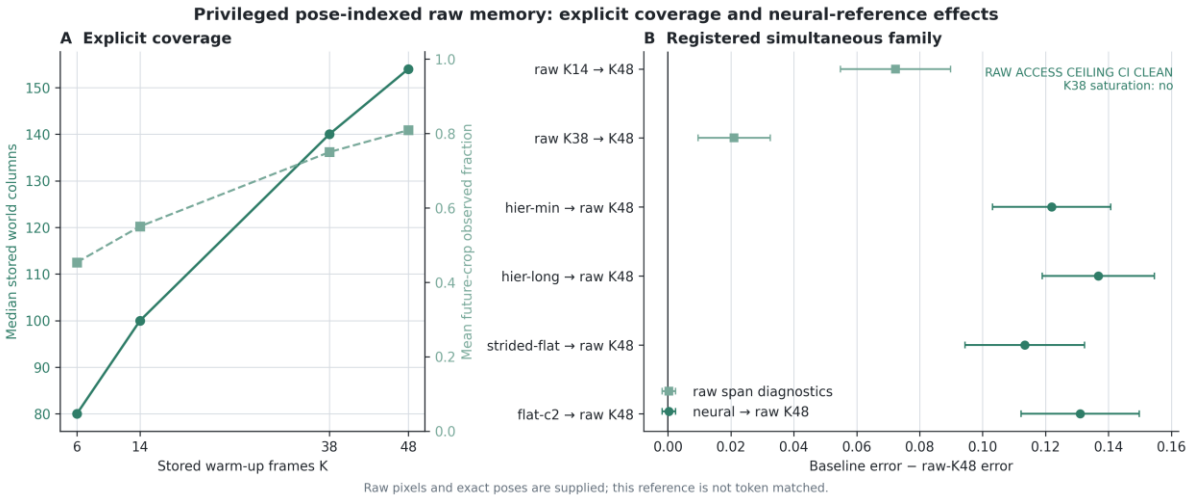



**Figure 4.21 — Pose-indexed raw-memory reference.** Coverage grows with the stored span and all six registered K48 effects have positive simultaneous lower bounds. *Source:* `artifacts/e8a_pose_mosaic/results.json`.

## 4.7 Implementation and Use of the Research Results

The results of the dissertation research have been implemented in:

- the **practical activities of ORNETIKA LLC** (Kyiv): the developed information technology was

used in designing intelligent systems for the analysis of spatio-temporal video data and anomaly detection in complex technical environments; the resource-efficient methods (the TempVerseFormer reversible-transformer method) and adaptive entropy-based tokenization substantially reduced the computational cost of modeling long temporal sequences (implementation act of October 14, 2025);

- the **educational process of VNTU** in the disciplines *Information Technologies for Monitoring

and Analysing the State of Complex Systems *and* Internet of Things and Intelligent Data Analysis* for PhD students of specialty 124 (implementation act of September 19, 2025).

Copies of the implementation acts are given in the appendices **[AUTHOR: assign the final appendix letters and insert the signed act copies]**.

## 4.8 Chapter Conclusions

1. In accordance with **Task 7** of the dissertation research, comprehensive experimental studies

of the developed methods and the intelligent information technology were conducted on the two-operating-point testbed family (§4.1) under an audit-hardened measurement protocol: paired clip-level cluster bootstrap confidence intervals (frame-iid intervals were shown to be 2–7× too narrow at lag-1 autocorrelation +0.86), exact analytic hardness stratification with an audit-corrected anchoring,

common random numbers, per-operating-point reference floors and ceilings (Table 4.1), and full disclosure of every correction and caveat (§4.1.5). The [A2] receipt datasets (2D/3D rotations, Moving MNIST, MMNIST-CIFAR, TaxiBJ, KTH) are retained as the published validations of the reversibility method; the OpenSTL bridge is **measured** (§4.6.2, Table 4.8): the protocol anchor is achieved, and the pre-registered "mid-pack aggregate MSE" clause is refuted and reported as such.

2. The matched-capacity and matched-NFE mechanism study (§4.2) established that **the generation-head ranking is

regime-dependent**: the diffusion core wins the single-step oracle-coverage regime on every column at matched capacity and NFE (fg-MSE 0.0512 vs 0.0664 overall; 0.1549 vs 0.1808 on the hard, genuinely generative stratum), while the regression core wins the deterministic best-of-1 rollout at equal sharpness (ALL delta CI [+0.0110, +0.0227]) — a refutation of the pre-registered fade expectation, reported as such. No statement of the form "one best head" survives the chapter; the memory claims were therefore verified under both viable heads.

3. The energy-based head was characterized by the two-sided capability study (§4.2.4) with the

official implementation and both published recipes at 20,000 converged steps: landscape ordering is recipe-dependent (S2 stably mis-ordered, 21/21 probes; S1 correctly ordered, 15/15 probes from step 6,000), while descent reachability fails under both recipes and under the paper's own inference procedure ($d_{\mathrm{tgt}} \approx 1.12$–$1.29$ against the mean's 0.964; foreground sharpness ≤ 0.035 against the 0.08 structure threshold). The result is a **characterized negative** — a capability boundary at 16 GB-reachable scale with the configuration gap stated (Table 4.4), not a reproduction failure — and it empirically justifies the diffusion selection of Chapter 3 jointly with conclusion 2.

4. The headline claim of the dissertation was confirmed (§4.3): **at an equal context-token

budget (1,152 tokens), the hierarchical long-coarse-memory model beats the flat control on 152-frame autoregressive rollouts on two seeds and under both heads** — seed 0 ALL +0.0189 [+0.0121, +0.0264] with the win concentrated from h ≥ 25 on hard re-entry bins (h 25–60 HARD +0.0344 [+0.0162, +0.0531]); seed 1 ALL +0.0354 [+0.0267, +0.0447], winning every cell; regression pairing ALL +0.0334 seed 0 and +0.0501 seed 1 — two-seed. The single-step memory gate showed the underlying retrieval capability emerging late in training (null at 6,000 steps, ≈13% hard-bin win at 20,000 on both seeds). The claim is honestly bounded: the onset horizon is seed-variable (the robust claim is the win and its mid-horizon concentration); sharpness fades for all arms at depth (the G2b retention clause is not met at h ≥ 81; the ceiling is owned by the frozen decoder); and flat-long at 2.7× the tokens is the outright Pareto frontier (0.1564) — at operating point A, hierarchy is an efficiency point, not a capability necessity.

5. The spatial-abstraction factorial (§4.4) returned an **economic, head-conditional** verdict.

At the fixed 18-frame window, coarse history costs diffusion −0.0082 [−0.0115, −0.0050] at 42% token savings and is statistically free for regression (+0.0008 [−0.0013, +0.0029]); its value is what the savings buy. At 48-frame reach, temporal full-resolution anchors beat spatial coarsening under diffusion on all five seeds, while regression has two temporal and three spatial material winners. The head × package interaction changes sign and its five-seed t95 crosses zero. The robust conclusion is a stable diffusion package result plus regression and cross-head training-seed heterogeneity, not a directional head law or one universally optimal package. The separate E7c inference study (§4.6.9) preserved the expected point-estimate direction in all 20 registered NFE/sampler variants, with 18 simultaneous lower bounds above zero. The 2,432-token inversion is family-wise unresolved at alternate sampler seeds 1/2; therefore the result is directionally robust rather than universally sampler-invariant. The independent-codec study E7a (§4.6.10) likewise preserves all four point-estimate directions: the 896 reach, 1,152 interior and package-shape contrasts remain simultaneous-CI clean, while the alternate-codec 2,432 inversion is +0.0079 [−0.0006, 0.0165]. Thus the reach–fidelity account is directionally robust to both inference allocation/randomness and one whole-codec intervention, while its upper tested edge remains the precision-limited boundary. The privileged explicit-access study E8a (§4.6.11) further gives a clean simultaneous reference: raw K48 foreground MSE is 0.0494, and all four registered neural-minus-K48 lower bounds are positive. This is evidence that geometric reach and reliable addressing are distinct, not a token-matched model comparison.

6. The temporal-capability study at operating point B (§4.5) closed with a characterized

single-step positive and a **registered negative replication of the proposed late trust boundary**: retrieval beyond the flat window exists on the position-unique wide world and is reach-causal (dose-response with the internal contiguous control), a memory span exceeding the effective loop $L_{\mathrm{eff}}$ (eq. 4.1) reliably enables single-step retrieval learning (2/2 span > loop runs CI-positive vs 0/4 span ≤ loop). The early rollout benefit is positive at all three training seeds, whereas the late inversion appears only at seed 0 and reverses at seeds 1–2. No stable trust boundary is therefore claimed. The negative replication, together with the three methodology lessons (effective-loop sizing; the autoencoder-reuse trap; evaluation power), constitutes the honest delimitation of this mechanism.

7. The generalization study (§4.6) carries the measured per-regime configuration selection —

regression head for the deterministic best-of-1 rollout, diffusion for the single-step coverage regime, hierarchical coarse-plus-fine context at the largest affordable window — onto the standard and synthetic suites. Its measured component, the OpenSTL Moving-MNIST bridge (Table 4.8), achieves the protocol anchor (bit-verified standard metrics; the autoencoder ceiling 3.30 shows the tokenizer is not the limiter) and returns the **predicted negative control**: flat ≈ hierarchical (180.91 vs 181.29 aggregate MSE) on a benchmark with no off-screen re-entry — hierarchy pays where the memory horizon exists (§4.3) and is a null where it does not, while the aggregate-MSE gap to the deterministic specialists

(22.06–29.80) is disclosed as the refutation of the pre-registered "mid-pack" clause, and no leaderboard claim is made. The valid persistent-object long Moving-MNIST study (§4.6.2a, Table 4.8a) returns a separate **mixed boundary**: hierarchy improves union error at all three seeds and trajectory/coverage cleanly at two, while flat context preserves digit identity at all three and foreground-only error is unresolved. It therefore supports a support/continuity-versus- identity-fidelity interpretation rather than universal superiority. TaxiBJ is a measured real-data transfer (§4.6.5); the unexecuted 2D-rotation, 3D-polycube, MMNIST-CIFAR and optional KTH-64 protocols are retained only as future-work designs and contribute no projected result to this conclusion. The token-budget frontier completion (§4.3.6; hier-rich at 3,072 tokens, flat-short at 512) is **measured**: the frontier inverts at both ends (hier-rich loses to flat-long −0.0131 [−0.0201, −0.0069] at equal budget and regime; flat-short beats hier-short −0.0036 [−0.0063, −0.0010]) — an outcome stricter than the pre-registered convergence expectation, reported as measured. The final bounded efficiency claim is therefore discrete and empirical: hierarchy is family-wise confirmed at the tested 896- and 1,152-token points, unresolved at 1,792, and full-resolution flat context is confirmed at 2,432 and 3,072; no untested interval or unique crossing is inferred.

8. The results of the research have been implemented in the practical activities of ORNETIKA LLC and in the educational process of VNTU (§4.7, per the implementation acts).

The results of this chapter have been published in [A4] (the mechanism, memory, factorial, and boundary studies) and are connected to [A2] through the published receipt summarized in §4.1.2. The stable labels are bound mechanically to the final numbered publication list.

9. The chapter's findings consolidate into **scoped empirical regularities and explicit boundaries **(single source of truth: `docs/CLAIM_LEDGER.md`; every statement names its testbed, head, representation and seed level):** (L1) reach **— long-span packaging wins at tested lower/interior budgets when it contains the relevant re-entry past and the comparator does not;** (L2) fidelity inversion **— the simultaneous family confirms hierarchy at 896/1,152, leaves 1,792 unresolved and confirms flat at 2,432/3,072, without interpolation;** (L3) stride-family configuration **— the 512-token stride-2 arm wins strongly but changes its target interface;** (L4) package shape **— sparse full-resolution history wins under diffusion on five seeds, while regression and the cross-head interaction are training-seed-heterogeneous;** (L5) depth — hypothesis, not law **— a depth-plus-objective-allocation probe on single-scale data;** (L6) self-fed reliability — negative replication, not law **— the seed-0 late inversion is not reproduced at seeds 1–2; and** (F4) inference robustness **— E7c retains all 20 NFE/sampler effect directions, with 18 simultaneous lower bounds positive and the upper inversion unresolved at alternate sampler seeds 1/2;** (F5) representation robustness **— E7a retains all four alternate-codec point directions, with three family-wise clean contrasts and the upper inversion again precision-limited;** (F6) explicit-access reference **— E8a gives a clean privileged raw-memory ceiling**

**while remaining outside token-matched model ranking; and** (X3) external persistent-object boundary** — E6 separates hierarchy-favoring spatial support/continuity from flat-favoring digit identity under one valid protected simultaneous family. Together, these confirmed regularities and explicit boundaries form a scoped, falsifiable theory of context allocation—the main scientific conclusion of the chapter.

# General Conclusions

In the dissertation research, the actual scientific and applied problem of systems analysis, mathematical modeling, and uncertainty resolution in complex multi-step systems of hierarchical video data generation has been solved: methods of systems analysis, mathematical models, and intelligent information technologies have been developed that improve the efficiency and the temporal and spatial consistency of hierarchical video generation processes. The purpose of the research has been achieved, and all seven of the set tasks have been fulfilled.

The resource contribution is best understood along two orthogonal memory axes. The hierarchical context-packaging model reduces the context/state memory needed to reach task-relevant history: distant frames can be stored at a cheaper spatial or temporal rate, while a short recent tail retains detail. TempVerseFormer reduces the activation memory needed to optimize long temporal computations by reconstructing intermediate states during the backward pass. These are complementary dissertation results, not one experimentally integrated architecture and not a claim that their savings multiply. Together they support one systems principle: **compress what must be remembered; reconstruct what need not be stored**.

The main scientific and practical results of the dissertation research are as follows.

**1.** *(Task 1)* The process of hierarchical video data generation has been characterized as a complex control system with inherent informational, structural-algorithmic, computational, and target uncertainties; structural, hierarchical, and temporal decompositions of the system have been performed, and the central systemic contradiction of the field has been identified — the resource blow-up (quadratic attention compute, linearly growing key–value memory, linearly growing training-activation memory) and the consistency degradation of flat long-context modeling, whereby content that leaves the field of view is not restored consistently upon its return. The analysis of existing approaches (Chapter 1) identified codebook-specific optimization risks in discrete vector-quantized hierarchies and delimited the implemented scope: hierarchical modeling in compact continuous hierarchical latent state spaces. Because the repository contains no controlled VQ-versus-continuous experiment, no representation-family superiority is claimed. Energy-based models are reviewed as literature and later characterized experimentally at the tested scale.

**2.** *(Task 2)* The criteria apparatus of systems analysis of generative processes has been improved by introducing the criteria of conceptual consistency (the ability to generate unknown combinations), hierarchical consistency (alignment of structural levels), and trajectory consistency, the latter operationalized through measurable autoregressive-rollout degradation metrics stratified by an exactly computable re-entry hardness label (the novel-foreground fraction $\nu_m$, eq. 3.18). The

operationalized criterion proved decision-grade in practice: it is what localized the hierarchical model's advantage to the hard re-entry strata and the beyond-flat-horizon bins (Table 4.6), and what turned the boundary study's negative into an exactly stated boundary rather than an unexplained failure.

**3.** *(Task 3)* A method for estimating the state of uncertainty of spatio-temporal data has been developed through their adaptive decomposition based on entropy metrics (the V-BLT architecture) with N-dimensional rotary positional embedding (RoPE-ND), which allows dynamic allocation of computational resources to high-information zones of the scene and mathematical alignment of coordinate systems between abstraction levels of hierarchical models. Per-pixel byte-level entropy maps were shown to act as a reliable diagnostic indicator of complex kinematics, and the RoPE-ND coordinate-scaling mechanism provides an instrument for cross-level attention localization in architectures that require spatial coordinate transfer. The measured Chapter 3 predictor uses learned level/role/frame embeddings in joint attention and therefore does not depend on RoPE-ND; the two results are complementary instruments rather than one being the implementation basis of the other.

**4.** *(Task 4)* A method for the computationally efficient optimization of multi-step iterative systems has been developed based on reversible transformations for time-agnostic backpropagation, ensuring constant activation-memory consumption independent of the temporal horizon (Lemma 1: $\mathrm{O(T \cdot L \cdot D) \to O(L \cdot D)}$), and extended to stochastic multi-level recurrences via the exactly invertible store-the-noise recurrence: state reconstruction is exact at numerical precision ($5.5 \times 10^{-15}$ in fp64 for the deterministic coupling; $8.4 \times 10^{-13}$ for the stochastic scheme), and reverse-replay gradients coincide with backpropagation through time (cosine similarity 1.000000). The measured memory frontier confirms the practical payoff on a consumer-grade 16 GB device: classical BPTT consumes 12.2 GB at rollout length $\mathrm{K} = 40$ and fails with out-of-memory at $\mathrm{K} = 120$, whereas the developed method remains at ≈1.1–1.2 GB through $\mathrm{K} = 160$, with gradient checkpointing a comparable alternative (1.47–1.97 GB). The method's applicability domain has been delimited by the three-axis reversibility taxonomy (depth — time — computation): the temporal-rollout axis pays and is the resource foundation of long-horizon training; the inner-optimization-loop axis provably cannot pay against checkpointing (measured saving 0.26% of peak at $\mathrm{K} = 128$, analytic band ≈0.6–1.0%; crossover at $\mathrm{K} \approx 5{,}000$–$30{,}000$ inner steps, far beyond practical refinement depths) — a characterized negative stated as a scoping theorem of the method. It has additionally been established by measurement that invertibility does not imply stability (perturbation growth from $333\times$ to $1.0\times$ over 100 steps purely as a function of the coupling Lipschitz bound), so trajectory stability is a separate design requirement.

**5.** *(Task 5)* A hierarchical mathematical model of multi-level video generation in compact hierarchical latent state spaces has been developed, supporting **multiple context streams at different temporal rates and spatial resolutions** over one shared attention backbone (the context-stream family of

§3.1/§3.3.5). Long-range temporal memory is carried by a long-span stream: the **coarse stream is the measured spatial implementation** (48-frame coarse context at 16 tokens per remembered frame versus the fine level's 64 — a 4× token and asymptotically ≈16× attention-cost advantage per memory frame), and **strided full-resolution history is the measured temporal implementation** (span 48 at 11 anchors; it beats the coarse form under the diffusion head on all five tested training seeds, while regression package ordering is seed-heterogeneous) — with unidirectional inter-level context transfer (gradient detachment at the representation/dynamics and past/future boundaries) and a per-frame iterative-refinement (diffusion) generation core selected by controlled mechanism analysis. The model demonstrably restores content re-entering the field of view beyond the horizon of a flat context window at a matched context-token budget: on 152-frame autoregressive rollouts the hierarchical model beats the token-matched flat control on two training seeds (seed 0: ALL +0.0189 [+0.0121, +0.0264], with the win concentrated from horizon 25 onward and peaking on hard re-entry bins, h 25–60 HARD +0.0344 [+0.0162, +0.0531]; seed 1: ALL +0.0354 [+0.0267, +0.0447], winning every horizon-hardness cell). This specific reach comparison also replicates under the regression head (every bin at both seeds, ALL +0.0334 [+0.0248, +0.0423] and +0.0501 [+0.0396, +0.0612]), and the underlying single-step retrieval capability reduces the hard re-entry error by ≈13% on both seeds (+0.0135 [+0.0080, +0.0196] and +0.0143 [+0.0083, +0.0208]), emerging only at the larger optimization budget (null at 6,000 steps, win at 20,000). The claim is honestly scoped: **the hierarchy is validated where its package reaches memory-relevant history under the tested budget, while full-resolution fidelity can recover at larger tested budgets**. At the primary operating point a flat model paying 2.7× the tokens remains the outright Pareto frontier (0.1564), so hierarchy there is an efficiency point, not a capability necessity; and at the capability operating point the single-step span-threshold capability (memory span exceeding the effective loop $L_{\mathrm{eff}} = \mathrm{world}/(\mathrm{speed}(1 - p_{\mathrm{hold}}))$ enables retrieval learning in 2/2 runs versus 0/4 below the threshold). In self-fed rollout, the hierarchy's early benefit is positive and CI-clean at all three training seeds, but the late inversion appears only at seed 0 and reverses to a positive late hierarchy effect at seeds 1–2. The registered E5a verdict is NOT_REPLICATED; no general trust boundary is claimed.

**6.** *(Task 6)* An intelligent information technology and a software complex (Python/PyTorch) have been created for the systems analysis of hierarchical video generation processes, integrating the developed methods of adaptive entropy-based tokenization, resource-efficient reversible optimization, and hierarchical generation, together with the purpose-built memory-horizon looping testbed and its re-entry-stratified evaluation protocol as a first-class measurement instrument: analytic (exactly computable) hardness labels from recorded control tracks, foreground-masked per-image metrics with verified thresholds and decoder-set ceilings, mandatory decoded-sample review, paired clip-level cluster bootstrap confidence intervals with common random numbers (frame-iid intervals were measured to be

2–7× too narrow at rollout autocorrelation +0.86), and per-operating-point reference floors and ceilings. The instrument operationalizes the trajectory-consistency criterion into a statistically disciplined measurement procedure, and the complete benchmark trains and evaluates on a single consumer-grade 16 GB GPU.

**7.** *(Task 7)* Comprehensive experimental verification of the developed methods and technology has been conducted (Chapter 4): a matched-capacity, matched-NFE comparison of prediction mechanisms established that the generation-head ranking is regime-dependent — diffusion wins the single-step oracle-coverage regime on every column (foreground error 0.0512 versus 0.0664 overall; 0.1549 versus 0.1808 on the hard, genuinely generative stratum), regression wins the deterministic best-of-1 rollout at equal sharpness (ALL delta CI [+0.0110, +0.0227]; a pre-registered expectation refuted and reported as such) — and included the two-sided characterized study of energy-based descent with the official implementation (landscape ordering recipe-dependent: stably mis-ordered under one published recipe at 21/21 probes, correctly ordered under the other at 15/15 probes from step 6,000; descent reachability failing under both recipes and under the paper's own sampler — a capability boundary at 16 GB-reachable scale, not a reproduction failure). The spatial-abstraction factorial isolated resolution from window length and returned an economic, head-conditional verdict (coarse history costs the diffusion head −0.0082 [−0.0115, −0.0050] at 42% token savings and is statistically free for the regression head, +0.0008 [−0.0013, +0.0029]; window extension at fixed representation is worth −0.027 and dwarfs the abstraction cost). The temporal-capability study at the second operating point closed with a registered negative replication of the seed-0 late trust inversion (conclusion 5) and three transferable methodology lessons (effective-loop span sizing; the autoencoder-reuse trap; evaluation power against plausible effect sizes). Memory-consumption measurements confirmed the constant-memory rollout capability (conclusion 4). The measured two-level spatial package outperforms the token-budget-matched flat control in the tested lower/interior memory-horizon regime (the shape comparison across packagings is conclusion 7's separate, seed-resolved statement), and the boundary conditions of every claim are stated explicitly. The published validation of the reversibility method ([A2]: procedurally generated 2-D rotating shapes under controlled temporal laws) is retained as its receipt; the bridge evaluation on the standard OpenSTL Moving-MNIST suite is a separate measured dissertation experiment and returns the predicted negative control — flat ≈ hierarchical (180.91 vs 181.29 aggregate pixel-sum MSE; autoencoder ceiling 3.30, so the tokenizer is not the limiter) on a benchmark with no off-screen re-entry and hence no memory horizon to carry, exactly complementing the operating-point-A win, with the aggregate-MSE gap to the deterministic specialist regressors (22.06–29.80) disclosed and no leaderboard claim made. A separate valid persistent-object 48→300 Moving-MNIST study resolves the long-rollout external boundary: hierarchy improves union error at all three seeds and trajectory/coverage cleanly at two, while flat context preserves digit identity at all three and

foreground-only error is unresolved. This `MIXED_BOUNDARY` result shows that spatial support/continuity and fine identity fidelity can favor different packages. The shape study (§4.4) established that the packaging axes are not interchangeable: the win band's lower edge follows each family's measured configuration (896 tokens for the stride-1 coarse family, two seeds; the stride-2 512-token configuration wins on both diffusion seeds but changes its target interface), and temporally subsampled full-resolution history beats the coarse packaging under the diffusion head on all five training seeds (effects +0.0046 to +0.0197), whereas regression has two temporal and three spatial material winners and the head × package interaction changes sign — the findings motivating the scoped empirical regularities of §4.8 (ledger: `docs/CLAIM_LEDGER.md`), whose conclusion is a testable conditional design rule rather than universal superiority: allocate context tokens to cover the task-relevant span with the least lossy family that achieves it; spend the remainder on resolution and recency; add depth only where the data contain a distinct slower factor. Extension to the remaining synthetic suites (2D rotations, 3D polycubes, MMNIST-CIFAR) is stated as **future work of the dissertation programme** (Chapter 4 §§4.6.3–4.6.5 carry the pre-registered protocols; per the shell end-state rule no projected result is claimed here). Measured in full, by contrast, is the **token-budget frontier completion** (hier-rich at 3,072 tokens loses to flat-long, −0.0131 [−0.0201, −0.0069] at seed 0 and −0.0107 [−0.0163, −0.0058] at seed 1 — a two-seed inversion — at equal budget and training regime; flat-short at 512 beats hier-short, −0.0036 [−0.0063, −0.0010] — the frontier inverts at both ends, an outcome stricter than the pre-registered convergence-at-saturation expectation and reported as measured) and the **measured lower edge of the win band** (a pre-registered geometric prediction confirmed the same day: at 896 tokens — the cheapest budget whose 48-slot coarse span covers the ≈48-frame loop, against a flat window of only 14 frames — the hierarchical arm wins every horizon × hardness cell, +0.0399 [+0.0299, +0.0501] at seed 0 and +0.0358 [+0.0267, +0.0454] at seed 1 — a two-seed result, the largest hierarchical advantage of the campaign), which together give the efficiency claim its final, tested-bracket form: **the simultaneous family confirms hierarchy at the tested 896- and 1,152-token lower/interior points, leaves 1,792 unresolved, and confirms flat fidelity at 2,432 and 3,072 tokens**. Thus 1,152 is the largest tested family-wise-confirmed hierarchy win and 2,432 the smallest tested confirmed flat win; monotonicity, a unique crossing and interpolation between tested budgets are not claimed. A separate frozen E7c study retained the expected point-estimate sign in all 20 NFE/sampler variants, with 18 simultaneous lower bounds positive; the upper inversion remains family-wise unresolved at alternate sampler seeds 1/2, so the result is directionally robust rather than universally sampler-invariant. The independent-codec E7a study retained all four registered directions: 896 reach +0.0521 [0.0374,0.0669], 1,152 interior +0.0439 [0.0310,0.0569], package shape +0.0112 [0.0039,0.0184], and the 2,432 inversion +0.0079 [−0.0006,0.0165]. Hence representation robustness is directional for this whole-codec intervention, while the upper edge remains family-wise precision-limited. The separate E8a explicit-access reference reaches foreground MSE 0.0494 with raw K48 memory; the four registered neural-minus-K48 effects

range from +0.1134 to +0.1368 and all simultaneous lower bounds are positive. Its `RAW_ACCESS_CEILING_CI_CLEAN` verdict shows that exact addressing and lossless storage can remove much of the remaining static-world error, but the exact future poses, raw-pixel payload and unmatched budget prohibit a neural-model or efficiency ranking. The E6 result is therefore not averaged into one architecture score: it is retained as the external reach–fidelity boundary showing what coarse long memory preserves and what it loses.

Beyond the numbered results, the dissertation deliberately records its **characterized negatives** as scientific results of equal epistemic rank: (i) the energy-based generation core is a two-sided characterized negative at 16 GB-reachable scale (ordering versus reachability dissociate); (ii) reversibility on the inner-optimization-loop axis cannot pay against gradient checkpointing (0.26% measured saving — the three-axis taxonomy closes this axis by analysis and measurement); (iii) spatial abstraction is not an intrinsically better representation — its verdict is economic and head-conditional; and (iv) the **combined-optimum readout** (regression head + coarse history + long window: 0.1682 at 1,152 tokens; at the short window, 0.2008 at 672 tokens versus the diffusion flat control's 0.2038 at 1,152) is reported as a *measured arm ordering* within the evaluated campaign, **not** as a factorial-interaction claim.

**The practical significance** of the obtained results lies in the development of a complex of algorithms and software-information tools implementing the developed intelligent information technology for the systems analysis of hierarchical video generation processes (Python/PyTorch; repository modules for the hierarchical model, the reversible recurrence, and the testbed generator). Unique synthetic spatio-temporal datasets have been created for testing generation and optimization systems: 2D shapes, 3D polycubes, and the looping memory-horizon testbed with recorded action tracks and analytic hardness labels (3,000 training clips × 100 frames; 500 test clips × 200 frames at the primary operating point). The developed technology has been implemented in the practical activities of ORNETIKA LLC (implementation act of October 14, 2025) and in the educational process of VNTU (implementation act of September 19, 2025), per the acts covering the reversible-transformer method and entropy-based tokenization. The proposed technology can be applied to state estimation and prediction of future states of complex systems in technical, medical, and environmental monitoring.

**Prospects for further research** follow directly from the measured boundary conditions rather than from speculation. First, **self-fed reliability across tasks and seeds**: single-step span-threshold retrieval exists, but the proposed late rollout inversion is training-seed-sensitive rather than a replicated boundary. A future study, if pursued, must be newly preregistered across multiple tasks, horizons and training seeds; it should compare teacher-forced, self-fed and controlled-refresh conditions instead of treating E5b as localization of an established boundary. Distribution-matching rollout objectives remain a possible intervention, with the constant-memory reversible method of Chapter 2 providing an implementation path [80]. Second, the **extension of the memory-horizon protocol to real video**: transferring the re-

entry-stratified evaluation methodology from the analytic testbed family to natural-video data of the KTH class (where hardness labels must be estimated rather than computed exactly), building on the completed OpenSTL bridge as the connectivity anchor and executing the future-work protocols of §4.6. These directions require no revision of the present results: they extend a boundary currently stated only for its measured testbeds and configurations.

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External sources are numbered by first appearance in the assembled dissertation.


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